ESE 4105 Complete Study Pack

Fuel Combustion and IC Engines · KUET · past papers 2020, 2021, 2022, 2023, 2024 · answers from the course slides

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 8 h for Part A and 6 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×5, 2023 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: in Part B, the rope brake numerical (B2, 20 marks in 2024) and the Otto efficiency derivation (B7, 2024) sit below the pass line only because they appeared once, in the newest paper. Both are pure method marks and take under an hour together: they are the best extra hour. B11 (CI combustion stages) is worth reading with B13/B14, because the knock answers lean on it.

Watch the section split: this pack groups topics by subject. In 2024 cooling (A13, A14) and emissions (A16–A18) were asked in Section B, and SI combustion stages (B10) in Section A; in 2021–2023 they sat as listed here. Study by the lists, then answer whichever 3 questions in each section you know best.

Section A: Fuels, combustion, mixture preparation, cooling, lubrication, emissions

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
A1 Fuel properties: octane, cetane, API gravity, RVP, aniline point, diesel index	Very likely 100%	Easy	9%
A9 Mixture requirements: idling, cruising, full throttle, acceleration, starting	Very likely 100%	Easy	16%
A13 Need for cooling and lubrication; effects of overcooling	Very likely 93%	Easy	21%
A14 Air vs liquid cooling; water cooling system, thermostat, thermosyphon	Very likely 80%	Easy	31%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A18 Catalytic converter: construction, reactions, temperature and equivalence ratio effects	Very likely 100%	Medium	39%

A2 Proximate vs ultimate analysis; HCV vs LCV	Very likely 93%	Medium	44%
A16 Pollutant formation in SI and CI engines; NOx mechanism; smog	Very likely 87%	Medium	52%
A11 Fuel injection system: purpose, components, types	Very likely 80%	Medium	58%

Block 4: likely · 2.6 h

Topic	Chance	Level	Covered
A4 Classification of fuels; storage and handling of solid, liquid and gaseous fuels	Likely 60%	Easy	62%
A8 Carburetion and the simple carburetor; compensating jet	Likely 60%	Easy	65%
A12 Injector nozzles: types, spray pattern, atomization, sac volume	Likely 67%	Medium	69%
A10 Venturi throat diameter numerical (compressible, isentropic)	Likely 60%	Hard	72%

Only if time is left · 3.2 h

Topic	Chance	Level	Covered
A17 Particulate emission, its measurement, smoke colours	Possible 53%	Easy	76%
✓ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 7.7 h)			
A3 Alternative fuels: ethanol, biodiesel, methanol, hydrogen	Possible 40%	Easy	83%
A15 Lubrication: parts lubricated, oil selection, classification, mist lubrication	Possible 33%	Easy	86%
A7 Producer gas air requirement numerical; fluidized bed combustion	Possible 33%	Easy	89%
A6 Combustion: definition, types, air-fuel ratio and its effect	Possible 33%	Easy	91%
A5 Petroleum refining: fractional distillation, cracking, reforming	Possible 13%	Easy	93%
A19 Emission control techniques; NOx reduction; SI vs CI emissions; blow-by	Possible 20%	Medium	100%

Section B: Performance, cycles, valve timing, SI/CI combustion, knock, supercharging

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B15 Supercharging and turbocharging: effects, SI vs CI suitability, limits, modifications	Very likely 100%	Medium	12%
B1 Performance parameters and performance curves (BP, torque, BSFC, ηv vs speed)	Very likely 87%	Medium	20%

Block 4: likely · 4.7 h

Topic	Chance	Level	Covered
B6 Valve timing diagram, valve overlap, why inlet valves are larger	Likely 73%	Easy	26%
B10 Stages of combustion in SI engine	Likely 73%	Easy	31%
B5 Stoichiometry, equivalence ratio, rich vs lean; why CI runs lean	Likely 67%	Easy	37%
B12 Surface ignition, pre- vs post-ignition, end gas, flame front	Likely 60%	Easy	41%
B13 Knock in SI vs CI; ignition delay SI vs CI; SI knock limited	Likely 67%	Medium	53%
B4 Indicator diagram: indicated and pumping work; pumping loss, WOT, turbocharged engines	Likely 67%	Medium	61%
B14 CI knock factors and ignition delay; good SI fuel is bad CI fuel; poor fuel in large slow CI; smoke limited	Likely 60%	Medium	72%
B9 SI mixture formation strategies; HCCI and DISI	Likely 60%	Medium	75%

✔ Pass line: 75% of Section B marks covered, enough to choose 3 of the 4 questions (about 6.2 h)

Only if time is left · 3.7 h

Topic	Chance	Level	Covered
B8 Scavenging; two-stroke vs four-stroke; why 2-stroke CI is common but 2-stroke SI is not	Possible 40%	Easy	83%
B11 Stages of combustion in CI engine (ROHR vs crank angle); flames; SIT	Possible 53%	Medium	90%
B3 Volumetric efficiency: significance, mean piston speed, valve timing and speed effects, quasi-static effect	Possible 33%	Medium	94%
B2 Rope brake engine test numerical (BP, IP, ηm, ηbth, BSFC)	Possible 33%	Medium	97%
B7 Air-standard cycles: Otto efficiency derivation; Diesel cut-off ratio and γ	Possible 33%	Hard	100%

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One-shot notes: ESE 4105 Fuel Combustion and IC Engines

Fuel properties

Octane = % iso-octane in iso-octane + n-heptane matching knock (resist auto-ignition; SI). **Cetane** = % n-hexadecane in hexadecane + 1-methylnaphthalene matching delay (ready to ignite; CI). ON 85 = knocks like 85/15 blend; CN 60 = good diesel.

API = 141.5/SG − 131.5 (hydrometer; light > 31.1, heavy < 22.3). **RVP**: 38 °C, air chamber 4× fuel chamber, volatility (start vs vapour lock). **Aniline point**: aniline-oil miscibility temperature, low = aromatic = poor diesel. **Diesel index** = AP(°F) × API / 100. Flash point, fire point (5 s), pour, cloud point.

Analysis and CV

Proximate: moisture 108-110 °C, VM 950 °C 7 min covered, ash 725 °C, FC by difference. **Ultimate**: C, H (KOH/$CaCl_2$), N (Kjeldahl), S ($BaSO_4$), O by difference.

HCV = 33 800C + 144 000(H − O/8) + 9 270S kJ/kg. LCV = HCV − 9H × 2442. Stoich air = (2.67C + 8H + S − O)/0.23 kg/kg.

Fuels: classes, storage, alternatives

Primary (wood, coal, crude, NG) vs secondary (coke, petrol, LPG, producer gas); solid/liquid/gas; fossil, by-product, chemical, nuclear.

Storage: solid = self-heating + dust; liquid = vapour, flash point, bunds, earthing; gas = leaks, pressure, odorant, ventilation.

Ethanol ON ~108, E10/E85, LCV 2/3. Biodiesel (FAME, transesterification) higher CN, less CO/HC/soot, bit more NOx. Methanol: half CV, toxic, corrosive, cold start, formaldehyde. H_2: SIT ~585 °C → dual fuel with diesel pilot / glow plug; 4-75 % flammable; only NOx.

Refining: distillation 300-350 °C column 20-30 m, cut every ~50 °C; cracking (heavy → light); reforming (Pt, 480-540 °C, naphthenes → aromatics, raises ON).

Combustion and A/F

Fire triangle: fuel + O_2 + ignition temperature. Types: complete, incomplete, rapid, spontaneous, explosion; premixed vs diffusion.

$\varphi = (F/A)/(F/A)_s = 1/\lambda$. Stoich A/F: petrol 14.7, diesel 14.5, CH_4 17.2. Best power ~12.5-13 (rich), best economy ~16-17 (lean).

Producer gas (H_2 25, CO 15, CH_4 5, CO_2 15, N_2 40): O_2 = 0.125 + 0.075 + 0.10 = 0.30 m³ → air = 0.30/0.21 = **1.43 m³/m³**.

Bubbling FBC: sand/limestone bed fluidised by air, 750-950 °C, drying → devolatilisation (volatiles burn in bubbles/freeboard) → char burns in bed; particle convection to immersed tubes + gas convection + radiation.

Carburetor and mixture

Venturi: V↑ p↓; float chamber at p_{atm}; Δp pushes fuel from jet; throttle after venturi; choke before. Simple carb richens with flow → **compensating jet** (leans with flow) / air-bleed emulsion tube keep A/F constant. Accelerator pump for transients.

Requirements: idle rich (~12, dilution by residual), cruise lean (best economy), full power rich (~12.5), cold start very rich, acceleration extra squirt.

Diluting: more expansion work (γ↑), less pumping, less heat loss and NOx.

Venturi numerical

$C_2 = C_v\sqrt{[2c_pT_1(1 - (p_2/p_1)^{0.286})]}$ → p_2; $\rho_1 = p_1/RT_1$; $\rho_2 = \rho_1(p_2/p_1)^{0.714}$; $A = \dot{m}/\rho_2C_2$; $d = \sqrt{(4A/\pi)}$.

2024: 0.1 kg/s, 1.02 bar, 298 K, 100 m/s, C_v 0.75 → p_2 = 0.918 bar, ρ_2 = 1.106, A = 9.04 cm², **d = 33.9 mm** (slide's simplified way: 37.8 mm).

2023 (take 102 m/s): 0.0833 kg/s, 1.05 bar, C_v 0.8 → **d = 30.1 mm**.

Fuel injection

Purpose: meter, time, rate, atomize, distribute, sharp cut-off. Parts: tank, feed pump, filters, HP pump, governor, HP lines, injector. Air injection (old) vs solid: individual pump, unit injector, distributor, common rail (1000-2000 bar, ECU). SI: TBI, MPFI, GDI.

Nozzles: single-hole (15°, dribbles), multi-hole (4-18 × 0.15-0.35 mm, DI), pintle (hollow cone 0-60°, low pressure, IDI), pintaux (auxiliary hole, cold start). **Sac volume** → late HC; VCO nozzle.

Cooling and lubrication

Need cooling: 2500 °C gas, oil film, seizure, thermal stress, knock, η_v. Overcooling: poor vaporization, oil dilution, corrosion, friction, low η.

Air (fins; bikes, aircraft, small) vs liquid (jackets, radiator, pump, fan, thermostat; cars, large; uniform). Circuit: jacket → thermostat → top tank → core + fan → bottom tank → pump. Thermostat wax/bellows, opens ~70-80 °C; stuck shut = overheat, stuck open = overcool. Thermosyphon: density difference, slow, stops if level low.

Lube: friction, cooling, sealing, cleaning, shock. SAE W at −18 °C, summer at 99 °C, multigrade 20W-50. Mist (petroil) 1:50 in 2-stroke: no pump, but smoke/deposits. Wet sump (splash, pressure), dry sump.

Pollutants

SI HC: crevices (38 %), quench, oil film, deposits, liquid fuel. CO: rich. NOx: Zeldovich O + N_2 → NO + N, N + O_2 → NO + O, N + OH → NO + H; > 1800 K, peaks slightly lean, grows with time. Also prompt (Fenimore) and fuel-N.

CI: over-lean spray edge HC, sac HC, soot in rich core, NOx in diffusion flame (NO_2 10-30 %), low CO.

Smog: NO_2 + hv → NO + O → O_3, PAN.

PM: filter-collected at ≤ 325 K: soot + organic fraction. Measure: dilution tunnel + filter, Bosch smoke, Hartridge opacity. CI >> SI (heterogeneous). Smoke: black = soot, blue = oil, white = fuel/water.

Catalytic converter and control

Steel shell, cordierite honeycomb, Al_2O_3 wash-coat, Pt/Pd (oxidise CO, HC), Rh (reduce NO), CeO_2 (O_2 store). 2CO + O_2 → 2CO_2; HC + O_2 → CO_2 + H_2O; 2NO + 2CO → N_2 + 2CO_2.

Light-off 250-300 °C (S-curve); > 800-900 °C sinters. φ window at 1: lean → NOx conversion drops; rich → CO/HC drop. Lead poisons.

Control: engine design/operation, after-treatment (thermal reactor > 600 °C, catalyst, DPF, SCR), fuel change, PCV, canister. NOx: EGR, retard, water injection, lean, SCR. Blow-by: gas past rings → PCV valve to intake.

Section B: performance, cycles, combustion, knock, supercharging

Performance

$\eta_v = 2\dot{m}_a/(\rho_a V_d N)$; power $\propto \eta_v$. BSFC = $\dot{m}_f$/BP. IMEP = as/l. BP = $2\pi NT/60$. FP = IP − BP; η_m = BP/IP; η_{bth} = $3600BP/(\dot{m}_f CV)$.

Curves vs speed: η_v and torque hump (max mid speed); BP peaks at higher speed; BSFC bowl (min near max-torque speed).

Mean piston speed 2LN = 8-15 m/s: above it choking, inertia, friction; below it low power.

Late IVC: ram effect at high speed, backflow at low speed. Quasi-static: fuel vapour partial pressure, p_e/p_i and r (residual).

Rope brake (2024 6b)

BP = $(W - S)\pi DN/60 = 350\pi(0.6)(600)/60$ = **6.60 kW**. p_{mi} = 420 × 1.1/60 = **7.7 bar**. IP = $p_{mi}LAn/60$, n = N/2 = 300 → **4.54 kW**. η_m = 1.45 (data inconsistent; say so). η_{bth} = 6.597 × 3600/(0.815 × 44 000) = 66.2 %. BSFC = 0.124 kg/kWh.

Pumping and lean CI

Upper loop = gross indicated work; lower loop = pumping work; net = A − B. SI throttle → low intake p → big pumping loss at part load; CI no throttle. Reduce: VVT/early IVC, lean/EGR, GDI stratified, downsizing, cylinder deactivation. WOT: throttle wide open, max torque, min pumping. Turbo: $p_{in} > p_{ex}$ → pumping work positive.

CI lean: quality governed (no throttle, full air), spray must find O_2, smoke limit φ ~0.7-0.8.

Valve timing

IVO 10-30° bTDC; IVC 30-60° aBDC (ram); EVO 30-55° bBDC (blow-down); EVC 10-20° aTDC (scavenging); spark 20-30° bTDC. Reasons: valves open gradually, gas inertia, blow-down, finite burn. Overlap: good scavenging at speed, backflow and charge loss at low speed.

Inlet valve larger: intake Δp only ~0.1-0.2 bar vs exhaust blow-down 3-5 bar; smaller exhaust valve easier to cool.

Cycles

Otto: $T_2/T_1 = T_3/T_4 = r^{\gamma-1}$ → **$\eta = 1 - 1/r^{\gamma-1}$** (r = 8: 56.5 %).

Diesel: $\eta = 1 - (\rho^{\gamma} - 1)/[\gamma\, r^{\gamma-1}(\rho - 1)]$; ρ↑ → η↓. γ↑ → η↑. Same r: Otto > Dual > Diesel; same p_{max}: Diesel > Dual > Otto.

2-stroke and scavenging

Scavenging = fresh charge pushes exhaust out (cross, loop, uniflow). Critical in 2-stroke (all exchange in ~130° around BDC, both ports open). 2-stroke SI scavenges with mixture → 20-30 % short-circuit, HC, oil smoke → limited use. 2-stroke CI scavenges with air → no fuel loss → large marine engines. 4-stroke: better economy, less pollution.

HCCI, DISI, SI mixture formation

Mixture: carburetor, port injection (TBI/MPFI), direct injection. HCCI: homogeneous lean charge auto-ignites; low NOx and soot, high η; timing hard, narrow load, HC/CO. DISI: in-cylinder petrol injection; charge cooling, higher r, stratified lean unthrottled; PM, NOx after-treatment.

SI and CI combustion

SI: I ignition lag A-B (~10-20°, chemical, 5-10 % burnt), II flame propagation B-C (80-90 %, 10-30 m/s, peak 10-15° aTDC), III after-burning C-D.

CI (ROHR): ignition delay → premixed spike → mixing-controlled hump → late tail.

SIT = lowest temperature for self-ignition. Premixed (SI, blue, kinetics) vs diffusion (CI, yellow soot, mixing).

Abnormal combustion

Surface ignition: hot plug, exhaust valve, deposits. Pre-ignition (before spark): uncontrolled advance, negative work, run-away, holed piston; post-ignition (after spark): only faster burn. End gas: unburnt charge ahead of flame. Flame front: thin reaction zone.

SI knock: end gas auto-ignites at END of combustion if $t_{flame} > t_{delay}$; pressure waves 5 kHz. CI knock: long delay → violent premixed burn at START. Factors opposite (SIT, delay, r, T_{in}, p_{in}, wall T, speed, size). SI knock-limited (r, boost, timing); CI not (smoke and stress limited).

CI fuel statements

CI delay down with: high CN, high r, high T_{in}/p_{in}, load, swirl, fine spray, optimum timing, pilot injection.

Good SI fuel (high ON, high SIT, long delay) = bad CI fuel (needs high CN, short delay).

Large slow CI: delay in ms is few crank degrees; big hot cylinders → can burn low-cetane heavy oil.

Smoke limited: more fuel → rich spray core can't find O_2 → black smoke caps max fuel and power.

Supercharging

Supercharger (mechanical: Roots, vane, centrifugal) vs turbocharger (exhaust turbine). Effects: power ↑ 30-50 %, η_m↑, positive pumping work, better combustion; thermal/mechanical load, SI knock, NOx.

CI more suitable (boost shortens delay, less knock, pushes smoke limit; air only). SI limited by knock.

Modifications: lower r, stronger parts, more overlap (up to 60°), cooling, lubrication, intercooler, bigger pump/nozzles, retarded timing, turbine manifold, waste-gate. Turbo limits: special manifold, cam overload, turbine sensitive to gas velocity, lag/range.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
A/F	air-fuel ratio
API	American Petroleum Institute (gravity scale)
BDC	bottom dead centre
BMEP	brake mean effective pressure
BP	brake power
BSFC	brake specific fuel consumption
CFR	Cooperative Fuel Research (standard test engine)
CI	compression ignition (diesel engine)
CN	cetane number
CNG	compressed natural gas
CO	carbon monoxide
CV	calorific value
DISI	direct injection spark ignition
DPF	diesel particulate filter
EGR	exhaust gas recirculation
EVC	exhaust valve closes
EVO	exhaust valve opens
F/A	fuel-air ratio
FAME	fatty acid methyl ester (biodiesel)
FBC	fluidised bed combustion
FC	fixed carbon
FP	friction power
GDI	gasoline direct injection
HC	hydrocarbons (unburnt fuel)
HCCI	homogeneous charge compression ignition
HCV	higher calorific value
IC	internal combustion
IMEP	indicated mean effective pressure
IP	indicated power
ISFC	indicated specific fuel consumption
IVC	inlet valve closes
IVO	inlet valve opens
LCV	lower calorific value
LPG	liquefied petroleum gas
MEP	mean effective pressure
NG	natural gas

NOx	nitrogen oxides (NO + NO_2)
ON	octane number
PCV	positive crankcase ventilation
PM	particulate matter
ROHR	rate of heat release
rpm	revolutions per minute
RVP	Reid vapour pressure
SAE	Society of Automotive Engineers (oil grade)
SCR	selective catalytic reduction
SI	spark ignition (petrol engine)
SIT	self-ignition temperature
SOF	soluble organic fraction
TDC	top dead centre
TWC	three-way catalytic converter
VM	volatile matter
WOT	wide open throttle

Section A: Fuels, combustion, mixture preparation, cooling, lubrication, emissions

A1 Fuel properties: octane, cetane, API gravity, RVP, aniline point, diesel index

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2024 Q1(c)** [8] Explain octane and cetane number. How does the octane number correlate with IC (internal combustion) engine performance?
- **2023 Q1(a)** [12] Briefly explain the measurement methods and significance of: (i) API (American Petroleum Institute (gravity scale)) gravity, (ii) Reid vapor pressure, (iii) Cetane number, (iv) Octane number.
- **2023 Q1(c)** [6] How does the octane number correlate with the performance of IC engines?
- **2022 Q1(a)** [15] Briefly explain the measurement methods and significance of: (i) API Gravity, (ii) Reid Vapor Pressure, (iii) Aniline Point and Diesel Index.
- **2021 Q1(b)** [10] What is octane and cetane number? Explain what it means by a cetane rating of 60 and an octane rating of 85?
- **2020 Q1(a)** [11] How can you use hydrogen as a fuel in CI (compression ignition (diesel engine)) engine? Explain the physical significances of octane number and cetane number with example.

Before you start: why fuels need ratings

A petrol (SI = spark ignition) engine wants a fuel that does **not** ignite by itself when squeezed, because self-ignition before the spark causes **knock** (a metallic ping that damages the engine). A diesel (CI = compression ignition) engine wants the opposite: fuel that ignites **quickly by itself** when sprayed into hot compressed air.

So petrol is rated by **octane number** (resistance to self-ignition) and diesel by **cetane number** (readiness to self-ignite). The other tests (API gravity, Reid vapour pressure, aniline point) measure density, how easily the fuel evaporates and how much aromatic content it has.

Octane number and cetane number (2024 1c, 2021 1b, 2020 1a)

Octane number (ON) of a petrol is the percentage by volume of **iso-octane** (2,2,4-trimethylpentane, ON (octane number) = 100) in a mixture with **normal heptane** (ON = 0) that gives the same knock intensity as the fuel in a standard CFR (Cooperative Fuel Research (standard test engine)) test engine (variable compression ratio, 600 rpm research / 900 rpm motor method). It measures the **resistance of an SI (spark ignition (petrol engine)) fuel to auto-ignition (knock)**.

Cetane number (CN) of a diesel is the percentage by volume of **n-hexadecane (cetane, CN (cetane number) = 100)** in a mixture with **1-methylnaphthalene** (CN = 0) that gives the same **ignition delay** as the fuel in a standard CFR diesel engine. It measures the **readiness of a CI fuel to self-ignite**.

Meaning of the ratings (2021 1b): an octane rating of 85 means the petrol knocks like a blend of 85 % iso-octane + 15 % n-heptane. A cetane rating of 60 means the diesel ignites with the same delay as a blend of 60 % cetane + 40 % 1-methylnaphthalene (a good, quick-igniting diesel; normal diesel is 40-55).

	Octane number	Cetane number
Used for	SI (petrol) engines	CI (diesel) engines
Wanted property	High self-ignition temperature, resist knock	Low self-ignition temperature, short delay
Reference fuels	iso-octane (100), n-heptane (0)	n-hexadecane (100), 1-methylnaphthalene (0)
High value gives	Higher allowable compression ratio, spark advance, boost	Short delay, smooth running, easy cold start, low CI knock
Typical	Petrol 87-98, ethanol ~108, methanol ~99	Diesel 40-55, biodiesel 50-65

How octane number correlates with engine performance (2024 1c, 2023 1c)

- SI efficiency $\eta = 1 - 1/r^{\gamma-1}$ rises with compression ratio, but higher r raises end-gas temperature and pressure → knock. A higher-ON fuel resists auto-ignition, so the engine can use **higher compression ratio** and **more spark advance** → more power and better fuel economy.
- Using a fuel with lower ON than the engine needs causes knock: noise, piston and head damage, overheating, loss of power; the knock sensor then retards timing, so power and economy fall.
- Using higher ON than needed gives **no extra power** by itself; the benefit only appears if r or timing is raised to use it.
- High ON is essential for supercharged / turbocharged SI engines because boost raises charge temperature and density.

Remember: Octane = how well petrol **resists** igniting by itself; cetane = how **easily** diesel ignites by itself. They are opposite, so a good SI fuel is a bad CI fuel.

Hydrogen as a fuel in a CI engine (2020 1a)

Hydrogen has a very high self-ignition temperature (~585 °C, near zero cetane), so it will not auto-ignite at normal CI compression temperatures. It can be used by: (1) **dual-fuel mode**: hydrogen inducted with the intake air (or port injected), ignited by a small **pilot injection of diesel**; (2) using a **glow plug / hot surface or spark plug** assist; (3) direct high-pressure hydrogen injection with ignition aid. Precautions: backfire and pre-ignition (very low ignition energy), high NOx (nitrogen oxides (NO + NO_2)) at high flame temperature (use lean mixture, EGR (exhaust gas recirculation), water injection).

Measurement methods and significance of fuel properties (2023 1a, 2022 1a)

Property	How it is measured	Significance
API gravity (ASTM D287)	Hydrometer (Baumé scale) gives specific gravity SG at 60 °F; **API = 141.5/SG − 131.5**	Heavy vs light oil. API > 10: lighter than water. Light crude > 31.1, medium 22.3-31.1, heavy < 22.3; ideal 40-45. Higher API = more volatile, higher CV (calorific value) per kg, more valuable; used for pricing and in diesel index
Reid vapour pressure (ASTM D323)	Chilled sample sealed in the fuel chamber, joined to an air chamber 4 times its volume (water-saturated air), immersed in a 38 °C (100 °F) bath and shaken; the absolute pressure read is the RVP (Reid vapour pressure)	Volatility of petrol: high RVP = easy cold start but vapour lock and evaporative losses in summer; low RVP = poor starting. Refiners blend summer/winter grades
Aniline point	Equal volumes of aniline and oil heated in a test tube while stirring until they become one phase, then cooled; temperature at which they separate is the aniline point	Indicates aromatic content: aromatics dissolve aniline at low temperature, so **low aniline point = high aromatic = poor ignition quality**; high aniline point = paraffinic, good diesel
Diesel index	Calculated: **DI = aniline point (°F) × API gravity / 100**	Quick estimate of ignition quality of diesel (correlates with cetane number); higher DI = better CI fuel
Cetane number	CFR engine: vary compression ratio until fuel shows standard delay (13° CA); match with cetane / 1-methylnaphthalene blend	Ignition quality of diesel
Octane number	CFR engine at standard knock intensity; compare with iso-octane / n-heptane blends (research method 600 rpm, motor method 900 rpm)	Anti-knock quality of petrol

A9 Mixture requirements: idling, cruising, full throttle, acceleration, starting

Must study · Very likely · 100% · Easy

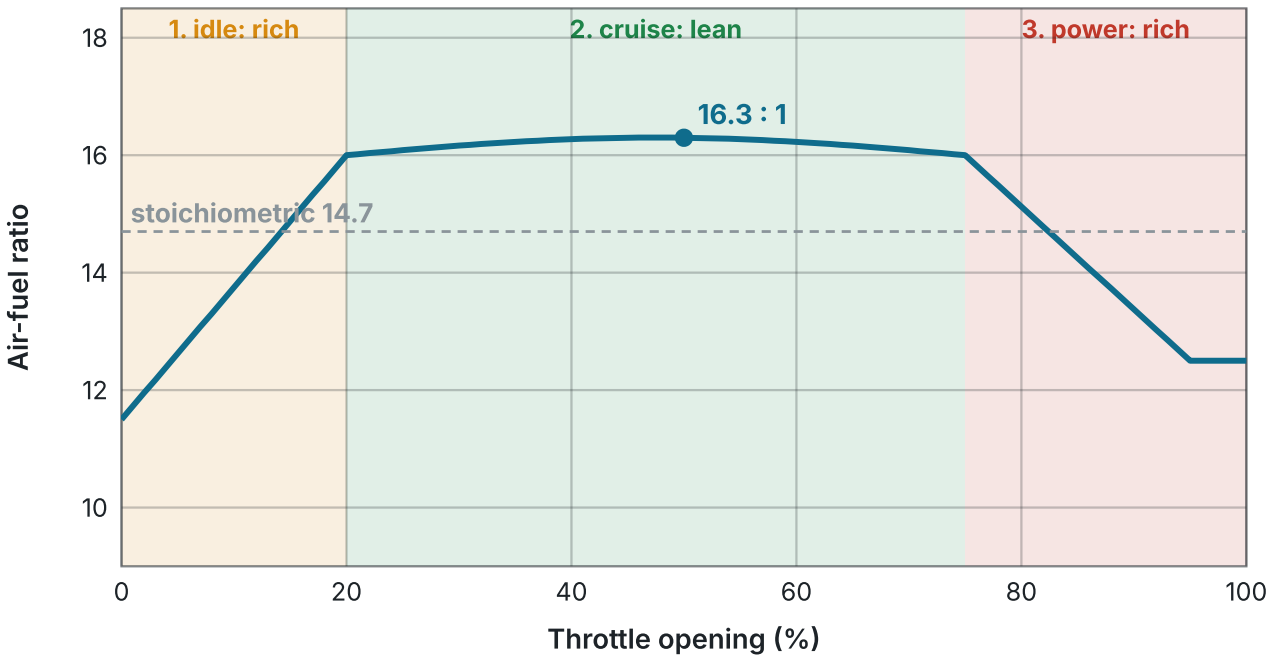
PAST-PAPER QUESTIONS

- **2024 Q3(b)** [9] How does the air-fuel mixture change with varying engine speeds from idling to high-speed operation?
- **2023 Q3(b)** [10] How does the air-fuel mixture requirement change between idling, cruising, and full throttle conditions in IC (internal combustion) engine?
- **2022 Q2(c)** [10] Why it is advantageous to dilute fuel-air mixture? How quasi-static effect influences the volumetric efficiency of an internal combustion engine.
- **2021 Q3(c)** [8] Explain the variation in mixture quality required by a SI (spark ignition (petrol engine)) engine running with different operating conditions.
- **2020 Q1(b)** [14] Why do you need extra fuel when the throttle is opened? (second part)

INTERACTIVE

Mixture needed vs throttle opening (SI engine)

Rich at idle, lean for cruising, rich again for full power. Move the throttle to see which range you are in and why.



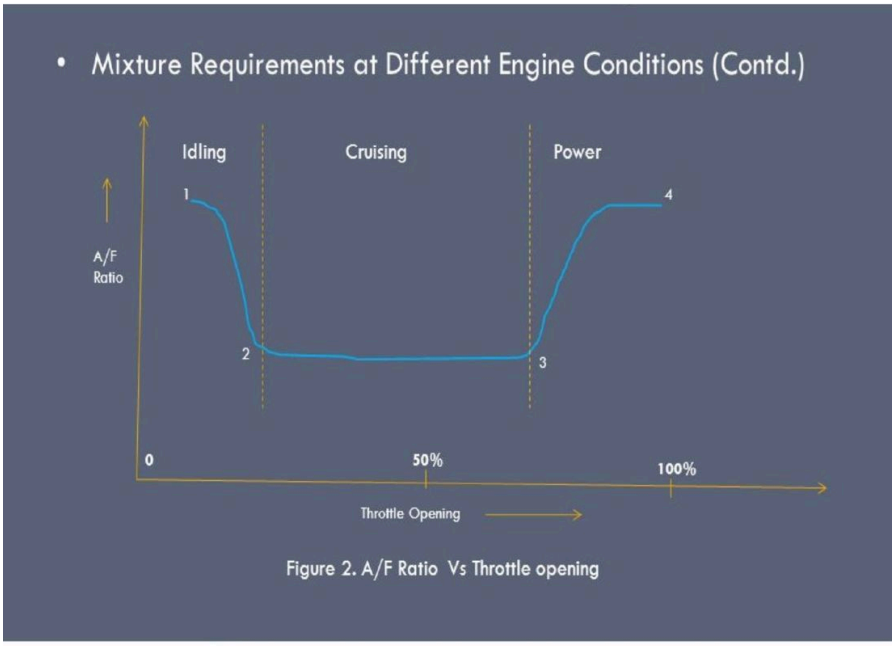
Cruising: lean (~16–17:1), best economy. Little dilution, so a slightly lean mixture burns completely and wastes no fuel.

Transients (not on the curve): cold start needs a very rich mixture (choke), acceleration needs a quick extra squirt (accelerator pump).

Before you start: rich and lean

Rich mixture = more fuel than the exact (stoichiometric) amount; **lean** = less fuel. An engine needs a rich mixture to start, idle and give maximum power, and a lean mixture for best economy when cruising.

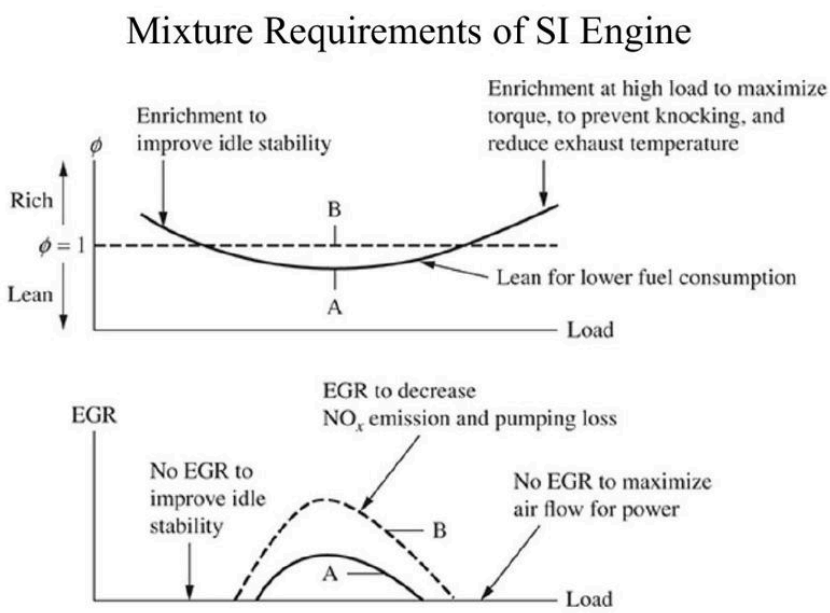
Mixture requirements at different operating conditions (2024 3b, 2023 3b, 2021 3c)



A/F (air-fuel ratio) ratio required vs throttle opening: idling (rich), cruising (lean), power (rich)

An automotive SI engine needs different A/F ratios in three steady ranges and two transient conditions:

Condition	A/F (petrol)	Why
Idling and low load (range 1-2)	Rich, ~10-12:1	Throttle nearly closed: very low manifold pressure, so exhaust residuals expand back and dilute the small fresh charge; rich mixture ensures ignition and smooth running. Enrichment reduces as throttle opens
Cruising / part load (range 2-3)	Lean, best economy ~15-17:1	Emphasis on fuel economy; dilution is small, so a slightly lean mixture burns completely and gives minimum BSFC (brake specific fuel consumption)
Full throttle / maximum power (range 3-4)	Rich, best power ~12-13:1	All air must be used to give maximum torque; extra fuel also cools the charge and valves and suppresses knock
Cold starting	Very rich, up to 3-5:1 (choke)	Fuel hardly evaporates on cold walls; only the light fraction vaporizes
Acceleration	Momentary extra fuel	When the throttle snaps open, air responds faster than fuel; accelerator pump squirts extra fuel to prevent a lean flat spot



Mixture and EGR (exhaust gas recirculation) requirements vs load (Heywood): enrich at idle and full load, lean in between

With speed (2024 3b): from idling to medium speed the mixture goes from rich to lean (best economy); at high speed and full load it is enriched again for maximum power. At higher speed the time for carburetion is shorter, so better atomization from the higher venturi depression is needed.

Why diluting the fuel-air mixture is advantageous (2022 2c)

- Burned-gas temperature falls, so γ of the gas is higher and dissociation is lower → more expansion work for the same expansion ratio (higher efficiency).
- For the same load the throttle is opened wider, so intake pressure rises and **pumping work falls**.
- Lower gas temperature → **lower heat loss** to walls and **lower NOx** (nitrogen oxides (NO + NO_2)).
- Excess air can be the diluent if NOx is not a concern; EGR is used when a 3-way catalyst needs φ = 1. Limit: combustion stability (cycle-to-cycle variation) must stay acceptable.

A13 Need for cooling and lubrication; effects of overcooling Must study

Very likely · 93% Easy

PAST-PAPER QUESTIONS

- **2024 Q7(a)** [12] Why cooling and lubrication systems are necessary for IC (internal combustion) engines?
- **2023 Q2(a)** [5] Why cooling and lubricating systems are necessary for IC engines?
- **2022 Q2(a)** [14] What are the effects of overcooling the IC engines? What are the advantages and disadvantages of different cooling systems?
- **2021 Q2(a)** [6] Why cooling and lubrication systems are necessary for IC engines?

Before you start: heat in an engine

Only about one-third of fuel energy becomes work; about one-third leaves with the exhaust and about one-third goes into the cylinder walls. Without cooling, walls and oil overheat; with too much cooling, fuel is wasted and parts corrode. Lubrication puts an oil film between moving parts so they slide without wearing.

Why cooling is necessary (2024 7a, 2023 2a, 2021 2a)

Peak gas temperature reaches ~2500 °C; about 25-35 % of the fuel energy flows into the cylinder walls, head and piston. Without cooling:

- The lubricating oil film on the cylinder wall breaks down (oil burns above ~200 °C) → wear and **piston seizure**.
- Metal strength falls and uneven heating gives **thermal stress, distortion and cracks** (head, valves, piston).
- Hot spots (exhaust valve, plug) cause **pre-ignition and knock** in SI (spark ignition (petrol engine)) engines.
- Hot intake charge lowers density → **lower volumetric efficiency** and power.

But cooling should not be excessive: keep walls hot enough (~180-200 °C) for good vaporization and efficiency.

Why lubrication is necessary

- **Reduce friction** and wear between moving parts (piston rings, bearings, cams), so less power lost.
- **Cooling**: oil carries heat away from piston, bearings.
- **Sealing**: oil film between rings and liner seals compression.
- **Cleaning**: carries away carbon and metal particles to the filter.
- **Shock absorption** in bearings, and protection against corrosion.

Effects of overcooling (2022 2a)

- Fuel vaporization poor, so combustion is incomplete; lower thermal and combustion efficiency.
- More heat lost to the coolant → lower power and economy.
- Fuel and water condense on cold walls: **oil dilution**, sludge, washing off of oil film → wear; sulphuric acid corrosion.

- Oil viscosity high → more friction loss; more HC (hydrocarbons (unburnt fuel)) and CO emission; hard starting.

Advantages and disadvantages of cooling systems (2022 2a)

	Air cooling	**Water (liquid) cooling**
Advantages	Simple, light, cheap; no radiator, pump, coolant; no freezing or leaks; warms up fast; good where water is scarce (deserts) or very cold	Uniform and effective cooling of all parts; closer clearances; higher volumetric efficiency; quieter (water jacket damps noise); engine can be placed anywhere; suits large, multi-cylinder engines
Disadvantages	Uneven cooling, higher operating temperature, lower ηv, noisy fins, fan absorbs power, limited to small engines	Heavier, costlier, pump power, needs maintenance; leaks, freezing, corrosion; failure can wreck the engine

A14 Air vs liquid cooling; water cooling system, thermostat, thermosyphon Must study Very likely · 80% Easy

PAST-PAPER QUESTIONS

- **2024 Q7(b)** [11] How does a liquid cooling system differ from an air cooling system in terms of applications in IC (internal combustion) engines?
- **2024 Q7(c)** [12] Describe the water cooling system of IC engine with a schematic diagram.
- **2023 Q2(b)** [10] How does a liquid cooling system differ from an air cooling system in terms of applications in IC engines?
- **2023 Q2(d)** [10] Describe the water cooling system of IC engine with a schematic diagram.
- **2023 Q2(c)** [10] How does the use of a thermostat in a water cooling system regulate engine temperature and what happens if the thermostat fails to function properly?
- **2021 Q3(a)** [15] Describe the water cooling system of IC engine with a schematic diagram.
- **2020 Q2(c)** [8] How is the circulation achieved in thermo-syphon system? What are the limitations of this system?

Before you start: two ways to cool

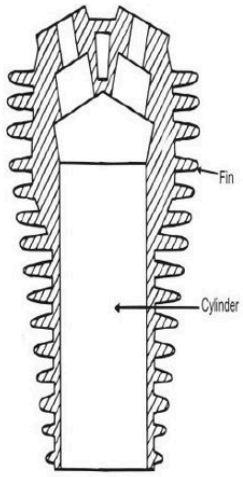
Air cooling: fins on the cylinder and a fan (motorbikes, small engines). **Liquid cooling**: water jacket around the cylinder, pump, radiator, thermostat (cars, big engines). A **thermostat** is a valve that stays shut until the water is hot, so the engine warms up quickly. **Thermosiphon** = water circulates naturally because hot water rises.

Liquid cooling vs air cooling, in terms of applications (2024 7b, 2023 2b)

Point	Air cooling	Liquid cooling
How	Fins on cylinder and head; air flow by vehicle motion or fan	Water/antifreeze circulates through jackets; heat rejected in a radiator
Applications	Motorcycles, scooters, small generators, lawn mowers, aircraft piston engines, small engines up to ~15-20 kW	Cars, buses, trucks, tractors, marine, stationary and large engines
Cooling uniformity	Uneven; hot spots	Even cooling of barrel, head and valve seats
Weight, cost	Light, cheap, compact	Heavier, costly (radiator, pump, coolant)
Maintenance	Little	Coolant level, leaks, antifreeze, corrosion
Noise	Fins ring, noisy	Jacket damps noise
Climate	No freezing; works in water-scarce areas	Needs antifreeze in cold, sensitive to climate
Power and efficiency	Lower η_v, higher wall temperature, lower compression ratio allowed	Higher η_v, allows higher compression ratio and output

Cooling system: Air Cooling

- Used in small engines (15-20 kW) and in aero plane engines.
- Fins or extended surfaces on the cylinder walls, cylinder head, etc.
- The amount of heat dissipated to air depends upon:
  - Amount of air flowing through the fins
  - Fin surface area
  - Thermal conductivity of metal used for fins



Air cooling: fins on cylinder and head

Water cooling system with schematic (2024 7c, 2023 2d, 2021 3a)

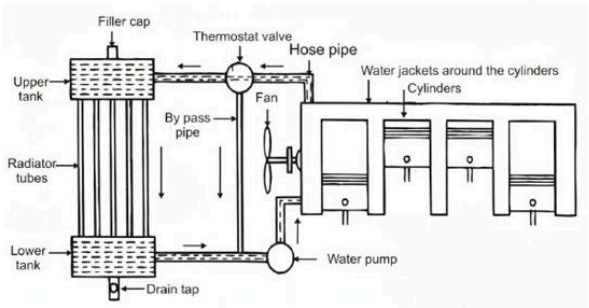
Cooling system: Water Cooling

(b) Forced or pump cooling:

- Used in large number of vehicles like cars, buses, trucks, and other heavy vehicles.
- Water or coolant is circulated through jackets around engine parts to be cooled.
- Circulation is maintained by a centrifugal pump, driven from the engine via a V-belt.

Limitations:

- Cooling is independent of temperature.
- Engine may be over-cooled.
- Can be overcome by using a thermostat.



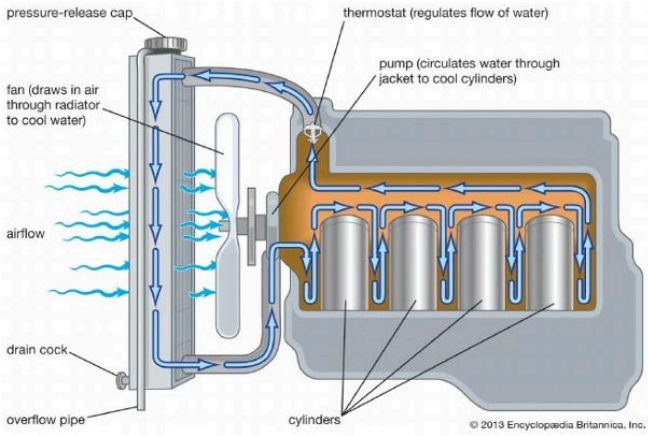
Forced (pump) circulation water cooling system: jackets, thermostat, radiator, fan, pump

Working: water (with antifreeze) flows through **water jackets** cast around the cylinders, head, valve seats and plug bosses, absorbing heat by conduction from the walls. The hot water leaves the top of the head, passes the **thermostat valve** into the **upper tank** of the **radiator**, flows down through the radiator **core** (tubes with fins) where air drawn by the **fan** (and ram air) cools it, collects in the **lower tank** and is pumped by the belt-driven centrifugal **water pump** back into the jackets. A **pressure cap** raises boiling point to ~120 °C; an overflow tank takes expansion.

Component	Function
Water jackets	Passages around cylinder and head carrying coolant to hot parts
Water pump	Centrifugal impeller driven by fan belt; forces circulation
Radiator (upper tank, core, lower tank)	Heat exchanger rejecting heat to air; core tubular or cellular
Fan	Draws air through radiator at low vehicle speed
Thermostat	Keeps engine at working temperature (~80-90 °C)
Antifreeze (ethylene glycol)	Lowers freezing point, raises boiling point, inhibits corrosion

Cooling system: Water Cooling

- Cooling water jackets surround the cylinder, cylinder head, and valve seats.
- Water circulates through these jackets, absorbing combustion heat.
- The hot water then cools in the radiator, aided by a fan and vehicle motion.
- Cooled water is recirculated through the water jackets.

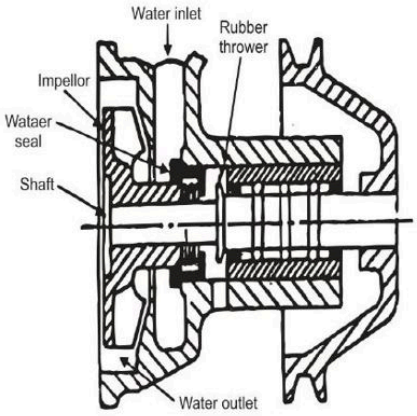


Water cooling layout

Cooling system: Water Cooling

Water Pump:

- Used to pump circulating water.
- Impeller type pump mounted at the front.
- Consists of an impeller on a shaft, enclosed in a casing with inlet/outlet.
- Driven by the engine output shaft via belts.



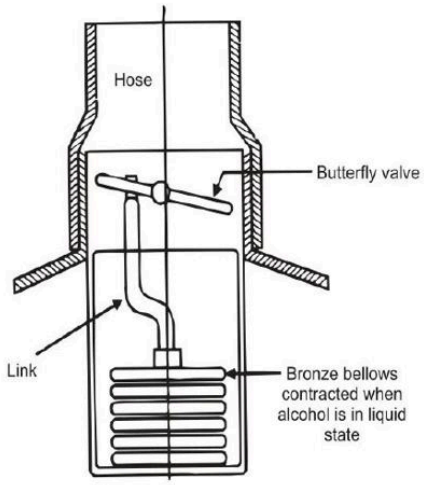
Water pump

Thermostat: how it regulates temperature and what happens if it fails (2023 2c)

Cooling system: Water Cooling

Thermostat Valve:

- Prevents water flow from the engine to the radiator until the engine reaches its maximum efficient operating temperature.
- Automatically opens to allow water flow once the optimal temperature is achieved.
- Typically restricts water flow when the temperature is below 70°C.



Bellows-type thermostat: butterfly valve opens as bellows expand

The thermostat is a temperature-sensitive valve between the engine outlet and the radiator. A **wax-pellet or bellows** element (filled with alcohol/wax) expands with temperature. Below ~70-80 °C it keeps the valve **closed**, so coolant circulates only through the engine (bypass) and the engine warms up quickly. As the coolant reaches its working temperature the element expands and **opens** the valve progressively, sending more coolant through the radiator; flow is modulated to hold a steady temperature regardless of load and weather.

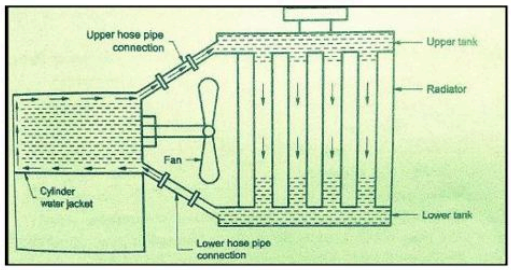
- **Stuck closed**: no flow to the radiator → overheating, boiling, head gasket failure, distortion, piston seizure, knock.
- **Stuck open**: engine runs **overcooled**: slow warm-up, poor vaporization, high fuel consumption, more HC (hydrocarbons (unburnt fuel))/CO, oil dilution and sludge, more wear, weak heater.

Thermosyphon circulation and its limitations (2020 2c)

Cooling system: Water Cooling

(a) Thermosyphon cooling

- Works on the principle that hot water is lighter and rises, while cold water is heavier and sinks.
- The radiator is positioned higher than the engine to facilitate water flow.
- Heat from the engine's water jackets is transferred to the circulating water through convection.
- Hot water rises to the top of the radiator.
- Cold water from the radiator replaces the rising hot water, creating a circulation.
- This continuous circulation maintains the engine at its optimal working temperature.



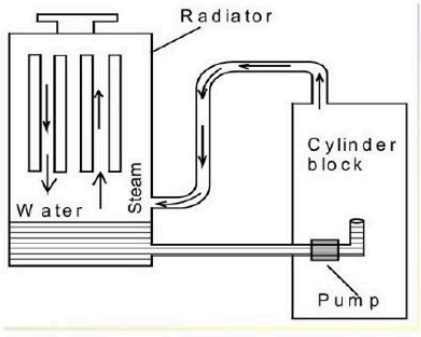
Thermosyphon cooling: radiator above engine, circulation by density difference

No pump: the radiator is placed higher than the engine. Water in the jackets is heated, becomes lighter and **rises** to the top of the radiator; cooled water in the radiator is denser and **sinks** to the bottom and back into the jacket. The density difference keeps a natural circulation going. **Limitations:** circulation is slow; it starts only after a large temperature difference develops; it **stops if the water level falls** below the top hose (delivery pipe); radiator must be high and large; cooling not controllable. Hence it is obsolete.

Cooling system: Water Cooling

(c) Evaporative cooling

- The engine is cooled by the evaporation of water into steam within the cylinder jackets.
- This system leverages the high latent heat of vaporization of water.
- Evaporative cooling is commonly used for various industrial engines.



Evaporative cooling (third type): latent heat of steam, used in stationary industrial engines

A18 Catalytic converter: construction, reactions, temperature and equivalence ratio effects

Must study | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2024 Q8(b)** [12] What is a catalytic converter? Explain how the effect of temperature and equivalence ratio on the conversion efficiency of catalytic converter.
- **2023 Q4(c)** [12] How do catalytic converter in IC (internal combustion) engines reduce the emissions of HC (hydrocarbons (unburnt fuel)), CO and NOx (nitrogen oxides (NO + NO_2)), and what are the chemical reactions involved?
- **2022 Q4(b)** [11] With proper sketch, discuss the construction and operation of a catalytic converter.
- **2021 Q4(a)** [10] How temperature and fuel equivalence ratio affect the conversion efficiency of a catalytic converter? Explain with necessary graphs.
- **2020 Q2(a)** [12] What is catalytic converter? With a schematic diagram explain how a 3-way catalytic converter does treat exhaust from engine?

LEARN IT FROM ZERO

What a catalyst does, in plain words

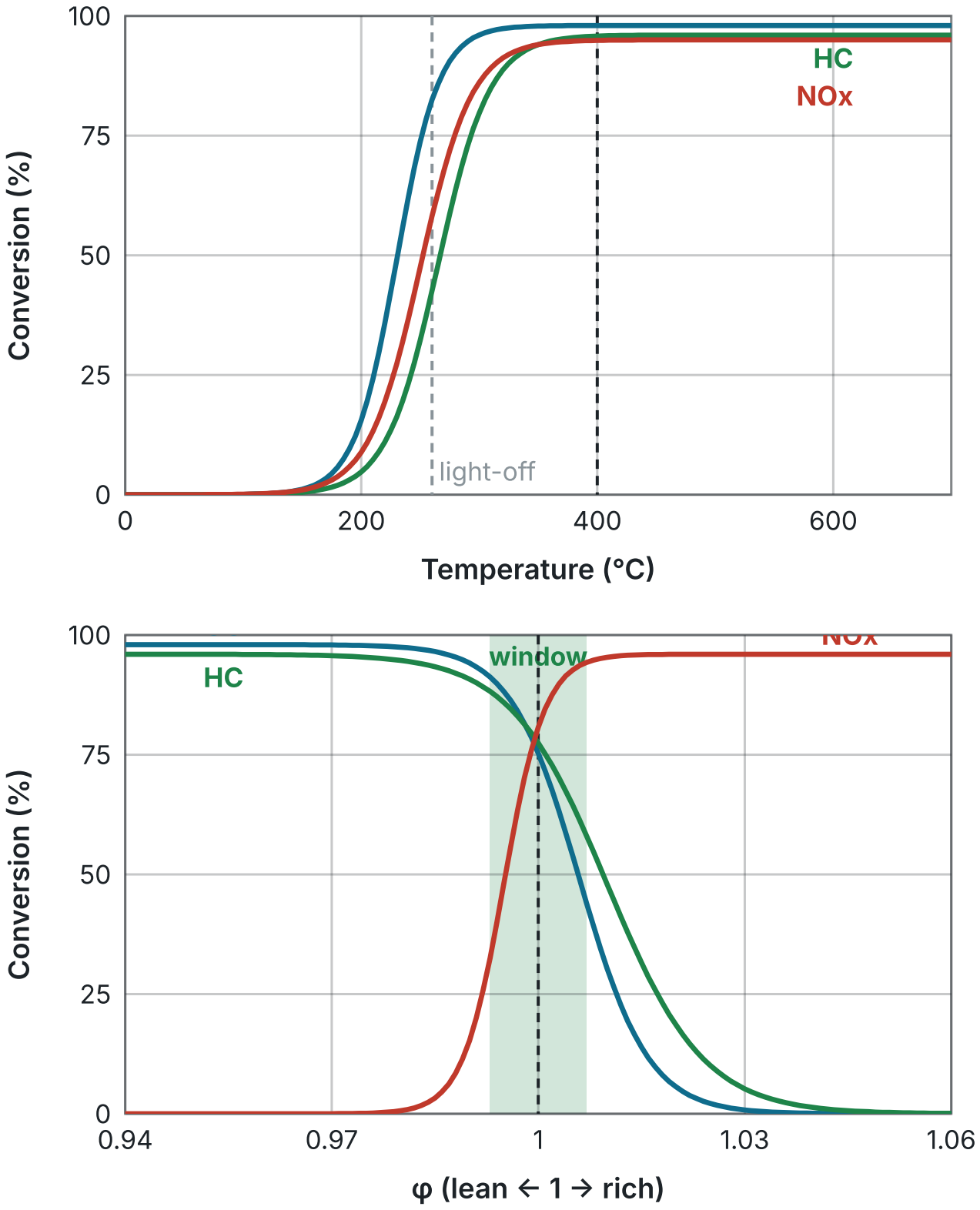
Exhaust contains three poisons: **CO** (carbon monoxide, half-burnt carbon), **HC** (unburnt fuel) and **NOx** (nitrogen that reacted with oxygen in the hot flame). Two of them need **more oxygen** to become harmless (CO → CO_2, HC → CO_2 + H_2O). The third needs oxygen **taken away** (NO → N_2). A catalyst is a surface (platinum, palladium, rhodium spread thinly over a honeycomb) that makes these reactions happen at exhaust temperature without being used up.

The clever part of a **three-way** converter: the CO and HC are used to pull the oxygen off the NO ($2NO + 2CO \rightarrow N_2 + 2CO_2$). That only works if there is **exactly** the right amount of oxygen: too much (lean) and O_2 burns the CO first, so NO survives; too little (rich) and there is no O_2 left for CO and HC. Hence the narrow **window at φ = 1** and the oxygen (lambda) sensor that keeps the engine there.

It also needs heat: below about 250 °C (**light-off**) almost nothing reacts, so a cold start gives most of a trip's pollution. Too hot (over ~850 °C) and the precious metal particles melt together (sinter) and die. Lead in fuel coats the surface (poisoning).

INTERACTIVE **Three-way catalytic converter: conversion efficiency**

Left: warm-up (light-off). Right: the narrow window at the stoichiometric mixture. Move the sliders and read the three pollutants.



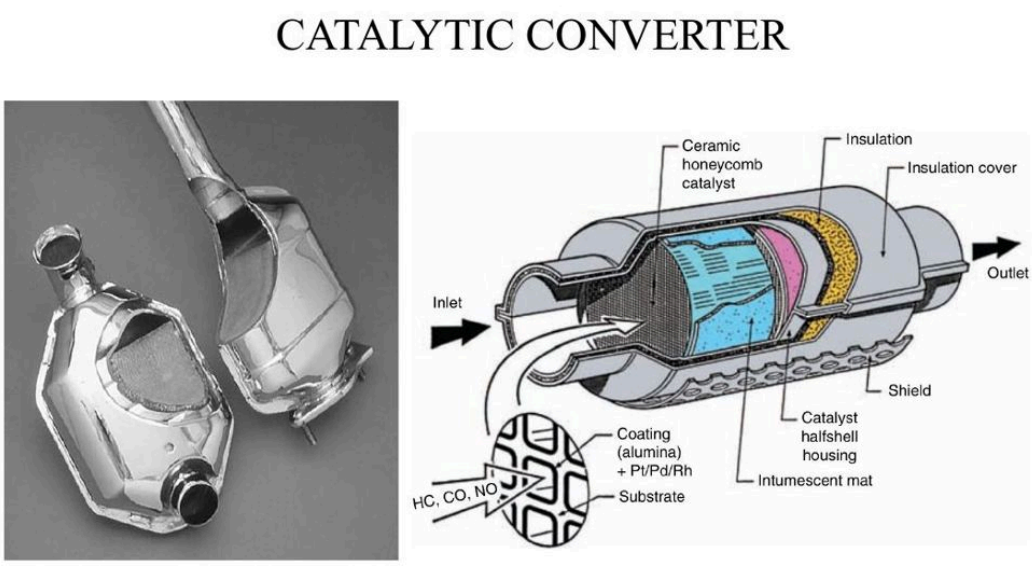
At 400 °C and φ = 1.000: CO removed **74 %**, HC **74 %**, NOx **77 %**.
Inside the window: all three are removed together, which is why a three-way catalyst needs an oxygen sensor and φ = 1 control. Above about 800–900 °C the coating sinters and the catalyst dies.

Before you start: what a catalytic converter does

A catalyst is a material (platinum, palladium, rhodium) that speeds up a reaction without being used up. In the exhaust it helps CO and HC burn to CO_2 and H_2O (oxidation) and turns NOx back into N_2 (reduction). It works only when hot (above about 250-300 °C, light-off) and a three-way converter needs the mixture very close to stoichiometric.

What a catalytic converter is; construction (2024 8b, 2022 4b, 2020 2a)

A **catalytic converter** is the most effective exhaust after-treatment device: a chamber in the exhaust pipe containing a **catalyst** that promotes oxidation of CO and HC and reduction of NOx to harmless CO_2, H_2O and N_2 at 250-600 °C, much lower temperatures than needed without catalyst. The catalyst is not consumed (only degraded by heat, age, poisoning).

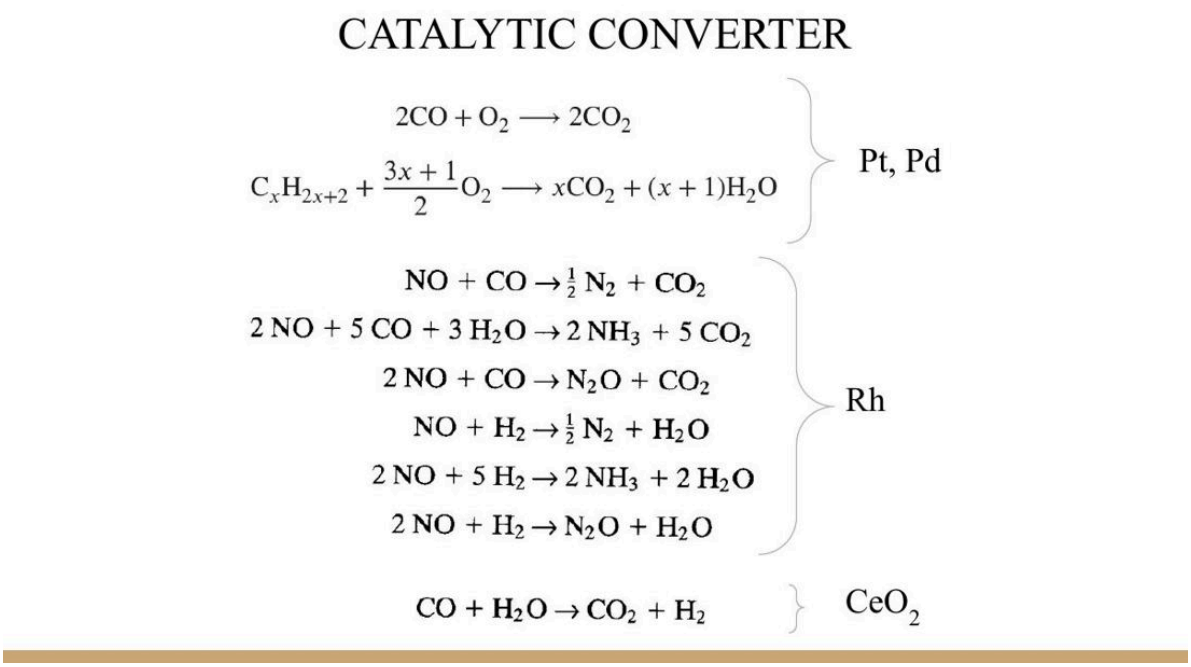


Catalytic converter construction: stainless shell, insulation, ceramic honeycomb substrate with alumina coating and Pt/Pd/Rh

- **Shell**: stainless steel container in the exhaust pipe, with heat shield and insulation (intumescent mat).
- **Substrate**: ceramic (cordierite) **honeycomb monolith** with ~60 cells/cm² (or pellets of alumina); large surface, low back pressure; volume about half the engine displacement.
- **Wash-coat**: porous γ-alumina (Al_2O_3) layer giving huge surface area, stable at high temperature.
- **Catalyst**: **platinum and palladium** promote oxidation of CO and HC; **rhodium** promotes NO reduction; **ceria (CeO_2)** stores and releases oxygen to buffer A/F (air-fuel ratio) swings.

Operation of a 3-way converter and the reactions (2023 4c, 2020 2a)

Exhaust from an engine running at stoichiometric ($\varphi \approx 1$, kept by the oxygen sensor and ECU) flows through the channels and the three pollutants are treated together, hence "three-way":



Catalytic reactions and the metal that promotes each

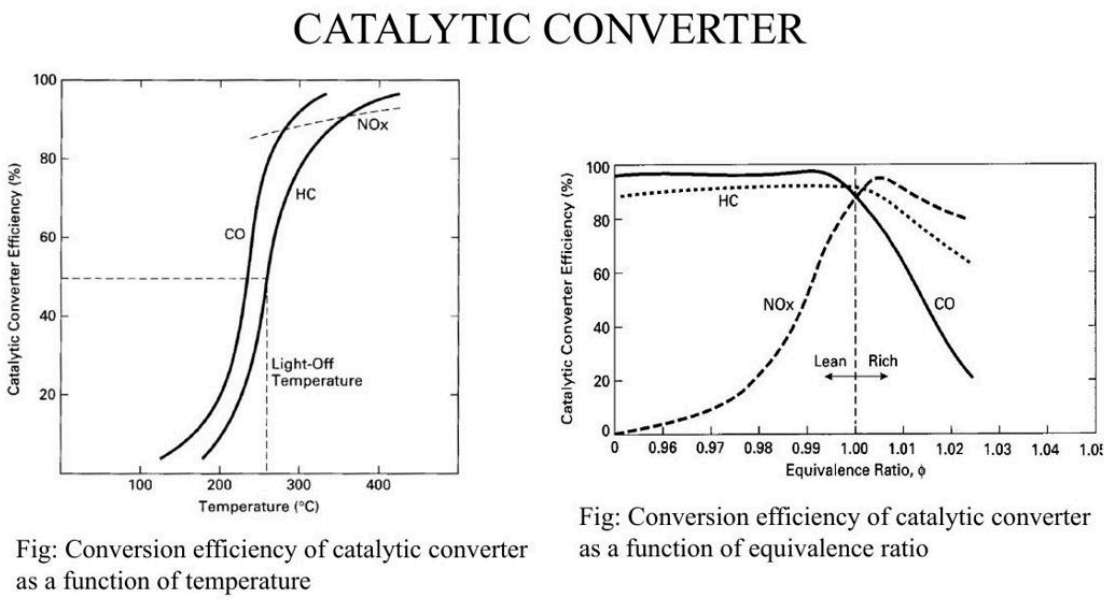
`Oxidation (Pt, Pd): 2CO + O2 → 2CO2;  CxH2x+2 + [(3x+1)/2] O2 → xCO2 + (x+1)H2O`

`Reduction (Rh): 2NO + 2CO → N2 + 2CO2;  2NO + 2H2 → N2 + 2H2O;  HC + NO → N2 + CO2 + H2O`

`Water-gas shift (CeO2): CO + H2O → CO2 + H2`

CO and HC are the reducing agents for NO. Undesired side reactions: $2NO + 5CO + 3H_2O \rightarrow 2NH_3 + 5CO_2$ and N_2O formation.

Effect of temperature and equivalence ratio on conversion efficiency (2024 8b, 2021 4a)



Left: conversion efficiency vs temperature (light-off). Right: conversion efficiency vs equivalence ratio (narrow window at φ = 1)

- **Temperature:** a cold converter does almost nothing. Efficiency rises sharply in an S-curve with temperature; the temperature at which it reaches 50 % is the **light-off temperature, ~250-300 °C** (CO lights off first, then HC). Above ~400 °C efficiency is 90 %+. Cold starts give 70-90 % of

total HC emission, so converters are placed close to the manifold or preheated. Very high temperature (> 800-900 °C) sinters the catalyst and reduces efficiency permanently.

- **Equivalence ratio:** on the **lean** side (φ < 1) excess O_2 makes CO and HC conversion ~100 %, but there are no reducing agents, so **NOx conversion collapses**. On the **rich** side (φ > 1) NOx conversion is high but there is not enough O_2, so **CO and HC conversion fall**. All three exceed ~80 % only in a narrow **window around φ = 1 (±0.5 %)**; hence closed-loop A/F control with the λ sensor.
- Other factors: lead (even small amounts poisons it, so unleaded fuel), sulphur, phosphorus/zinc from oil, ageing, space velocity.

Remember: Temperature: S-curve, light-off 250-300 °C. φ: lean kills NOx conversion, rich kills CO/HC conversion, all three work only at φ ≈ 1.

A2 Proximate vs ultimate analysis; HCV vs LCV Must study Very likely · 93% Medium

PAST-PAPER QUESTIONS

- **2024 Q1(d)** [10] Explain how ultimate analysis contributes to understanding the combustion behavior and energy potential of a fuel.
- **2023 Q1(b)** [6] What are the key differences between proximate analysis and ultimate analysis of fuel?
- **2022 Q1(b)** [8] Compare and contrast ultimate analysis and proximate analysis.
- **2021 Q1(c)** [12] Explain the differences between: (i) Ultimate analysis and proximate analysis, (ii) Higher calorific value and lower calorific value.

Before you start: two ways to analyse a fuel

Proximate analysis = a quick, practical test with an oven: how much moisture, volatile matter (gases given off when heated), ash and fixed carbon (the solid carbon that burns slowly).
Ultimate analysis = exact chemistry: % of carbon, hydrogen, nitrogen, sulphur, oxygen by weight.

Calorific value = heat released by burning 1 kg of fuel. The hydrogen in fuel burns to water. If that water is condensed back to liquid, its latent heat is recovered: that is the **higher** calorific value (HCV). If it leaves as steam (as in an engine exhaust), we get the **lower** calorific value (LCV).

Proximate vs ultimate analysis (2023 1b, 2022 1b, 2021 1c)

Proximate analysis is a quick, simple test of fuel quality: it gives **moisture, volatile matter, ash and fixed carbon** as percentage by weight. **Ultimate analysis** gives the **elemental composition: C, H, N, S, O** (and ash) by weight.

Point	Proximate analysis	Ultimate analysis
Gives	Moisture, volatile matter, ash, fixed carbon	C, H, O, N, S (+ ash)
Method	Moisture: 1 g at 108-110 °C, loss of weight. VM (volatile matter): 1 g in covered crucible at 950 °C for 7 min (minus moisture). Ash: burn 1 g in air at ~725 °C, weigh residue. FC (fixed carbon) = 100 − (M + VM + A)	C and H: burn ~0.2 g in O_2, CO_2 absorbed in KOH, H_2O in $CaCl_2$. N: Kjeldahl (boil with conc. H_2SO_4). S: bomb washings as $BaSO_4$. O by difference
Equipment	Simple oven, furnace, balance	Laboratory chemical analysis, more costly
Use	Grading/buying coal, furnace and boiler design, ignition and burning behaviour	Theoretical air, flue gas composition, calorific value (Dulong), combustion calculations
Type	Practical, empirical	Scientific, fundamental

How ultimate analysis explains combustion behaviour and energy potential (2024 1d)

- **Carbon:** main heat source (33.8 MJ/kg); more carbon = higher calorific value and more CO_2.
- **Hydrogen:** 144 MJ/kg; raises CV (calorific value) and flame speed, but forms water, so it decides the gap between HCV (higher calorific value) and LCV (lower calorific value).

- **Oxygen:** already combined; reduces CV (available hydrogen = H − O/8) and reduces the air needed.
- **Sulphur:** small heat (9.27 MJ/kg) but forms SO_2/SO_3: corrosion, acid rain, dew-point limits on stack temperature.
- **Nitrogen:** inert but forms fuel-NOx.
- From the analysis we calculate: **stoichiometric air** = (2.67C + 8H + S − O)/0.23 kg per kg fuel, the expected **flue gas composition**, excess air from flue gas, and the **calorific value by Dulong's formula**.

```
HCV = 33 800 C + 144 000 (H - O/8) + 9 270 S kJ/kg   (C, H, O, S as mass fractions)
```

Higher vs lower calorific value (2021 1c)

Calorific value = heat released by complete combustion of unit mass (or volume) of fuel. **HCV (gross)**: products cooled to room temperature so the water vapour condenses and its latent heat is counted (measured by bomb calorimeter). **LCV (net)**: water leaves as vapour, so latent heat is not available; this is what engines and boilers actually get.

```
LCV = HCV - m_water × h_fg = HCV - 9H × 2442 kJ/kg   (or HCV - 0.09 H% × 587 kcal/kg)
```

Remember: Proximate = **P**ractical (M, VM, A, FC). **U**ltimate = **U**nits of elements (C, H, O, N, S). LCV is HCV minus the steam's latent heat.

A16 Pollutant formation in SI and CI engines; NOx mechanism; smog Must study Very likely · 87% Medium

PAST-PAPER QUESTIONS

- **2024 Q8(a)** [12] Explain the pollutant formation mechanism in CI (compression ignition (diesel engine)) Engine.
- **2023 Q4(a)** [11] Describe the pollutants formation mechanisms in SI (spark ignition (petrol engine)) engine with appropriate diagrams.
- **2022 Q4(a)** [12] What is smog? Discuss the mechanisms of NOx (nitrogen oxides (NO + NO_2)) formation? How do reduce this emission?
- **2022 Q3(c)** [9] What is the relationship between NOx formation and combustion time? (second part)
- **2020 Q3(a)** [10] What is meant by total emission control packages? How NOx and PM (particulate matter) are generated in CI engines?

Before you start: where pollutants come from

Perfect combustion gives only CO_2 and H_2O. Real engines also give: **CO** (not enough oxygen), **HC** (hydrocarbons (unburnt fuel)) (unburnt fuel left near cold walls and crevices), **NOx** (nitrogen and oxygen of the air combine at very high flame temperature, above about 1800 K), and **particulates/soot** (rich pockets in diesel sprays).

Pollutant formation in SI engines (2023 4a)

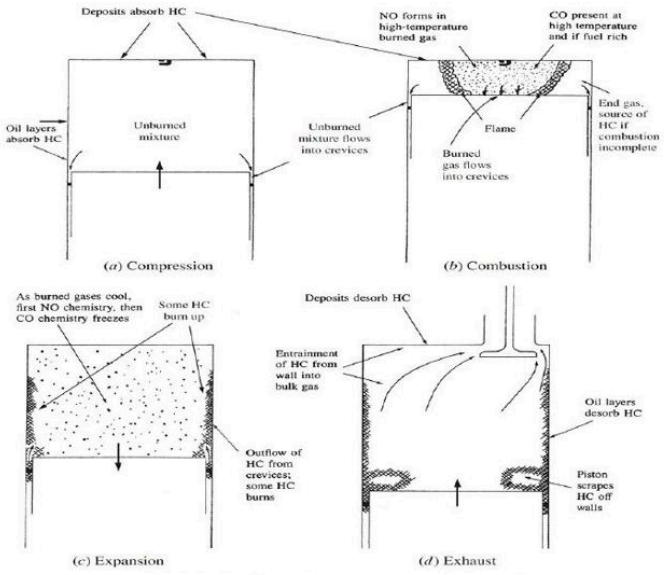


Fig: Pollutant formation mechanisms in SI engine

HC formation in an SI engine: crevices, quench layer, oil film absorption/desorption, deposits, exhaust

The three main pollutants are HC, CO and NOx.

Pollutant	Mechanism
Unburned HC	(1) **Crevice volumes**: mixture pushed into ring crevices, gasket and plug threads during compression escapes the flame and comes out in expansion (~38 %); (2) **flame quenching** at cold walls leaves a thin unburnt layer; (3) **oil film absorption/desorption**: fuel vapour dissolves in oil during compression and is released later (~16 %); (4) **deposits** absorb fuel (~16 %); (5) liquid fuel, incomplete combustion at misfire, valve overlap short-circuiting, exhaust valve leakage
CO	Mainly from **rich mixture** (lack of O_2 to finish CO + OH → CO_2 + H); also dissociation of CO_2 at high temperature, frozen during expansion; poor mixing / cylinder-to-cylinder A/F (air-fuel ratio) variation
NOx	Not from incomplete combustion: atmospheric N_2 oxidised in the hot burnt gas behind the flame by the **Zeldovich mechanism**; significant above 1800 K; maximum slightly lean of stoichiometric (high T and some free O_2)

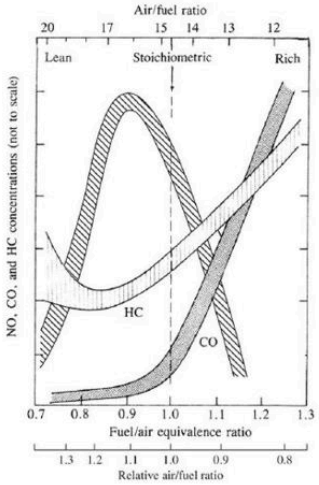


Fig: Variation of HC, CO, and NO concentration in the exhaust of a conventional spark-ignition engine with relative air-fuel ratio and fuel/air equivalence ratio.

HC, CO and NO in SI exhaust vs fuel/air equivalence ratio

Pollutant formation in CI engines (2024 8a)

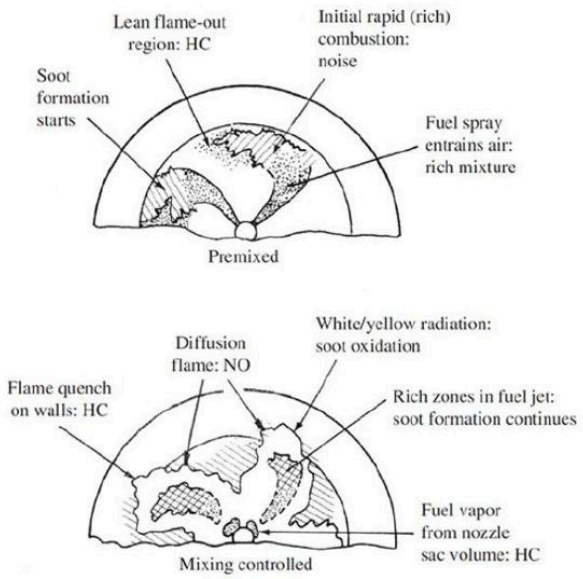


Fig: Pollutant formation mechanisms in a direct injection diesel engine

Pollutant formation zones in a DI diesel spray: lean flame-out HC, rich core soot, diffusion flame NO, sac volume HC

- **Overall lean**, so CO is low and HC is about 1/5 of SI; but mixture is **heterogeneous**.
- **HC**: (a) **over-lean** region at the spray edge (outside the lean flammability limit) during the ignition delay, (b) **over-rich** pockets and under-mixing, (c) fuel from the **nozzle sac volume** and holes evaporating late, (d) flame quench at walls, wall wetting. Heavy HC condense on soot.
- **Soot / PM**: formed in the **fuel-rich core** of the spray ($\varphi > 2$) by pyrolysis at high temperature; most is later oxidised in the diffusion flame, the remainder leaves as smoke, especially at full load (rich).
- **NOx**: formed in the high-temperature, near-stoichiometric **diffusion flame zone** and the gas burnt early in premixed combustion; NO_2 can be 10-30 % of NOx in diesels (vs ~1-2 % in SI).
- CO: only where mixture is locally rich or quenched; small.

NOx formation mechanisms (2022 4a, 2020 3a)

Zeldovich (thermal): $O + N_2 \rightleftarrows NO + N$ (slow, rate-limiting, $E \approx 75$ kcal); $N + O_2 \rightleftarrows NO + O$; $N + OH \rightleftarrows NO + H$

- **Thermal (Zeldovich)**: most important in engines; strongly dependent on **temperature**, **O_2 availability** and **residence time**.
- **Prompt (Fenimore)**: in the thin flame front in rich zones, CH radicals attack N_2 → HCN → NO; small.
- **Fuel-bound**: N in the fuel (0.5-2 % in coal/heavy oil) → HCN, NH_3 → NO.
- NO converts to NO_2 by $NO + HO_2 \rightarrow NO_2 + OH$ in the flame and in air.

How to reduce NOx: lower peak temperature and O_2: EGR (exhaust gas recirculation), retarded injection/spark timing, water injection, charge cooling (intercooler), lean burn or rich-lean staging, and after-treatment: 3-way catalyst (SI), SCR (selective catalytic reduction) with urea / lean NOx trap (CI). See A19.

Smog (2022 4a)

Smog (smoke + fog) is a brown haze of air pollutants. **Photochemical smog** forms when NOx and HC from vehicle exhaust react in **sunlight**: NO_2 + hv (< 415 nm) → NO + O; O + O_2 → O_3 (ozone); O, O_3, HC and NO form **PAN** (peroxyacyl nitrate) and aldehydes. These secondary pollutants irritate eyes and lungs, damage plants and reduce visibility. (Classical sulphurous smog: SO_2 + soot + fog from coal burning.)

A11 Fuel injection system: purpose, components, types Must study

Very likely · 80% Medium

PAST-PAPER QUESTIONS

- **2024 Q4(a)** [7] What is the purpose of a fuel injection system in an IC (internal combustion) engine?
- **2024 Q4(b)** [8] Describe the main components of a fuel injection system and their functions.
- **2023 Q3(c)** [10] Describe the different types of fuel injection systems and injector nozzles.
- **2022 Q3(a)** [16] Using proper sketch, discuss various fuel injection systems and injector nozzles. Compare their relative merits and demerits.

Before you start: why diesel needs injection

A diesel engine compresses only air. Fuel must be sprayed in at the right moment at very high pressure (hundreds of bar) so that it breaks into tiny droplets, evaporates and ignites in the hot air. The injection system pumps, meters, times and sprays the fuel.

Purpose of a fuel injection system (2024 4a)

1. **Meter** the exact quantity of fuel demanded by speed and load, and the same quantity to every cylinder every cycle.
2. **Time** the injection correctly in the cycle (start of injection), and inject at the correct **rate** (rate shaping) for the desired heat release.
3. **Atomize** the fuel into fine droplets and give the **spray pattern and penetration** that the combustion chamber needs, so it mixes with air quickly.
4. **Distribute** the fuel uniformly through the air; begin and end injection **sharply without dribbling** or secondary injections.
5. In SI (spark ignition (petrol engine)) engines: replace the carburetor for precise A/F (air-fuel ratio) control ($\varphi = 1$ for the catalyst), better distribution, better cold start and economy.

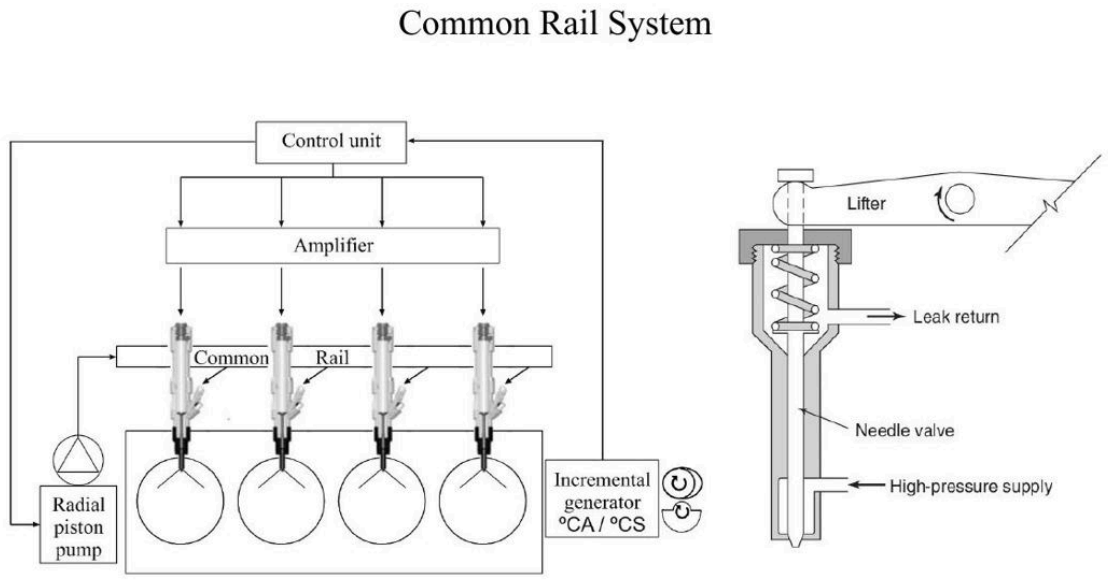
Main components and functions (2024 4b)

Component	Function
Fuel tank	Stores fuel
Feed (lift / transfer) pump	Low-pressure supply (1-3 bar) from tank to the injection pump
Fuel filters (primary, secondary)	Remove dirt and water that would damage the precision pump and nozzles
High-pressure injection pump (or common rail pump)	Raises pressure to 100-2000 bar and meters fuel for each cylinder (plunger and helix / ECU-controlled)
Governor	Controls fuel quantity to regulate speed under changing load
High-pressure pipes	Carry fuel to the injectors (equal length to keep timing equal)
Injector with nozzle	Opens at set pressure (spring-loaded needle) or by solenoid/piezo, atomizes and sprays fuel into the chamber
Return line / ECU and sensors (modern)	Leak-off back to tank; ECU sets timing and quantity from speed, load, temperature sensors

Types of fuel injection systems (2023 3c, 2022 3a)

CI (compression ignition (diesel engine)) engines: (a) **Air injection** (old): fuel blasted in by compressed air at ~70 bar from an engine-driven compressor; good atomization but bulky and costly, obsolete. (b) **Solid (airless) injection**: fuel pressurised by a pump and sprayed directly. Sub-types:

System	Description	Merits	Demerits
Individual pump (jerk pump)	Separate metering and pressure pump for each cylinder	Accurate metering, simple	Many pumps, timing per cylinder
Unit pump / unit injector	Pump and injector in one unit per cylinder driven by cam	Very high pressure, no long HP pipes	Costly, cam drive in head
Distributor	One pump meters and pressurises, a rotor distributes to cylinders	Compact, cheap, equal distribution	Pressure limited, small engines
Common rail	High-pressure pump keeps a rail at constant 1000-2000 bar; ECU opens electronic injectors	Pressure independent of speed, multiple injections, flexible timing, low noise and emission	Expensive, needs clean fuel and electronics



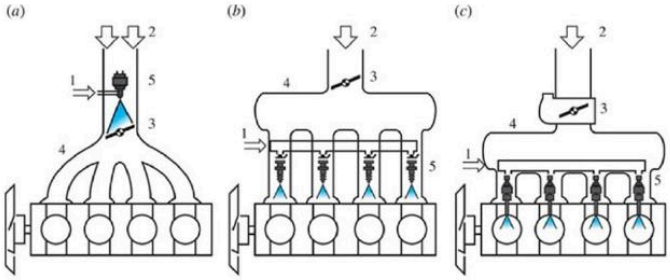
Common rail system: radial piston pump, rail, ECU-controlled injectors

SI engines: throttle-body (single-point) injection, **multipoint port fuel injection** (one injector per intake port), and **gasoline direct injection** into the cylinder. Direct vs indirect in CI: DI sprays into an open chamber in the piston at high pressure; IDI sprays into a pre-chamber/swirl chamber at lower pressure.

Fuel Injection System

Fuel injection system can be divided into two basic types:

1. Manifold: (a) Throttle body & (b) Port
2. (c) Gasoline direct injection (GDI)



Schematics of different fuel-injection system arrangements: (*a*) single point, (*b*) multipoint (port), and (*c*) direct in-cylinder injection (DI). (1) Fuel; (2) Air; (3) Throttle; (4) Intake manifold; (5) Fuel injector. [2] (*Courtesy Robert Bosch GmbH and SAE.*)

SI injection arrangements: (a) single point, (b) multipoint port, (c) direct in-cylinder

A4 Classification of fuels; storage and handling of solid, liquid and gaseous fuels Good to have Likely · 60% Easy

PAST-PAPER QUESTIONS

- **2024 Q1(a)** [8] What are the primary classifications of fuels, and how are they differentiated based on their origin?
- **2024 Q1(b)** [9] What factors influence the safe storage and handling of solid, liquid, and gaseous fuels?
- **2023 Q1(d)** [11] How do the storage and handling requirement of solid, liquid and gaseous fuels differ and what safety considerations are associated with each?

Before you start: what a fuel classification means

Fuels are sorted by **state** (solid, liquid, gas) and by **origin** (primary = found in nature, like coal and natural gas; secondary = made from a primary fuel, like coke, petrol, LPG (liquefied petroleum gas)). Storage problems differ: solids self-heat and make dust, liquids give flammable vapour, gases leak and can explode.

Classification of fuels by origin (2024 1a)

Basis	Classes	Examples
Origin / nature	**Primary (natural)**: used as found in nature	Wood, peat, coal, crude petroleum, natural gas
	Secondary (derived): made from primary fuels; either manufactured for a purpose or by-products	Charcoal, coke, petrol, diesel, kerosene, LPG, producer gas, coke oven gas, blast furnace gas
Physical state	Solid, liquid, gaseous	Coal / petrol / natural gas
General	Fossil, by-product, chemical, nuclear	Coal; coke oven gas; hydrazine; uranium
Renewability	Renewable vs non-renewable	Biomass, biodiesel, H_2 vs coal, oil, gas

They differ in origin: **fossil fuels** come from plant and animal remains buried for millions of years (coal from plants; oil and gas from marine organisms under heat and pressure); **biomass fuels** from recent plant matter; **manufactured fuels** from processing (carbonization, refining, gasification).

Factors for safe storage and handling (2024 1b, 2023 1d)

	Solid (coal, biomass)	Liquid (petrol, diesel, furnace oil)	Gaseous (natural gas, LPG, H_2)
Main hazards	Spontaneous combustion (slow oxidation heats the pile), dust explosion (pulverized coal), weathering loss, moisture	Fire from vapour at flash point, leaks and spills, static electricity, water contamination, waxing at pour point	Leaks form explosive mixtures within flammability limits, high pressure, asphyxiation, cryogenic burns
Storage	Covered sheds or compacted heaps, limited height, drainage, avoid hot spots, first-in-first-out	Cylindrical tanks above/below ground with vents, flame arrestors, bunds (dykes), >10 days' supply, heating for heavy oil	High-pressure cylinders or spheres, cryogenic tanks (LNG, liquid H_2), gas holders, pipeline
Handling	Conveyors, crushers with dust suppression, ventilation, temperature monitoring	Pumps, filters, earthing/bonding, no-smoking, fire extinguishers, regular tank cleaning	Leak detectors, odorant (mercaptan), pressure relief valves, ventilated rooms, flame-proof electricals
Key property to watch	Volatile matter, moisture, particle size	Flash point, viscosity, pour point, vapour pressure	Flammability limits, density relative to air (LPG sinks, H_2 rises)

Remember: Solid: **self-heating and dust**. Liquid: **vapour and spills**. Gas: **leaks and pressure**.

A8 Carburetion and the simple carburetor; compensating jet Must study

Likely · 60% Easy

PAST-PAPER QUESTIONS

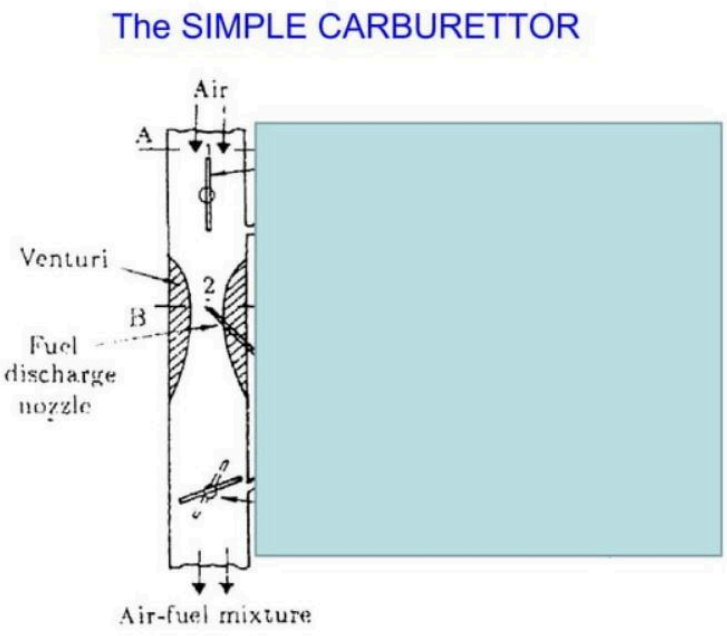
- **2024 Q3(a)** [8] Briefly explain the working principle of a carburetor in IC (internal combustion) engines with neat sketches.
- **2024 Q4(d)** [10] Explain the function of the compensating jet in a carburetor and how it helps maintain proper combustion.
- **2023 Q3(a)** [8] What is carburation in an IC engine? Describe the working principle of a carburator in IC engine.

Before you start: what a carburetor does

A petrol engine needs air and petrol mixed in the right ratio before entering the cylinder. A carburetor does this using the **venturi effect**: air speeds up in a narrow throat, its pressure drops below atmospheric, and that low pressure sucks petrol out of a small jet into the air stream.

Carburetion (2023 3a)

Carburetion is the process of forming a combustible air-fuel mixture of correct strength outside the cylinder by mixing the right amount of fuel with air in a device called a carburetor. It involves **metering** the fuel, **atomizing** it into droplets, **vaporizing** and **distributing** it uniformly to the cylinders. Factors: engine speed (time available), fuel volatility, inlet air temperature and pressure, carburetor design.



Simple carburettor: float chamber, fuel nozzle in the venturi, throttle valve

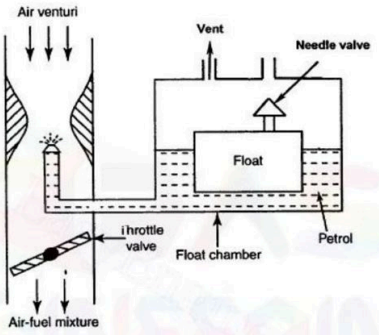
Working principle of a simple carburetor (2024 3a, 2023 3a)

1. **Float chamber:** float and needle valve keep fuel at a constant level, a little ($z \approx 5$ mm) below the nozzle tip, at atmospheric pressure (vented).
2. **Venturi:** during suction the piston draws air through the air filter and the venturi. At the narrow throat **velocity rises and pressure falls** (Bernoulli: $p + \frac{1}{2}\rho V^2 = \text{constant}$).

3. **Fuel discharge:** the pressure difference ($p_{atm} - p_{throat}$) pushes fuel through the metering jet and out of the discharge nozzle at the throat.
4. **Atomization and mixing:** the high-speed air breaks the fuel into fine droplets that start to evaporate and mix with air.
5. **Throttle valve** (butterfly, after the venturi) controls the quantity of mixture, and hence speed and load (quantity governing). Choke valve before the venturi enriches the mixture for cold start.

Principle of Operation of Carburetor

A carburetor mainly works on the principle of Bernoulli. So, as and when the velocity increases, the pressure for the same will decrease. Mainly air and gasoline go to the combustion chamber of the engine, passing through the carburetor.



Carburetor principle sketch (slide)

Remember: Narrow passage → higher air velocity → lower pressure → fuel is sucked into the airflow.

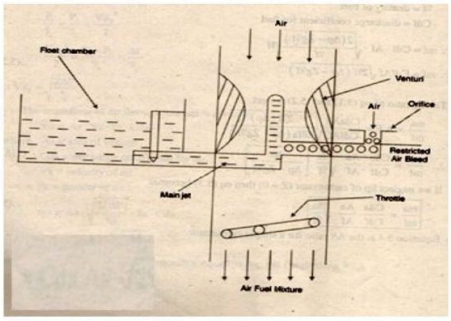
Compensating jet: function and how it keeps combustion correct (2024 4d)

In a simple carburetor, as air flow (throttle opening/speed) rises, fuel flow rises **faster** than air flow (air density at throat falls, fuel is incompressible), so the mixture becomes **richer** at high speed and leaner at low speed. The **compensating system** corrects this.

Principles of a Compensating System:

Main Jet and Air Bleed System:

- **Main Jet:** This is the primary pathway through which fuel is drawn into the air stream. The size of the main jet determines the maximum amount of fuel that can flow through it.



Carburetor with main jet and compensating jet (slide)

- **Compensating jet**: a second jet fed from a small well (compensating well) open to atmosphere, which is itself supplied from the float chamber through a restricted jet. At low air flow it adds fuel; at high air flow the well empties faster than the restricted orifice refills it, so its contribution becomes **constant** and the mixture produced by it gets **leaner**. The main jet richens, the compensating jet leans, and together they give a nearly **constant A/F** (air-fuel ratio).
- **Air bleed / emulsion tube** (modern form): air admitted into the fuel well through small holes; as suction rises more holes uncover, more air bleeds into the fuel, reducing the pressure difference on the jet and preventing over-richening. It also pre-atomizes the fuel (emulsion).
- Result: correct mixture over the cruising range → complete combustion, good economy, low CO/HC, no flooding at high speed and no lean misfire at low speed.

A12 Injector nozzles: types, spray pattern, atomization, sac volume Must study Likely · 67% Medium

PAST-PAPER QUESTIONS

- **2024 Q4(c)** [10] Explain how fuel injection nozzles control the spray pattern and atomization of fuel for efficient combustion.
- **2022 Q3(c)** [9] What is sac volume? What is the relationship between NOx (nitrogen oxides (NO + NO_2)) formation and combustion time?
- **2021 Q3(b)** [12] Describe the different types of injection nozzles with their relative merits and demerits.

Before you start: what a nozzle does

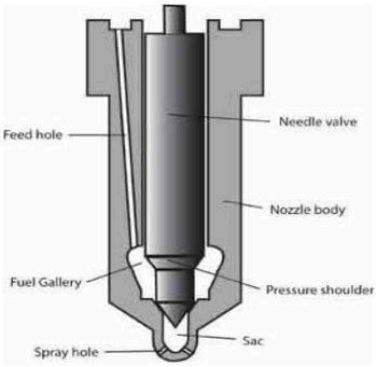
The nozzle is the tip of the injector. Its holes break fuel into a fine spray (**atomisation**), send it in the right direction (**penetration**) and spread it through the air (**distribution**). **Sac volume** = small space below the needle where fuel is left after injection; it dribbles out and causes smoke and HC (hydrocarbons (unburnt fuel)).

Types of injection nozzles, merits and demerits (2021 3b, 2023 3c, 2022 3a)

Nozzle	Construction / spray	Merits	Demerits / use
Single-hole	One hole > 0.2 mm at the centre or angled; narrow jet ~15°	Simple, good penetration	Needs very high pressure, tends to dribble, narrow spray gives poor mixing unless high air swirl; open chambers
Multi-hole	4-18 holes of 0.15-0.35 mm drilled in the tip; several fine sprays	Good distribution and atomization in open (DI) chambers with low air motion	High pressure, small holes clog/carbonize; DI engines
Pintle	Needle tip extends through the hole as a pin (two truncated cones); hollow conical spray 0-60°	Lower injection pressure, self-cleaning, good atomization with reduced penetration	Needs swirl/pre-combustion chamber (IDI)
Pintaux	Pintle nozzle plus a small auxiliary hole that sprays pilot fuel upstream into the hottest air	Very good cold starting: at low speed most fuel goes through the auxiliary hole; delay reduced	More complex; IDI engines

Injection Nozzles

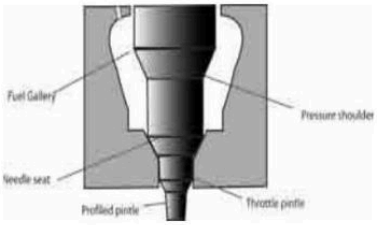
(b) Multi-Hole Nozzle: in order to mix the fuel properly even with slow air movement available with many types of open combustion chambers a multi-hole nozzle is used. The numbers of holes varies from 4 to 18 and the size varies from 0.15 mm to 0.35 mm.



Multi-hole nozzle

Injection Nozzles

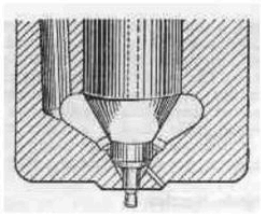
(c) Pintle Nozzle: the Pintle nozzle has the form of two truncated cone with their smaller base together or cylindrical shape. The spray obtained by the Pintle nozzle is hollow conical spray and the cone angle can be varied from 0 to 60 by giving different tapers to the Pintle. This type of nozzle is used with pre-combustion chamber, or high swirl chamber.



Pintle nozzle

Injection Nozzles

(d) Pintaux Nozzle (Pintle with Auxiliary Hole): to improve cold starting performance without determinate effect on efficiency a Pintaux nozzle is used (if the fuel is injected in a direction upstream the direction of air, the delay period is reduced due to increasing heat transfer between air and fuel). This is a development of Pintle nozzle and has auxiliary hole drilled in the nozzle body. This hole allows a small amount of pilot fuel injection in the upstream at a time slightly in advance of the main downstream injection. At low speeds, the needle valve does not left fully and most of the fuel is injected through the side hole resulting in very good cold starting performance.



Pintaux nozzle with auxiliary hole

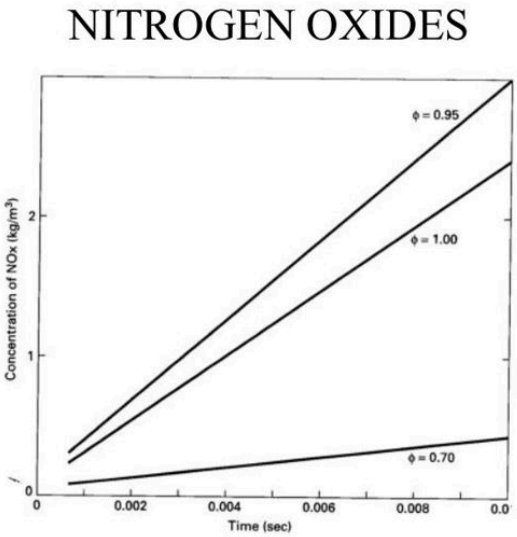
How nozzles control spray pattern and atomization (2024 4c)

- **Atomization:** fuel is forced through small orifices by a large pressure difference (injection minus cylinder pressure); jet velocity $V = C_d\sqrt{(2\Delta p/\rho_f)}$ of 200-400 m/s breaks the jet into fine droplets (Sauter mean diameter 10-30 µm). Higher pressure and smaller holes → finer droplets.
- **Penetration** grows with pressure, hole size and fuel density, and falls with air density; it must reach the air in the chamber but not wet the walls.
- **Spray cone angle and pattern** are set by the number, size and angle of holes (multi-hole), and by the pintle shape (hollow cone). More holes give better air utilisation.
- **Needle valve and spring** give sharp opening at the set pressure and sharp closing to prevent dribbling; electronic injectors give pilot, main and post injections.
- **L/d of the hole, sac shape, and nozzle cavitation** also improve break-up.

Sac volume (2022 3c)

The **sac** is the small volume in the nozzle tip below the needle seat, from which the spray holes branch. After the needle closes, the fuel left in the sac (and holes) evaporates slowly late in expansion and leaves as **unburnt HC** (and smoke). Reduced by mini-sac or **valve-covered-orifice (VCO)** nozzles where the needle covers the holes directly.

NOx formation and combustion time (2022 3c)



NOx as a function of combustion time

NO concentration rises with time at high temperature (thermal NO is rate-controlled)

Thermal NO forms by the Zeldovich reactions only above ~1800 K and its rate is slow compared with combustion, so NO concentration **increases with the time** the burnt gas spends at high temperature: the longer the residence time (early burning gas, slow combustion near peak temperature, low engine speed) the more NO. The first gas to burn is compressed further and stays hottest longest, so it produces most NO. Reducing time at peak temperature (retarded timing, faster expansion) reduces NO.

A10 Venturi throat diameter numerical (compressible, isentropic) Good to have Likely · 60% Hard

PAST-PAPER QUESTIONS

- **2024 Q3(d)** [8] An engine requires 6 kg of air per minute from its carburetor. The air is at a pressure of 1.02 bar and a temperature of 25 °C. Determine the throat diameter of the venturi when the air flow velocity at the throat is 100 m/s with a velocity coefficient of 0.75. Assume isentropic flow and consider air as compressible. R = 287 J/kg·K.
- **2023 Q3(d)** [7] An engine requires 5 kg of air per minute from its carburator. The air is at a pressure of 1.05 bar and a temperature of 25 °C. Determine the throat diameter of the venturi when the air-flow velocity at the throat is 1.02 m/s [sic] with a velocity coefficient of 0.8. Assume isentropic flow and consider air as compressible.

LEARN IT FROM ZERO

What the venturi question is really asking

Picture the carburetor throat as a **narrow neck in a pipe**. The same kilograms of air per second must pass through it, so in the narrow part the air must move **faster**. Energy to speed the air up comes from its own pressure and temperature, so in the throat the **pressure drops**. That low pressure is what sucks petrol out of the jet.

The question gives you how much air flows (kg/min), the air at the inlet (p_1, T_1) and how fast it moves in the throat (C_2). It asks: **how wide must the throat be?** Width comes from **mass flow = density × area × velocity**, so you need the air density in the throat. Density depends on pressure, so first find the throat pressure. That is the whole chain:

velocity → throat pressure p_2 → throat density ρ_2 → area A_2 → diameter d_2

Why the long square-root formula? It is just "kinetic energy gained = enthalpy lost": $C_2^2/2 = c_p(T_1 - T_2)$. For frictionless (isentropic) flow $T_2/T_1 = (p_2/p_1)^{(\gamma-1)/\gamma} = (p_2/p_1)^{0.286}$. Put that in and solve for the velocity: $C_2 = \sqrt{[2c_pT_1(1 - (p_2/p_1)^{0.286})]}$. Real air rubs on the walls, so the real velocity is a bit less: $C_{actual} = C_v \times C_{ideal}$. That is why you **divide** the given velocity by C_v before using the formula.

Why "compressible"? At about 100 m/s the pressure drop is about 10 %, so air density also falls about 7 %. Ignoring this (the slide's quick way) gives a wrong, too-large diameter.

Symbols used here

Symbol	Name and meaning	Unit
$\dot{m}$	Mass flow rate of air	kg/s (given in kg/min, divide by 60)
p_1, T_1	Pressure and temperature at the venturi inlet (outside air)	bar ($\times 10^5$ for Pa), K
p_2	Pressure at the throat (lowest pressure)	bar
C_2	Actual air velocity at the throat	m/s
C_v	Velocity coefficient = actual / ideal velocity (friction)	no unit, 0.75–0.9
c_p	Specific heat of air at constant pressure	1005 J/kg K
R	Gas constant of air	287 J/kg K
γ	c_p/c_v for air	1.4 (so $(\gamma-1)/\gamma$ = 0.286, $\gamma/(\gamma-1)$ = 3.5, $1/\gamma$ = 0.714)
ρ_1, ρ_2	Air density at inlet and throat	kg/m³
A_2, d_2	Throat area and throat diameter (the answer)	m², mm

UNDERSTAND **Read this to understand; you do not need to write it**

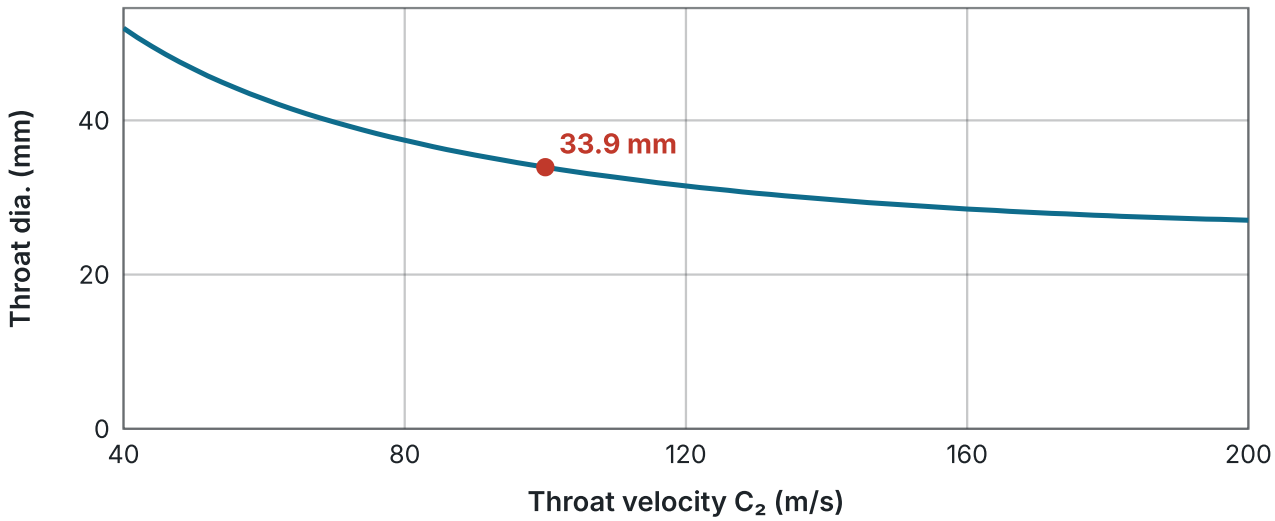
The three powers you need come from γ = 1.4: **0.286** (pressure ratio → temperature ratio), **3.5** (undo it to get the pressure ratio back) and **0.714** (pressure ratio → density ratio). Write them once in the given data.

Use the **actual** velocity C_2 (not the ideal one) in $\dot{m} = \rho_2 A_2 C_2$: the air really moves at that speed.

A sensible answer is 25–40 mm. If you get centimetres or metres, you used the wrong units somewhere (bar not converted to Pa is the usual mistake).

INTERACTIVE **Venturi throat calculator (compressible, isentropic)**

Type the numbers from any paper; every step updates. The graph shows how the throat diameter changes with throat velocity.



1. $\dot{m}$ = 6/60 = **0.1000 kg/s**; T_1 = 298 K
2. Ideal velocity = C_2/C_v = 100/0.75 = 133.33 m/s
3. $1 - (p_2/p_1)^{0.286}$ = 133.33^2 / (2 × 1005 × 298) = 0.02968
4. $p_2/p_1 = 0.97032^{3.5} = 0.8999 \Rightarrow$ **p_2 = 0.918 bar**
5. ρ_1 = 102000 / (287 × 298) = 1.193 kg/m³
6. $\rho_2 = 1.193 \times 0.8999^{0.714}$ = **1.106 kg/m³**
7. A_2 = 0.1000 / (1.106 × 100) = 9.041×10^{-4} m²
8. $d_2 = \sqrt{(4A_2/\pi)}$ = **33.9 mm**

Before you start: the physics behind the venturi numerical

When air speeds up, its pressure falls (energy conservation). For gas at high speed the density also changes, so we treat it as **compressible, isentropic** (no friction, no heat loss) flow.

Three tools: (1) energy equation $C_2 = \sqrt{[2c_pT_1(1 - (p_2/p_1)^{0.286})]}$ gives the throat pressure; (2) density $\rho = p/RT$ and $\rho_2 = \rho_1(p_2/p_1)^{1/1.4}$; (3) mass flow = ρ A C gives the area, then diameter.

C_v (velocity coefficient) = actual velocity / ideal velocity; it accounts for friction.

Venturi throat diameter, compressible isentropic flow (2024 3d)

Given: $\dot{m}_a$ = 6 kg/min = 0.1 kg/s, p_1 = 1.02 bar, T_1 = 25 °C = 298 K, actual throat velocity C_2 = 100 m/s, C_v = 0.75, R = 287 J/kg K, c_p = 1005 J/kg K, γ = 1.4. Inlet velocity $C_1 \approx 0$.

Steady flow energy, isentropic: $C_2 = C_v \sqrt{[\ 2\ c_p\ T_1\ (1 - (p_2/p_1)^{(\gamma-1)/\gamma})\ ]}$

$100 = 0.75 \sqrt{[\ 2 \times 1005 \times 298\ (1 - x)\ ]} \Rightarrow (133.33)^2 = 598\ 980\ (1 - x)$

$1 - x = 17\ 778 / 598\ 980 = 0.02968 \Rightarrow x = (p_2/p_1)^{0.2857} = 0.97032$

$p_2/p_1 = 0.97032^{3.5} = 0.8999 \Rightarrow$ **$p_2 = 0.918$ bar**

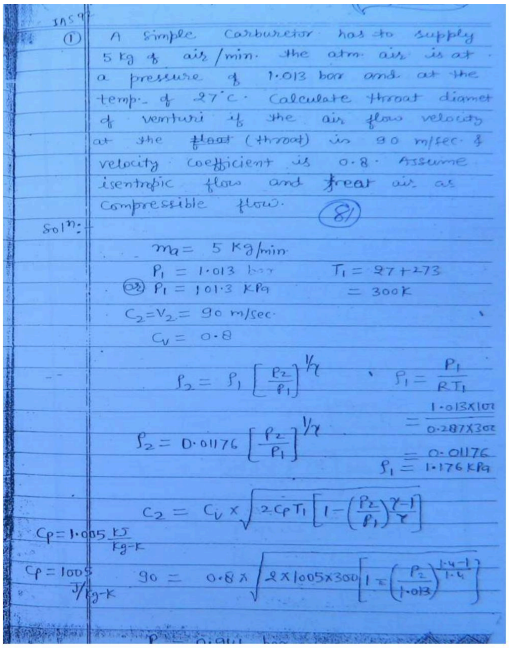
$\rho_1 = p_1/RT_1 = 102\ 000 / (287 \times 298) = 1.193\ kg/m^3$

$\rho_2 = \rho_1(p_2/p_1)^{1/\gamma} = 1.193 \times 0.8999^{0.714} = 1.106\ kg/m^3$

$A_2 = \dot{m}/(\rho_2 C_2) = 0.1 / (1.106 \times 100) = 9.04 \times 10^{-4}\ m^2$

$d_2 = \sqrt{(4A_2/\pi)} = \sqrt{(4 \times 9.04 \times 10^{-4}/\pi)} = 0.0339\ m =$ **33.9 mm**

Note on the class slide (m01, Problem 2): the slide solves this exact problem in a simplified way: it multiplies $100 \times 0.75 = 75$ m/s and uses the inlet density 1.188 kg/m³, giving $A = 0.1/(1.188 \times 75) = 1.122 \times 10^{-3}$ m² and **d = 37.8 mm**. Because the question says "isentropic, compressible", write the method above; if time allows add one line saying the incompressible estimate gives 37.8 mm.



Class notes: the same method for 5 kg/min, 1.013 bar, 27 °C, 90 m/s, $C_v = 0.8$ (answer 32.5 mm)

2023 3(d): 5 kg/min, 1.05 bar, 25 °C, $C_v = 0.8$

The paper prints the throat velocity as "1.02 m/s", which is a misprint (a carburetor throat runs at 90-100+ m/s; with 1.02 m/s the throat would be 29 cm wide). Take **$C_2 = 102$ m/s** and say so in the script.

$\dot{m} = 5/60 = 0.0833$ kg/s; $C_2/C_v = 102/0.8 = 127.5$ m/s

$1 - x = 127.5^2 / (2 \times 1005 \times 298) = 0.02714 \Rightarrow x = 0.97286 \Rightarrow p_2/p_1 = 0.97286^{3.5} = 0.9082 \Rightarrow p_2 = 0.954$ bar

$\rho_1 = 105\ 000/(287 \times 298) = 1.228\ kg/m^3$; $\rho_2 = 1.228 \times 0.9082^{0.714} = 1.146\ kg/m^3$

$A_2 = 0.0833/(1.146 \times 102) = 7.13 \times 10^{-4}\ m^2 \Rightarrow$ **$d_2 = 30.1$ mm**

Remember: Four lines: (1) velocity equation → p_2; (2) $\rho_1 = p/RT$; (3) $\rho_2 = \rho_1(p_2/p_1)^{1/1.4}$; (4) $A = \dot{m}/\rho_2 C_2$, $d = \sqrt{(4A/\pi)}$. Typical answer 30-35 mm.

Same answer, step by step with reasons

Step (write this)	Why / what it means
$\dot{m} = 6$ kg/min $= 0.1$ kg/s; $T_1 = 298$ K; $p_1 = 1.02$ bar	Convert to SI (spark ignition (petrol engine)) units first.
Ideal velocity = actual / C_v = 100/0.75 = 133.3 m/s	The energy equation gives ideal (frictionless) velocity, so divide by C_v.
$133.3^2 = 2 \times 1005 \times 298 \times (1 - x) \Rightarrow 1 - x = 0.0297$	Kinetic energy gained = enthalpy lost: $C^2/2 = c_p(T_1 - T_2)$, and $T_2/T_1 = x = (p_2/p_1)^{0.286}$.
$p_2/p_1 = 0.9703^{3.5} = 0.900 \Rightarrow p_2 = 0.918$ bar	Undo the power: 1/0.2857 = 3.5.
$\rho_1 = 102{,}000/(287 \times 298) = 1.193$ kg/m^3	Ideal gas law.
$\rho_2 = 1.193 \times 0.900^{0.714} = 1.106$ kg/m^3	Isentropic: $\rho \propto p^{1/\gamma}$, 1/1.4 = 0.714.
$A_2 = 0.1/(1.106 \times 100) = 9.04 \times 10^{-4}$ m^2	Mass flow = density × area × ACTUAL velocity.
$d_2 = \sqrt{(4A/\pi)} =$ **33.9 mm**	Circle area = $\pi d^2/4$.

Memory trick: "Velocity → pressure → density → area → diameter".

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2024 3(d) Given: $\dot{m} = 6$ kg/min $= 0.1$ kg/s, $p_1 = 1.02$ bar, $T_1 = 298$ K, $C_2 = 100$ m/s, $C_v = 0.75$; air $c_p = 1005$ J/kg K, $R = 287$ J/kg K, $\gamma = 1.4$

Ideal throat velocity: $C_{2,ideal} = C_2 / C_v$

$= 100 / 0.75$

$=$ **133.3 m/s**

Throat pressure: $C_{2,ideal} = \sqrt{[2c_pT_1\{1 - (p_2/p_1)^{0.286}\}]}$

$= 133.3^2 = 2 \times 1005 \times 298 \{1 - (p_2/p_1)^{0.286}\} \Rightarrow (p_2/p_1)^{0.286} = 0.9703 \Rightarrow p_2/p_1 = 0.9703^{3.5} = 0.900$

$=$ **$p_2 = 0.918$ bar**

Inlet density: $\rho_1 = p_1 / RT_1$

$= 1.02 \times 10^5 / (287 \times 298)$

$=$ **1.193 kg/m³**

Throat density: $\rho_2 = \rho_1(p_2/p_1)^{1/\gamma}$

$= 1.193 \times 0.900^{0.714}$

$=$ **1.106 kg/m³**

Throat area: $A_2 = \dot{m} / (\rho_2 C_2)$

$= 0.1 / (1.106 \times 100)$

= **9.04×10^{-4} m²**

Throat diameter: $d_2 = \sqrt{(4A_2/\pi)}$

$= \sqrt{(4 \times 9.04 \times 10^{-4} / \pi)}$

= **$d_2 = 33.9$ mm**

2023 3(d) (throat velocity printed as 1.02 m/s, taken as 102 m/s): $\dot{m} = 0.0833$ kg/s, $p_1 = 1.05$ bar, $C_v = 0.8$ ⇒ $C_{2,ideal} = 127.5$ m/s, $p_2/p_1 = 0.9082$, $p_2 = 0.954$ bar, $\rho_1 = 1.228$, $\rho_2 = 1.146$ kg/m³, $A_2 = 7.13 \times 10^{-4}$ m², **$d_2 = 30.1$ mm**

Memory trick: V-P-D-A-D: Velocity → Pressure → Density → Area → Diameter.

A17 Particulate emission, its measurement, smoke colours Must study

Possible · 53% Easy

PAST-PAPER QUESTIONS

- **2023 Q4(b)** [12] What is particulate emission? How can particulate emission be measured? Why does particulate emission from CI (compression ignition (diesel engine)) engines differ significantly than that of SI (spark ignition (petrol engine)) engines?
- **2022 Q4(c)** [12] What is particulate emission? How particulate emission can be measured? Why does particulate emission from CI engines differ significantly than that of SI engines?
- **2020 Q3(c)** [8] What is smoke? "Smoke colors are indicative of dominant source of particulate" – Explain.

Before you start: particulates

Particulate matter (PM) = tiny solid/liquid particles in exhaust, mostly soot (carbon) with absorbed hydrocarbons, sulphates and water. Diesels produce much more PM (particulate matter) than petrol engines. Smoke colour hints at the cause: black = soot (too much fuel), blue = burning lubricating oil, white = unburnt fuel droplets or water at cold start.

Particulate emission (2023 4b, 2022 4c)

Particulate matter (PM) is defined (US EPA) as any substance, other than condensed water, that is collected on a filter from exhaust **diluted with air to a temperature at or below 325 K (52 °C)**. It has two parts: (a) **solid carbon soot** (spherical particles 10-80 nm agglomerating into chains, up to ~4000 spheres per particle) and (b) the **soluble organic fraction**: hydrocarbons and their partial oxidation products (25-75 % from lubricating oil) condensed or adsorbed on the soot, plus sulphates and ash. PM_{10} and $PM_{2.5}$ penetrate deep into the lungs.

Measurement of PM

- **Dilution tunnel + filter (gravimetric)**: exhaust is diluted with filtered air in a tunnel to ≤ 325 K, drawn through a weighed Teflon filter; mass gain / volume gives g/km or g/kWh. This is the legal method.
- **Filter paper smoke meter (Bosch)**: a fixed volume of exhaust drawn through filter paper; blackening read by reflectance (Bosch smoke number 0-10).
- **Light absorption / opacity meter (Hartridge)**: light beam through an exhaust column; percentage of light blocked = opacity.
- Particle counters and size analysers (condensation particle counter, electrical mobility).

Why CI engines emit much more PM than SI engines

- CI combustion is **heterogeneous**: fuel is injected as a spray and burns as a diffusion flame; the spray core is very rich ($\varphi > 2$) at high temperature → soot forms by pyrolysis. SI burns a **premixed homogeneous** vapour mixture, so there are no rich pockets.
- Diesel fuel has heavier molecules, higher boiling point and more aromatics; heavy HC (hydrocarbons (unburnt fuel)) condense on soot.
- Diesels run near full load with little time for mixing at high speed; soot rises steeply as more fuel is injected.
- Unleaded petrol and SI combustion make PM very small (except GDI (gasoline direct injection) engines with wall wetting).

Smoke and its colours (2020 3c)

Smoke is visible exhaust: a suspension of solid or liquid particles (soot, oil or fuel droplets, water) that absorbs or scatters light. Its colour tells the dominant source:

Colour	Source	Cause
Black / grey	Carbon soot	Over-fuelling, rich mixture, poor atomization, dirty air filter, overload, wrong injection timing
Blue / bluish-white	Burning **lubricating oil**	Worn rings, liner or valve guides; oil passing into the chamber
White	**Unburnt fuel** droplets or water vapour	Cold start, misfire, low compression, retarded timing (fuel condenses); coolant leak (steam)

A3 Alternative fuels: ethanol, biodiesel, methanol, hydrogen Must study

Possible · 40% Easy

PAST-PAPER QUESTIONS

- **2022 Q1(c)** [12] Describe the possible pathways of using alcohol based fuels and hydrogen fuel for IC (internal combustion) engines.
- **2021 Q1(a)** [13] What is meant by "Alternative fuels"? Discuss the prospects of ethanol and biodiesel as a fuel for IC engines.
- **2020 Q1(a)** [11] How can you use hydrogen as a fuel in CI (compression ignition (diesel engine)) Engine? (first part)
- **2020 Q1(b)** [14] What are the problems of methanol as alternative fuel? Why do you need extra fuel when the throttle is opened?

Before you start: why alternative fuels

Petrol and diesel come from crude oil, which is limited and gives CO_2. Alternative fuels (ethanol from sugar cane/corn, biodiesel from vegetable oil, methanol from natural gas/biomass, hydrogen) can be renewable and cleaner, but each has a weakness: less energy per litre, cold-start problems, corrosion or storage difficulty.

Alternative fuels (2021 1a)

Alternative fuels are fuels other than conventional petroleum petrol and diesel that can run IC engines, ideally renewable, domestically produced and cleaner: alcohols (ethanol, methanol, butanol), biodiesel and vegetable oils, natural gas (CNG/LNG), LPG (liquefied petroleum gas), biogas, hydrogen, and electricity/fuel cells.

Prospects of ethanol

- Made by fermenting sugar/starch (sugarcane, corn) or cellulosic waste; renewable, biodegradable, sulphur and aromatic free.
- High octane (~108) and high latent heat (charge cooling) → allows higher compression ratio; burns cleaner (less CO and HC (hydrocarbons (unburnt fuel))).
- Used as blends: **E10** in normal cars, up to **E85** in flexible-fuel vehicles; Brazil runs E100 engines.
- Problems: LCV (lower calorific value) ~27 MJ/kg (about 2/3 of petrol) so more fuel per km, absorbs water, corrodes some rubbers and aluminium, cold start difficulty, food vs fuel debate.

Prospects of biodiesel

- Fatty acid methyl ester (FAME) from vegetable oils, waste cooking oil or animal fat by **transesterification** with methanol + catalyst (NaOH/KOH).
- Higher cetane number (50-65), better lubricity, non-toxic, no sulphur or aromatics; can run unmodified diesel engines as B5-B20 (B100 with minor changes).
- Reduces CO, HC and soot and net CO_2; but NOx (nitrogen oxides (NO + NO_2)) rises slightly, viscosity and cloud point are higher (cold flow problems), LCV ~10 % lower, oxidation stability poor.

Pathways for using alcohol fuels and hydrogen in IC engines (2022 1c)

Fuel	SI (spark ignition (petrol engine)) engine pathway	CI engine pathway
Alcohols (ethanol, methanol)	Blends E10-E85 / M15-M85 in petrol; neat alcohol with higher compression ratio (11-13) and richer jets; flexible-fuel vehicles with sensor	Low cetane, so: dual-fuel (alcohol fumigated into intake + diesel pilot), alcohol-diesel emulsions with additives, ignition improver, glow plug or spark assisted ignition
Hydrogen	Port or direct injection with spark ignition; lean burn to control NOx and backfire; or as an additive to CNG (HCNG)	Dual-fuel: H_2 inducted with air, ignited by diesel pilot; or glow plug / spark assist
Hydrogen (indirect)	Fuel cells: H_2 + O_2 → electricity for an electric drive (no combustion)	

Hydrogen facts: produced by steam reforming of natural gas, gasification or electrolysis; very high CV (120 MJ/kg) and flame speed, wide flammability limits (4-75 %), very low ignition energy; stored as cryogenic liquid or high-pressure gas (bulky, costly); no CO_2, HC or CO, only NOx.

Problems of methanol, and why extra fuel is needed when the throttle opens (2020 1b)

- Low heating value (~20 MJ/kg, about half of petrol) → larger fuel flow and tank.
- Very high latent heat and single boiling point → **cold-start difficulty**, poor vaporization.
- Toxic (poisonous by skin and vapour), invisible flame; **corrosive** to aluminium, zinc, rubber and plastics.
- Absorbs water, phase separation in blends; produces **formaldehyde** in exhaust; mostly made from natural gas, so not truly renewable.

Extra fuel on opening the throttle: when the throttle is opened suddenly, air (low density) accelerates at once, but liquid fuel (dense, with inertia) and fuel film on manifold walls lag; manifold pressure rises so fuel evaporation falls. The mixture leans momentarily and the engine hesitates or stalls, so an **accelerator pump** squirts extra fuel. With alcohols this is worse because of their high latent heat.

A15 Lubrication: parts lubricated, oil selection, classification, mist lubrication Good to have Possible · 33% Easy

PAST-PAPER QUESTIONS

- **2022 Q2(b)** [11] What are the various components to be lubricated in an engine? Write the selection criteria of lubricating oils in various weather conditions.
- **2021 Q2(b)** [14] How lubricating oils are classified? Discuss the mist lubrication system with diagram.

Before you start: what lubrication is

Lubrication = keeping a thin film of oil between surfaces that rub (piston and cylinder, crankshaft and bearings) so they do not touch. Oil also cools, cleans, seals and protects against rust. **Viscosity** = thickness of oil (SAE number).

Components to be lubricated (2022 2b)

Parts which require lubrication

(i) cylinder walls and piston

(ii) piston pin

(iii) crankshaft and connecting rod bearings

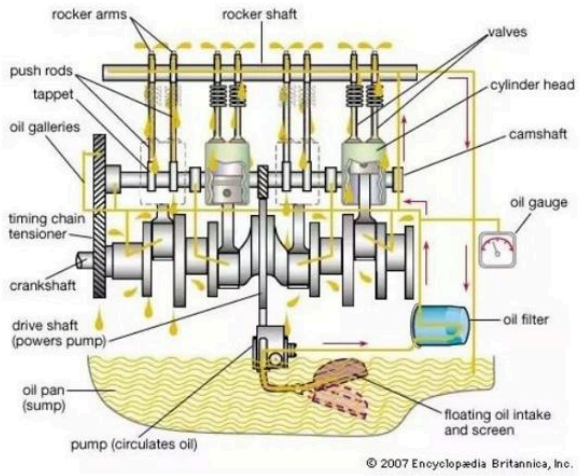
(iv) camshaft bearings

(v) valves and valve operating mechanism

(vi) cooling fan

(vii) water pump and

(viii) ignition mechanism.



Parts requiring lubrication

1. Cylinder walls, piston and piston rings
2. Piston (gudgeon) pin
3. Crankshaft main bearings and connecting rod big-end bearings
4. Camshaft bearings and cams
5. Valves, valve guides, tappets, rocker arms (valve operating mechanism)
6. Timing gears / chain
7. Water pump and fan bearings, ignition distributor

Selection criteria of lubricating oil for weather conditions (2022 2b)

- **Viscosity**: high enough to keep a film at hot running conditions but low enough to flow at cold start. SAE (Society of Automotive Engineers (oil grade)) grades: **winter** grades 5W, 10W, 20W rated at −18 °C (0 °F); **summer** grades SAE 20, 30, 40, 50 rated at 99 °C (210 °F).
- **Cold climate**: low-viscosity W grade (5W/10W), low **pour point and cloud point** so the oil flows to bearings at start-up.

- **Hot climate / heavy duty**: higher viscosity (SAE 40-50), high **flash and fire point**, good oxidation and thermal stability.
- **All-season**: multigrade such as **SAE 20W-50** (behaves like 20W cold and SAE 50 hot) with high **viscosity index** (viscosity changes little with temperature).
- Also: detergent-dispersant additives, low carbon residue, anti-corrosion, anti-foam, specific gravity 0.85-0.96.

Classification of lubricating oils (2021 2b)

- **By state**: solid (graphite, mica, MoS_2), semi-solid (grease), liquid (oils).
- **By origin**: animal and vegetable oils (castor), mineral oils (from petroleum), synthetic oils (PAO, esters), blended/compounded oils.
- **By viscosity (SAE)**: single grade winter (W) and summer, multigrade (e.g. 10W-30, 20W-50).
- **By service (API)**: S-series for petrol (SN, SP), C-series for diesel (CI-4, CK-4).

Mist (petroil) lubrication system (2021 2b)

Lubrication system

(a) Mist lubrication system

➢ This system is used in 2 stroke cycle engines

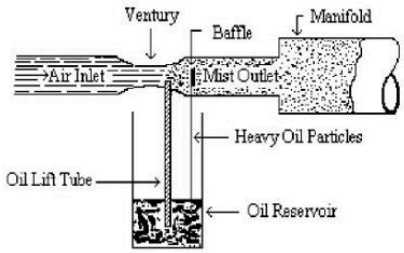
➢ The lubrication oil (2% to 3%) is mixed with the petrol in the fuel tank

➢ The oil and the fuel mixture is inducted through carburetor

➢ The optimum fuel oil ratio used is 50:1

➢ Petrol gets evaporated and the oil lubricates the main parts of cylinder

➢ Fuel oil ratio used is important for the good performance of engine



Mist lubrication: 2-3 % oil mixed in petrol, inducted through the carburetor

Used in small **two-stroke** petrol engines where the crankcase is used for scavenging and cannot hold an oil sump. Lubricating oil (2-3 %, about **1:50 oil:petrol**) is mixed with the petrol in the fuel tank. In the carburetor and crankcase the petrol evaporates; the oil remains as a fine **mist** that settles on the main bearings, big end, piston pin, cylinder walls and rings, then is burnt with the charge or exhausted. (Modern versions meter oil separately with a pump.)

Advantages	Disadvantages
No separate oil pump, sump, filter; simple, cheap, light; no maintenance	Too little oil → seizure; too much → smoke, carbon deposits, plug fouling, HC (hydrocarbons (unburnt fuel)) emission; oil burnt continuously; poor control at closed throttle (engine braking downhill starves bearings)

Other systems for comparison: **wet sump** (splash, pressure, splash + pressure) and **dry sump** (oil in external tank, scavenging pump).

A7 Producer gas air requirement numerical; fluidized bed combustion

Skip if short | Possible · 33% | Easy

PAST-PAPER QUESTIONS

- **2024 Q2(d)** [10] A producer gas used as a fuel has the volumetric composition H_2 25 %, CO 15 %, CH_4 5 %, CO_2 15 % and N_2 40 %. Find the volume of air required for complete combustion of 1 m^3 of the gas. Air contains 21 % by volume of oxygen.
- **2024 Q2(c)** [9] Explain the combustion mechanism and heat transfer process involved in bubbling fluidized bed combustion of solid biomass.

UNDERSTAND **Read this to understand; you do not need to write it**

For gases, **volume ratio = mole ratio** (equal volumes of gases hold equal numbers of molecules). So the balanced equation $H_2 + ½O_2 \rightarrow H_2O$ also says: 1 m^3 of H_2 needs ½ m^3 of O_2.

Work on **1 m^3 of fuel gas**. Multiply each burnable gas by its O_2 multiplier, add them, and divide by 0.21 because air is 21 % oxygen by volume. CO_2 and N_2 are already burnt or inert, so they need nothing.

INTERACTIVE **Air needed to burn a gaseous fuel**

Volume % of each gas in 1 m^3 of fuel (2024 2d producer gas by default). O_2 multipliers: H_2 × ½, CO × ½, CH_4 × 2, C_2H_4 × 3; O_2 already in the fuel is subtracted.

1. $H_2 + ½O_2 \rightarrow H_2O$: 0.250 × 0.5 = 0.125 m^3 O_2
2. $CO + ½O_2 \rightarrow CO_2$: 0.150 × 0.5 = 0.075 m^3 O_2
3. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$: 0.050 × 2 = 0.100 m^3 O_2
4. Total O_2 = **0.300 m^3** per m^3 of gas
5. Theoretical air = 0.300/0.21 = **1.429 m^3/m^3**

Before you start: gas combustion with volumes

For gases at the same temperature and pressure, **volume ratio = mole ratio** (Avogadro). So in $H_2 + ½O_2 \rightarrow H_2O$, 1 m^3 of hydrogen needs 0.5 m^3 of oxygen. Air is 21 % O_2 by volume, so air needed = O_2 needed / 0.21.

Fluidised bed: if you blow air up through sand fast enough, the sand starts to bubble like boiling water. Fuel burns inside this hot, moving sand bed.

Producer gas numerical (2024 2d)

Basis: 1 m^3 of gas. Combustion equations (volumes of gases react in mole ratio):

$H_2 + ½O_2 \rightarrow H_2O$: 0.25 m³ H_2 needs 0.25 × 0.5 = **0.125 m³ O_2**

$CO + ½O_2 \rightarrow CO_2$: 0.15 m³ CO needs 0.15 × 0.5 = **0.075 m³ O_2**

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$: 0.05 m³ CH_4 needs 0.05 × 2 = **0.100 m³ O_2**

CO_2 (0.15) and N_2 (0.40) do not burn.

Total O_2 required = 0.125 + 0.075 + 0.100 = **0.300 m³**

Air required = 0.300 / 0.21 = **1.43 m³ of air per m³ of producer gas**

Remember: For gases, volume ratio = mole ratio. Take O_2: H_2 × ½, CO × ½, CH_4 × 2, then divide by 0.21.

Bubbling fluidized bed combustion of biomass (2024 2c)

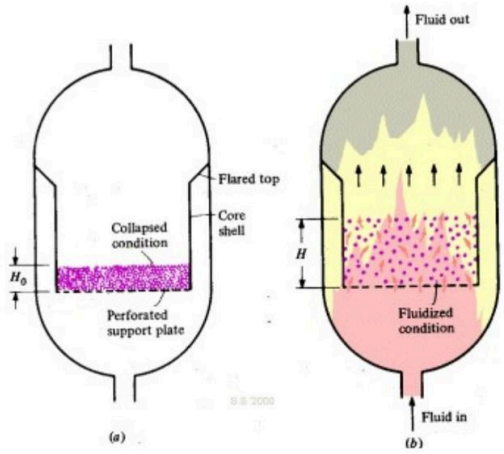


Figure: Fluidized-bed combustion

Fluidized bed combustion: air blown up through a bed of sand/limestone holding the burning fuel

Principle: air is blown upward through a distributor plate under a bed of inert particles (sand, ash, or limestone/dolomite to capture sulphur). Above the **minimum fluidization velocity** the bed behaves like a boiling liquid; in a **bubbling** bed air in excess of that passes as bubbles (velocity ~1-3 m/s) and the bed has a clear surface. Biomass (1-3 % of bed mass) is fed into the hot bed at **750-950 °C**.

Combustion mechanism: (1) **drying**: particles heat very fast in contact with hot sand; (2) **devolatilization/pyrolysis**: volatiles (70-80 % of biomass) are released and burn mostly in the bubbles and the freeboard above the bed, so secondary air is often added there; (3) **char combustion**: the remaining char burns inside the bed, controlled by O_2 diffusion; the strong mixing keeps temperature uniform. Low temperature avoids ash melting (agglomeration) and keeps thermal NOx (nitrogen oxides (NO + NO_2)) low; long residence time handles wet, low-grade, varied fuels.

Heat transfer: (1) from burning particles to bed sand by **particle-to-particle conduction and gas convection**; (2) from bed to **immersed tubes and walls** by **particle convection** (moving solid particles touch the tube, give up heat and are replaced, the main mode, h ~ 200-500 W/m²K, much

higher than gas alone), **gas convection** through bubbles, and **radiation** at bed temperature; (3) freeboard surfaces by convection and radiation from flue gas.

Same answer, step by step with reasons

Step (write this)	Why / what it means
Basis: 1 m^3 producer gas: H_2 0.25, CO 0.15, CH_4 0.05, CO_2 0.15, N_2 0.40	Write the composition as m^3 in 1 m^3 of gas.
H_2: 0.25 × ½ = 0.125 m^3 O_2	Each H_2 needs half its volume of O_2.
CO: 0.15 × ½ = 0.075 m^3 O_2	Each CO needs half its volume.
CH_4: 0.05 × 2 = 0.100 m^3 O_2	Methane needs twice its volume (CH_4 + $2O_2$).
CO_2, N_2: 0	Already burnt / inert.
Total O_2 = 0.300 m^3; air = 0.300 / 0.21 = **1.43 m^3**	Air is 21 % oxygen by volume.

Memory trick: Oxygen multipliers **"half, half, two"** for H_2, CO, CH_4; then divide by 0.21.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

Basis: 1 m³ of producer gas (H_2 0.25, CO 0.15, CH_4 0.05, CO_2 0.15, N_2 0.40 m³)

$H_2 + ½O_2 \rightarrow H_2O$ $O_2 = 0.25 \times 0.5 = 0.125$ m³

$CO + ½O_2 \rightarrow CO_2$ $O_2 = 0.15 \times 0.5 = 0.075$ m³

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ $O_2 = 0.05 \times 2 = 0.100$ m³

CO_2, N_2: no O_2 needed

Total O_2: 0.125 + 0.075 + 0.100

= 0.300 m³

= **0.30 m³ O_2 / m³ gas**

Air required: O_2 / 0.21

= 0.300 / 0.21

= **1.43 m³ air / m³ gas**

A6 Combustion: definition, types, air-fuel ratio and its effect

Good to have

Possible · 33% Easy

PAST-PAPER QUESTIONS

- **2024 Q2(a)** [7] What is combustion? Briefly explain the different types of combustion.
- **2024 Q2(b)** [9] Explain the concept of air-fuel ratio and describe how it affects the efficiency and completeness of combustion.

Before you start: combustion in one line

Combustion = fast chemical reaction of fuel with oxygen that releases heat: $C + O_2 \rightarrow CO_2$, $2H_2 + O_2 \rightarrow 2H_2O$. Air is only 21 % oxygen (by volume; 23 % by mass), so a lot of air is needed. **Air-fuel ratio (A/F)** = kg of air per kg of fuel; about 14.7 for petrol is the exact (stoichiometric) amount.

Combustion and its types (2024 2a)

Combustion is a rapid, exothermic chemical reaction between a fuel and an oxidizer (usually oxygen of air) that releases heat and usually light. It needs the **fire triangle**: fuel, oxygen, and heat to reach the ignition temperature (plus the chain reaction, fire tetrahedron). E.g. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat.

Type	Meaning	Example
Complete	Enough oxygen; all C → CO_2, H → H_2O; blue flame, maximum heat	Gas burner with proper air
Incomplete	Not enough O_2 or too low temperature; CO, soot and HC (hydrocarbons (unburnt fuel)) formed, less heat	$2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$; smoky candle
Rapid	Burns fast with heat and light, usually after external heating	LPG (liquefied petroleum gas), petrol, natural gas
Spontaneous	Starts by self-heating with no external flame	Coal stockpile, haystack, phosphorus
Explosion	Very rapid combustion with sudden pressure, heat, sound, gas	Fuel-air or coal-dust explosion, firecracker
Premixed / diffusion	Fuel and air mixed before the flame / mix at the flame	SI (spark ignition (petrol engine)) engine, Bunsen burner / CI (compression ignition (diesel engine)) engine, candle

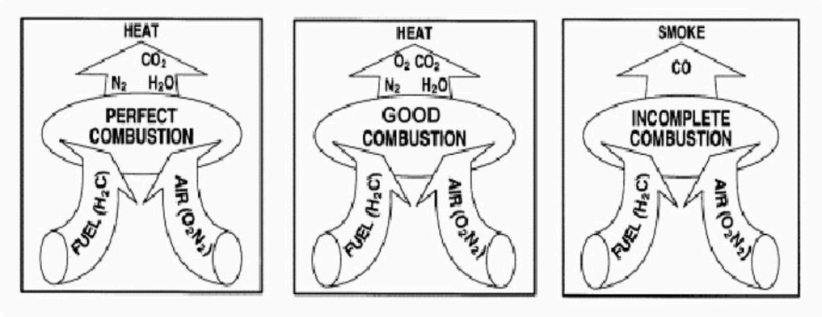


Fig. Perfect, good and incomplete combustion

Perfect, good and incomplete combustion: effect of air supply

Air-fuel ratio and its effect on efficiency and completeness (2024 2b)

A/F (air-fuel ratio) **ratio** = mass of air / mass of fuel supplied. The **stoichiometric** A/F (about 14.7 for petrol, 14.5 for diesel, 17.2 for methane) supplies exactly the oxygen for complete combustion. **Lean**: more air than stoichiometric; **rich**: less air.

Mixture	Completeness	Efficiency and effects
Rich (A/F < stoich)	Incomplete: CO, HC, soot	Highest power (best-power ~12.5:1 for petrol) because all air is used, but fuel wasted, low efficiency, high CO/HC, carbon deposits
Stoichiometric	Theoretically complete; in practice some CO/O_2 due to poor mixing and dissociation	Highest flame temperature, highest NOx (nitrogen oxides (NO + NO_2)); needed for 3-way catalyst
Slightly lean (~16-17:1)	Complete, excess O_2	Best economy: highest thermal efficiency (more expansion, higher γ, lower heat loss and dissociation)
Too lean	Slow flame, misfire, HC rises	Power and efficiency fall; beyond lean limit no ignition

In furnaces some **excess air** (10-20 % for gas/oil) is used to ensure complete combustion, but too much excess air carries heat up the stack and lowers efficiency.

A5 Petroleum refining: fractional distillation, cracking, reforming

Skip if short

Possible · 13% Easy

PAST-PAPER QUESTIONS

- **2021 Q2(c)** [15] Why petroleum refining is necessary? Explain fractional distillation and reforming process.

Before you start: what refining does

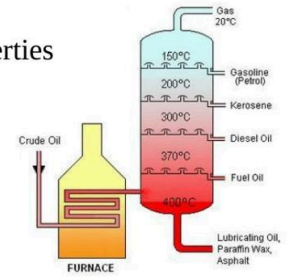
Crude oil is a mixture of thousands of hydrocarbons with different boiling points. A refinery heats it and separates it by boiling point (**fractional distillation**), then breaks big molecules into smaller, more useful ones (**cracking**) and rearranges molecules to raise octane (**reforming**).

Why refining is necessary (2021 2c)

Crude oil is a mixture of thousands of hydrocarbons (paraffins, naphthenes, aromatics) with water, salts, sulphur and gases. It cannot be used directly in engines; refining **separates** it into fractions of suitable boiling range (petrol, kerosene, diesel, lubricating oil), **converts** heavy low-value fractions into lighter ones (cracking), **improves quality** such as octane number (reforming), and **removes impurities** (sulphur, water, gum).

❑ **Fractional Distillation**

- A number of bubble caps are fitted into the column
- The heavier fractions which have higher boiling points gets liquefied first
- The remaining fractions rise and condense in order of their boiling points
- Various fractions are taken out at an approximate temperature difference of 50°C
- Uncondensed gas leave from the top of the tower and is sent for absorption
- Treated further for improved properties



Fractional distillation column: crude heated in a furnace, vapours condense at different heights

Fractional distillation

1. Crude is dewatered (centrifuge), gases removed, then heated to 300-350 °C in a tube furnace.
2. Hot vapour enters near the bottom of a fractionating column, 1.5-2 m diameter and 20-30 m high, fitted with trays and bubble caps.
3. Vapour rises and cools; heavier fractions (high boiling point) condense low down, lighter ones higher up. Fractions are drawn off at about 50 °C intervals: gas (top, <30 °C), petrol (40-120 °C), naphtha, kerosene (180-250 °C), diesel (250-320 °C), fuel oil, lubricating oil and residue (bottom).
4. Residue goes to vacuum distillation; products are treated further.

Reforming

Reforming changes the molecular structure of low-octane naphtha/gasoline to raise its octane number (it is not done on crude). **Thermal reforming**: 500-600 °C and ~80 kg/cm², produces iso-paraffins and aromatics; quenched with cold oil then fractionated. **Catalytic reforming**: 480-540 °C, 14-21 kg/cm², platinum catalyst (~0.75 %) on alumina; main reaction is **dehydrogenation of naphthenes to aromatics**, plus isomerisation of paraffins and desulphurisation; gives high-octane reformate and hydrogen. (Cracking, by contrast, breaks heavy molecules into lighter ones: thermal at ~540 °C and 7000 kPa, or catalytic with alumino-silicates at lower T and p.)

A19 Emission control techniques; NOx reduction; SI vs CI emissions; blow-by

Must study · Possible · 20% · Medium

PAST-PAPER QUESTIONS

- **2021 Q4(c)** [13] What are the common emission control techniques? Discuss the strategies to reduce NOx (nitrogen oxides (NO + NO_2)) emission from IC (internal combustion) engines?
- **2021 Q4(b)** [12] Compare the emissions from SI (spark ignition (petrol engine)) engine with that from CI (compression ignition (diesel engine)) engine.
- **2020 Q3(b)** [12] What is blow-by loss? How can you achieve effective removal of blow by? Explain with schematic diagram.
- **2020 Q3(a)** [10] What is meant by total emission control packages? (first part)

Before you start: controlling emissions

Two approaches: (1) stop pollutants forming inside the engine (better design, exhaust gas recirculation EGR (exhaust gas recirculation) to cool the flame and cut NOx, correct timing); (2) clean the exhaust afterwards (catalytic converter, particulate filter, SCR (selective catalytic reduction) with urea for NOx). **Blow-by** = gas leaking past the piston rings into the crankcase; PCV (positive crankcase ventilation) sends it back to the intake.

Common emission control techniques (2021 4c) and total emission control packages (2020 3a)

A **total emission control package** is the combination of all measures on a vehicle that together meet the emission limits: engine design changes + fuel system control + after-treatment + crankcase and evaporative controls. The methods are:

1. **Modification of engine design and operating parameters**: combustion chamber shape (less crevice and quench area), compression ratio, spark/injection timing, EGR, fuel injection with closed-loop A/F (air-fuel ratio) control, multi-valve, high-pressure injection, turbocharging with intercooling.
2. **Treatment of exhaust products**: after-burner / **thermal reactor** (enlarged exhaust manifold with secondary air, > 600 °C, oxidises CO and HC (hydrocarbons (unburnt fuel))), **catalytic converter** (2-way or 3-way), diesel oxidation catalyst, **diesel particulate filter**, **SCR**, lean NOx trap.
3. **Modification of the fuel**: unleaded petrol, low-sulphur diesel, oxygenates (ethanol), alternative fuels (CNG, LPG (liquefied petroleum gas)).
4. **Crankcase (PCV) and evaporative (charcoal canister) emission control**.

ENGINE EMISSION CONTROL

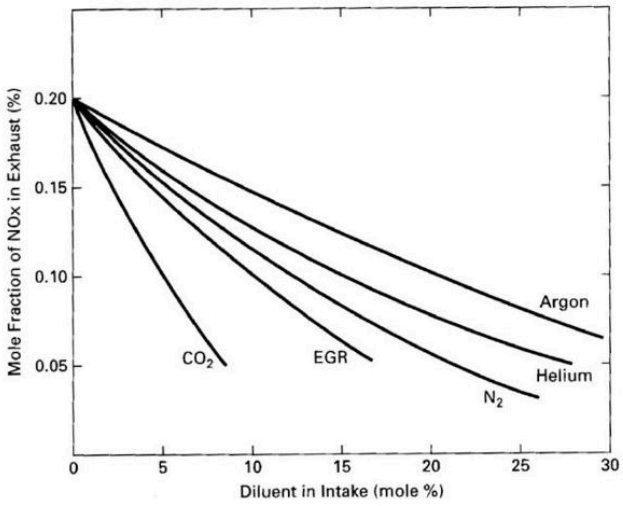
- **Post-combustion reactions:** Components in the cylinder gas mixture that haven't fully burned continue to react during the expansion stroke, exhaust blowdown, and exhaust process.
- **Reduction of remaining hydrocarbons (HC):** Up to 90% of the HC remaining after combustion reacts during this time, either in the cylinder, near the exhaust port, or in the upstream part of the exhaust manifold.
- **Formation of CO2 and H2O:** Carbon monoxide (CO) and small component hydrocarbons react with oxygen to form carbon dioxide (CO2) and water (H2O), reducing undesirable emissions.
- **Importance of exhaust temperature:** Higher exhaust temperatures promote these secondary reactions, resulting in lower engine emissions.
- **Factors affecting exhaust temperature:** Several factors can contribute to higher exhaust temperatures, including stoichiometric air-fuel combustion, high engine speed, retarded spark timing, and a low expansion ratio.

After-burner / exhaust manifold reactor with secondary air

Strategies to reduce NOx (2021 4c, 2022 4a)

- **Exhaust gas recirculation (EGR)**: 5-25 % of exhaust re-introduced to intake; inert CO_2 and H_2O absorb heat and dilute O_2 → lower peak temperature → NOx falls strongly.
- **Retarding** spark timing (SI) or injection timing (CI): lower peak pressure and temperature.
- **Water / steam injection**, charge cooling with intercooler, lower compression ratio.
- **Lean burn** or very rich operation, staged combustion; reduce residence time at peak temperature.
- **After-treatment**: 3-way catalyst (SI), SCR with urea ($4NO + 4NH_3 + O_2 \rightarrow 4N_2 + 6H_2O$) or lean NOx trap (CI).

EXHAUST GAS RECYCLE (EGR)



Reduction of NOx generation by adding non combustible diluent gas to intake mixture

Reduction of NOx by adding diluent (EGR, CO_2, N_2) to the intake

SI vs CI emissions (2021 4b)

Pollutant	SI engine	CI engine
CO	High (runs near stoichiometric or rich at idle/full load)	Very low (overall lean)
HC	High (crevices, quench, oil film)	Lower (~1/5); from over-lean spray edge, sac volume
NOx	High near $\varphi = 1$; NO_2 small	High in diffusion flame; NO_2 10-30 %
PM (particulate matter) / smoke	Negligible (except GDI (gasoline direct injection))	High: soot from rich spray core
SO_2	Low	Depends on fuel sulphur
Aldehydes, odour	Low	Odour, noise, aldehydes
Control	3-way catalyst at $\varphi = 1$	Oxidation catalyst + DPF (diesel particulate filter) + SCR/EGR (3-way cannot work in lean exhaust)

Blow-by and its removal (2020 3b)

Blow-by is the leakage of unburnt mixture and combustion gases past the piston rings into the crankcase during compression and expansion. It is a loss of charge (power) and, if vented, ~20 % of the engine's HC emission; it also contaminates and dilutes the oil and forms sludge and acids.

Removal: Positive Crankcase Ventilation (PCV): fresh filtered air enters the crankcase from the air cleaner; the blow-by gases are drawn through a **PCV valve** (flow-control valve) into the intake manifold and burned in the cylinders. The valve meters flow according to manifold vacuum: small flow at idle (high vacuum) so the mixture is not upset, larger flow at part load, and it closes on a backfire. Closed systems have no vent to atmosphere at all.

Sketch: air cleaner → hose → rocker cover / crankcase → oil separator → PCV valve → intake manifold → cylinders.

Section B: Performance, cycles, valve timing, SI/CI combustion, knock, supercharging

B15 Supercharging and turbocharging: effects, SI vs CI suitability, limits, modifications

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

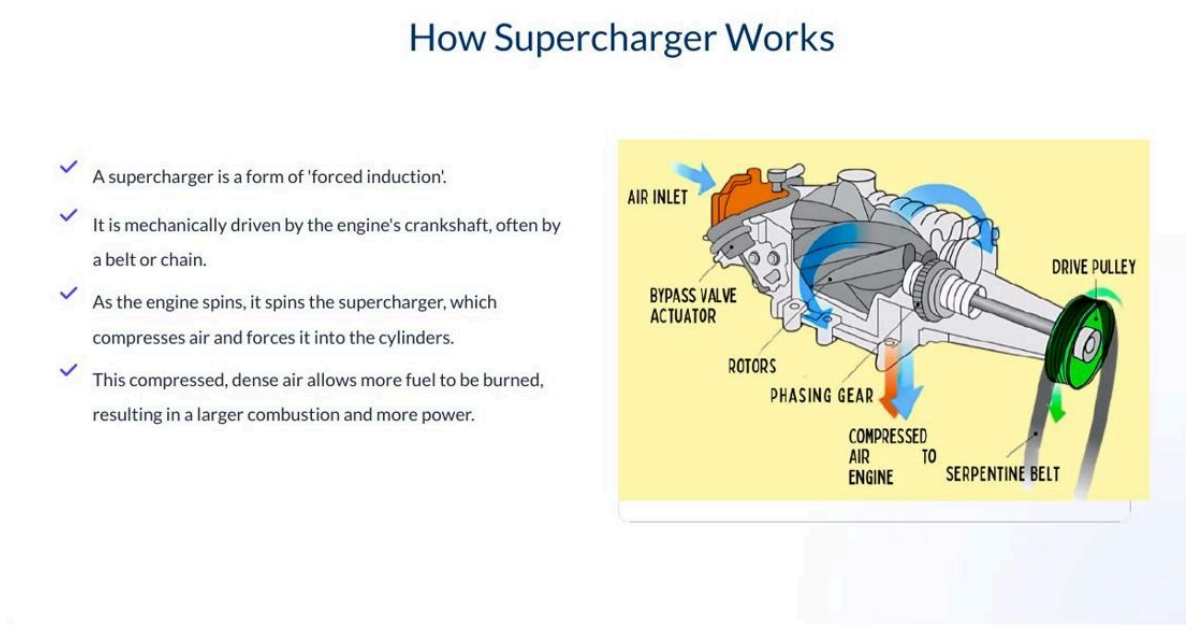
- **2024 Q8(c)** [11] Explain the concept of supercharging and how it affects engine power and efficiency.
- **2023 Q8(a)** [11] Describe the effect of supercharging. Which engine is more suitable for supercharging among SI (spark ignition (petrol engine)) engine or CI (compression ignition (diesel engine)) engine? Explain why?
- **2023 Q8(b)** [8] What types of engine modifications are required if supercharging is to be employed? Explain.
- **2022 Q8(a)** [14] Discuss the limits of supercharging in both SI engine and CI engine.
- **2021 Q6(d)** [7] What are the benefits of supercharging? (first part)
- **2020 Q6(b)** [10] Describe the effects of supercharging. Which engine is more suitable for supercharging – SI engine or CI engine? Why?
- **2020 Q6(a)** [10] Discuss the supercharging limits for both SI engine and CI engine.
- **2020 Q6(c)** [10] How can you modify an engine for supercharging? What are the limitations of turbocharging?

Before you start: supercharging

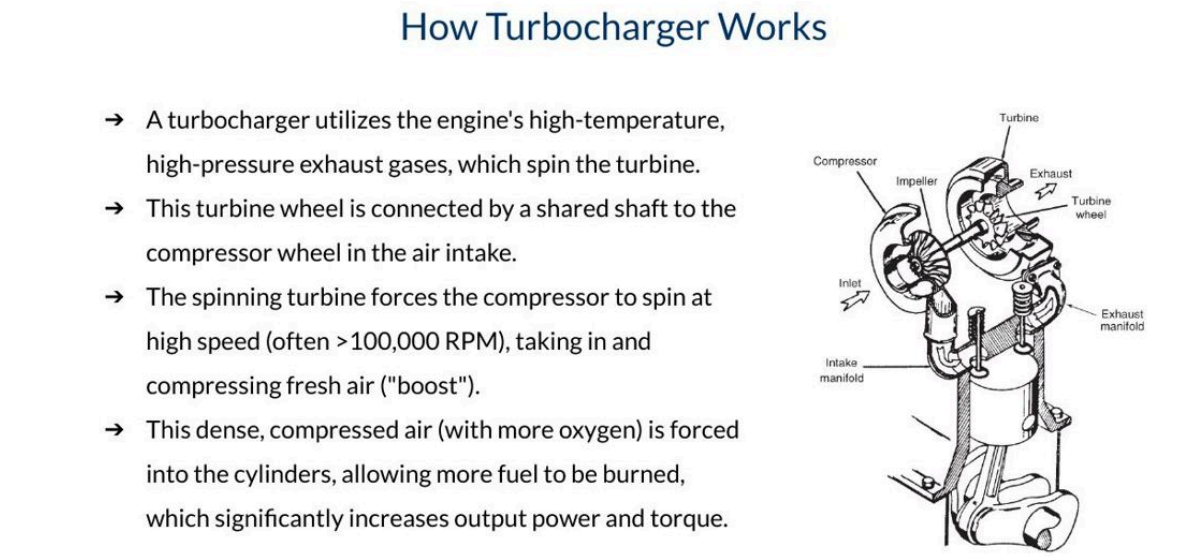
Supercharging = pushing more air into the cylinder than it would suck in by itself, using a compressor. More air allows more fuel, so more power from the same engine size. A **supercharger** is driven by the crankshaft (belt/gears); a **turbocharger** is driven by a turbine in the exhaust gas.

Supercharging and turbocharging: concept (2024 8c)

Supercharging is the process of supplying air (or mixture) to the engine at a pressure, hence density, higher than atmospheric, so that a larger mass of charge is inducted per cycle, more fuel can be burnt and more power produced from the same engine size. A **supercharger** is a compressor driven mechanically from the crankshaft (belt, gear, chain): reciprocating, or rotary (Roots lobe, vane, centrifugal). A **turbocharger** is a centrifugal compressor driven by an exhaust gas turbine on a common shaft (>100 000 rpm).



Mechanically driven supercharger (Roots type)



Turbocharger: exhaust-driven turbine drives compressor

Objects: increase power output (racing), reduce weight per kW (aircraft), maintain power at altitude (aero engines), reduce engine size (marine), and lower oil consumption per kW.

Effects on power and efficiency (2024 8c, 2023 8a, 2020 6b, 2021 6d)

- **Power**: higher charge density → more air and fuel per cycle → BMEP (brake mean effective pressure) and power rise roughly in proportion to boost (30-50 %+); power at altitude is maintained.
- **Mechanical efficiency** improves: friction stays about the same while power rises.
- **Pumping work** becomes positive (boost pressure on the intake stroke).
- **Combustion**: better mixing and higher temperature/pressure shorten ignition delay in CI → smoother, more complete combustion, less smoke; better scavenging with larger overlap.
- **Thermal efficiency / BSFC** (brake specific fuel consumption): turbocharging recovers exhaust energy, so overall efficiency improves (BSFC falls); a mechanical supercharger absorbs shaft

power (up to 10-15 %), so BSFC may rise slightly.
- **Drawbacks**: higher peak pressure and thermal load on pistons, valves, head; higher charge temperature (intercooler needed); **knock in SI**; more NOx (nitrogen oxides (NO + NO_2)).

Which engine is more suitable: SI or CI? (2023 8a, 2020 6b)

CI engines. In a CI engine supercharging raises compressed-air temperature and pressure, which **shortens the ignition delay and reduces knock**; it also supplies the excess air that pushes back the smoke limit. Only air is compressed, so there is no pre-ignition risk, and the excess air during overlap cools pistons and valves. In an SI engine the higher temperature and density of the end gas **increases knock tendency**, so boost is limited unless compression ratio is reduced, spark retarded, or high-octane fuel and intercooling used; scavenging with mixture during overlap wastes fuel. Hence almost all diesels are turbocharged, while SI boost is modest.

Limits of supercharging in SI and CI engines (2022 8a, 2020 6a)

SI engine	CI engine
Knock is the main limit: boost raises end-gas temperature and pressure; boost must be matched by lower compression ratio, retarded timing, high octane, intercooling	**Thermal and mechanical loading**: peak pressure and heat flux to pistons, rings, head, exhaust valves, bearings limit boost
Thermal load and pre-ignition from hot exhaust valve / plug	Exhaust temperature limit for turbine blades (~700-750 °C)
Fuel loss during valve overlap (scavenging with mixture)	Compressor and turbine efficiency, surge and matching; turbo lag
Cost and fuel economy of mechanical drive	Higher NOx; need for intercooler

General limitations (slides): detonation tendency increases in SI engines; heat losses from turbulence and thermal stresses rise; valve overlap increases up to 60°; better lubrication and better cooling of piston and valves needed; cost increases.

Engine modifications required for supercharging (2023 8b, 2020 6c)

- Reduce compression ratio (SI, to avoid knock; CI, to limit peak pressure).
- Stronger, heavier pistons, rings, connecting rods, crankshaft, bearings and head gasket to carry higher pressures.
- Increased valve overlap (up to ~60° CA) for scavenging and cooling; sodium-cooled exhaust valves, heat-resistant materials.
- Better cooling of pistons (oil jets), valves and head; larger radiator; intercooler (aftercooler) for the compressed air.
- Better lubrication (higher oil flow, better oil).
- Fuel system: larger pump elements and nozzles to inject more fuel per unit time; boost-compensated fuel control; retarded injection/spark timing.
- Special exhaust manifold (pulse/constant pressure) for the turbine; waste-gate or variable geometry turbine.

Limitations of turbocharging (2020 6c)

1. Needs special exhaust manifolds and careful matching of turbine and compressor to the engine.
2. Larger pumping elements or nozzles are needed to inject more fuel per unit time, which overloads cams and other components.
3. Turbine blade efficiency is very sensitive to gas velocity; poor at low exhaust flow.

4. Difficult to maintain satisfactory air charging over the complete speed range: **turbo lag** at low speed and sudden load, over-boost at high speed (waste-gate needed).
5. High exhaust temperature stresses turbine; more back pressure; lubrication of high-speed bearings; extra cost.

B1 Performance parameters and performance curves (BP, torque, BSFC, ηv vs speed)

Must study | Very likely · 87% | Medium

PAST-PAPER QUESTIONS

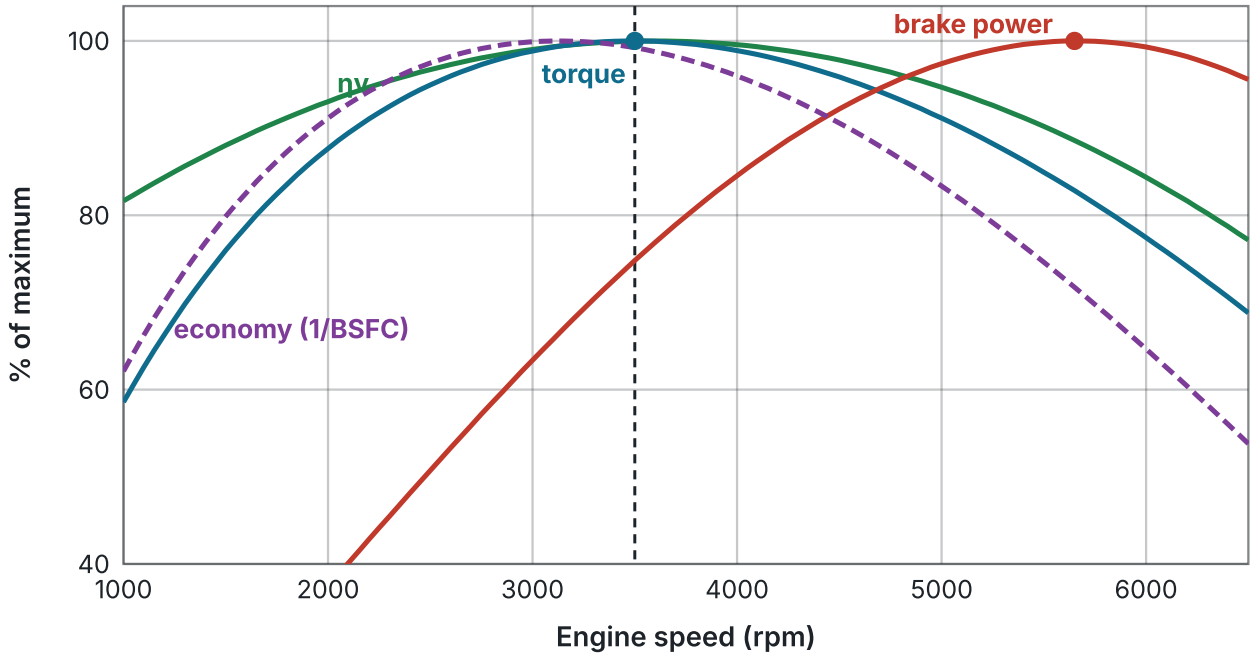
- **2024 Q5(a)** [12] Define the following engine performance parameters: (i) Volumetric efficiency, (ii) Brake specific fuel consumption, (iii) Indicated mean effective pressure, (iv) Brake power.
- **2023 Q5(a)** [14] What is the physical significance of volumetric efficiency? Draw and explain the following engine performance curves: (i) Brake torque, (ii) BSFC (brake specific fuel consumption), (iii) volumetric efficiency.
- **2022 Q5(a)** [12] Discuss the following engine performance parameters with respect to engine speed using necessary diagrams: (i) Brake power, (ii) Brake torque, (iii) Brake specific fuel consumption.
- **2020 Q4(a)** [9] Discuss the following engine performance parameter with necessary diagrams: (i) Brake power (ii) Specific fuel consumption (iii) Volumetric efficiency.

Symbols used here

Symbol	Name and meaning	Unit
BP (brake power)	Brake power: useful power at the crankshaft	kW
IP (indicated power)	Indicated power: power the gas gives the piston	kW
T	Brake torque (turning moment at the shaft)	N m
N	Engine speed	rpm (rev/min)
η_v	Volumetric efficiency: air actually drawn in / air that would fill the swept volume	% (75–90 for petrol engines)
$\dot{m}_a$, $\dot{m}_f$	Mass flow rate of air / fuel	kg/s or kg/h
$\rho_{a,i}$	Density of air at the intake	kg/m³
V_d	Displacement (swept) volume of all cylinders	m³
BSFC	Brake specific fuel consumption: fuel per unit brake energy	kg/kWh (or g/kWh)
CV (calorific value)	Calorific value of the fuel	kJ/kg
p_{mi}	Indicated mean effective pressure (IMEP)	bar

INTERACTIVE **Performance curves at full throttle vs speed**

Typical shape for a car SI engine. Drag the speed and read where each curve peaks. Every curve is drawn as a % of its own maximum (BSFC: lower is better).



At 3500 rpm: torque **100 %**, power **75 %**, η_v **90 %** (absolute), BSFC **101 %** of the minimum. Torque peaks near 3500 rpm (where η_v is best), power peaks later near 5650 rpm (BP = 2πNT/60 keeps rising while torque falls slowly), then falls as friction grows with N^2. BSFC is lowest near the torque peak: a bowl shape.

Before you start: power words

Indicated power (IP) = power developed by the gas on the piston. **Brake power (BP)** = power at the output shaft (measured by a brake/dynamometer). Friction power = IP − BP. **Torque** = turning force × radius; BP = 2πNT/60. **BSFC** (brake specific fuel consumption) = kg fuel per kWh of brake output; lower is better.

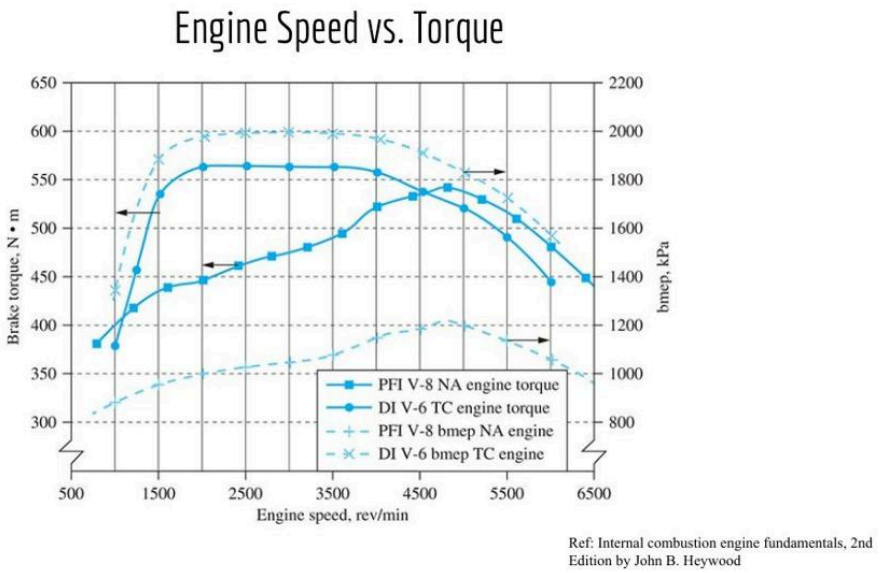
Definitions (2024 5a)

Parameter	Definition	Formula
Volumetric efficiency η_v	Ratio of the mass of fresh air actually inducted per cycle to the mass of air that would fill the swept volume at inlet (or atmospheric) density	$\eta_v = 2\dot{m}_a / (\rho_{a,i} V_d N)$ (4-stroke); 75-90 % for NA engines
Brake specific fuel consumption BSFC	Fuel mass consumed per unit brake power per hour; inverse measure of efficiency	BSFC = $\dot{m}_f$ / BP (kg/kWh); η_{bth} = 3600 / (BSFC × CV)
Indicated mean effective pressure IMEP (indicated mean effective pressure)	The constant pressure which, acting on the piston over one full stroke, would do the same work as the actual indicated work per cycle; from the indicator diagram	$p_{mi} = W_i / V_s$ = a s / l (area × spring scale / length)
Brake power BP	Useful power at the crankshaft, measured by a dynamometer (prony, rope, hydraulic, eddy current); BP = IP − FP (friction power)	BP = 2πNT / 60 (W, N in rpm, T in N m)

Physical significance of volumetric efficiency (2023 5a)

η_v measures how well the engine **breathes**. Power = $\eta_f \eta_v \rho_a V_d N Q_{LHV} (F/A) / 2$, so for a given fuel and A/F (air-fuel ratio) the power and torque an engine can give are directly proportional to η_v: only as much fuel can be burnt as there is air in the cylinder. It is reduced by flow restrictions (air filter, throttle, valves), heating of the charge, residual gas, valve timing, and choking at high speed.

Performance curves vs engine speed at full throttle (2023 5a, 2022 5a, 2020 4a)



Brake torque and power vs engine speed (Heywood): torque peaks at mid speed, power peaks at higher speed

- **Volumetric efficiency**: low at low speed (backflow because of late IVC (inlet valve closes), heat transfer), **maximum at medium speed** where valve timing and intake tuning match, then falls at high speed (flow friction rises with speed², choking at the inlet valve).
- **Brake torque**: follows η_v; rises to a **maximum at medium speed** (maximum-torque speed) and then falls because η_v drops and friction rises. At low speed heat losses also reduce it.
- **Brake power** = $2\pi NT$: rises almost linearly with speed at first, reaches a **maximum at a higher speed** than maximum torque (rated speed), then falls because torque drops faster than speed rises (friction power rises roughly with N^2).
- **BSFC**: a bowl-shaped curve. High at low speed (more time for heat loss per cycle), **minimum at medium speed** (slightly below maximum-torque speed), rising at high speed because friction losses increase. Versus load, BSFC is high at light load (friction is a large fraction) and lowest near 75-90 % load.

> **Sketch to draw:** x-axis speed (rpm). Curves: η_v and torque humps peaking around the middle; BP rising and peaking at ~80-90 % of maximum speed; BSFC U-shape with minimum near maximum-torque speed.

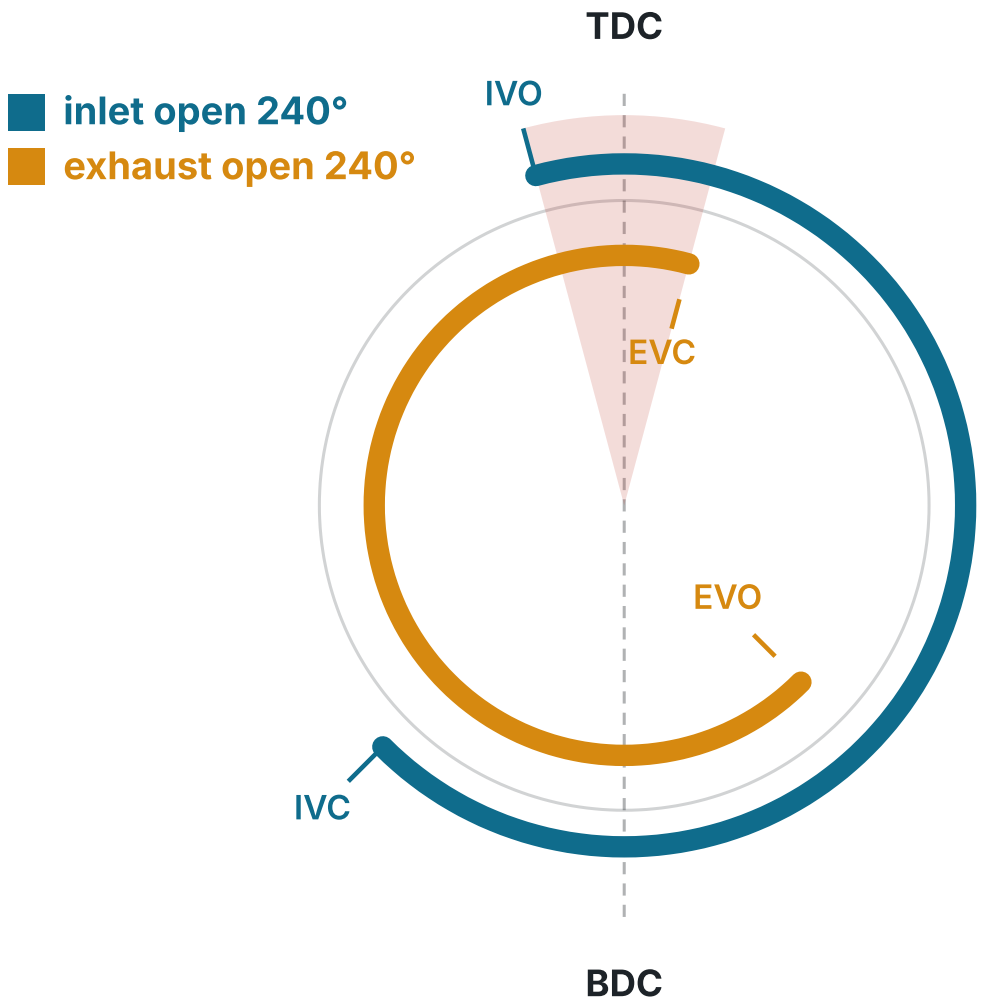
B6 Valve timing diagram, valve overlap, why inlet valves are larger Must study Likely · 73% Easy

PAST-PAPER QUESTIONS

- **2024 Q5(b)** [10] Briefly explain valve timing diagram and its functions of a typical four stroke engine with proper sketches.
- **2022 Q7(b)** [14] Why do engine inlet valves and exhaust valves open earlier and close later? Why are inlet valves made larger than exhaust valves?
- **2021 Q6(b)** [12] Discuss why actual valve timing differ from theoretical valve timing. Also, write the consequences of valve overlapping.
- **2020 Q1(c)** [5] Why are the intake valves larger than exhaust valves?

INTERACTIVE **Valve timing diagram (four-stroke)**

TDC at the top, crank turning clockwise. Outer arc = inlet valve open, inner arc = exhaust valve open. Red wedge = valve overlap.



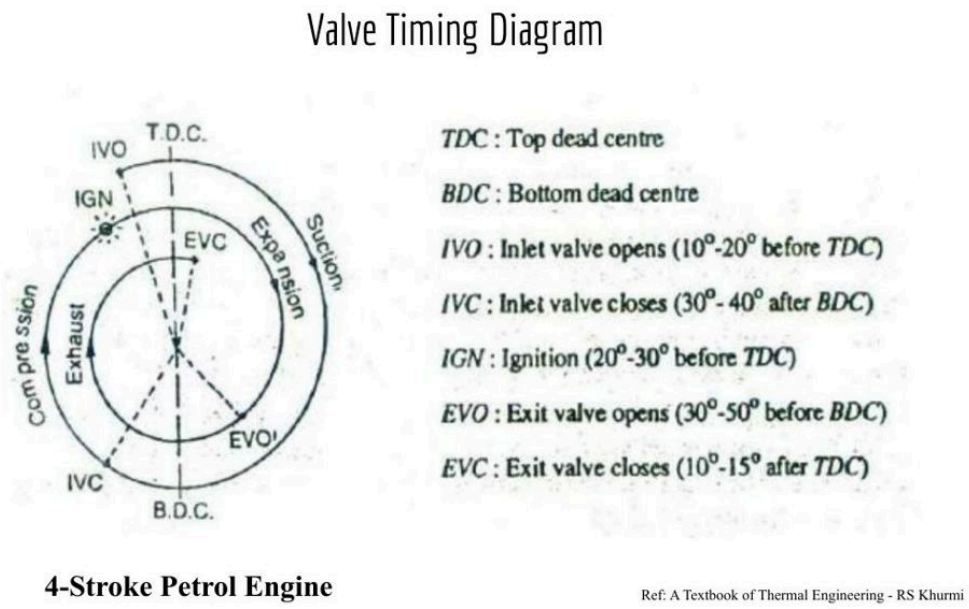
Inlet open for **240°**, exhaust open for **240°**, overlap **30°**.
Moderate overlap: a normal road-engine compromise.

Before you start: valve timing

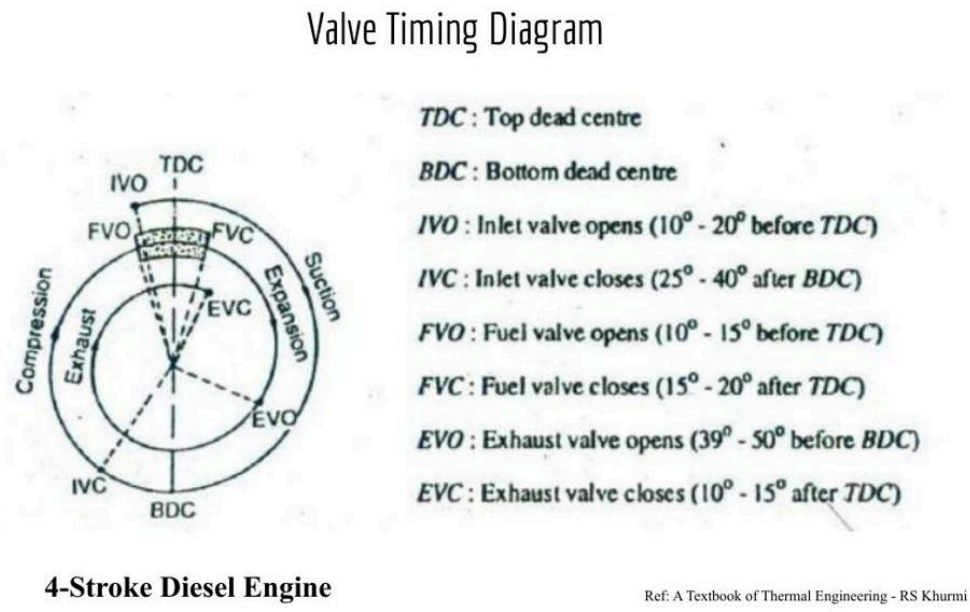
Valves do not open and close exactly at top or bottom dead centre (TDC/BDC). The inlet opens a little before TDC (top dead centre) and closes well after BDC (bottom dead centre); the exhaust opens before BDC and closes after TDC. This uses the momentum of the moving gases to fill and empty the cylinder better. The period when both are open is the **valve overlap**.

Valve timing diagram of a four-stroke engine (2024 5b)

The valve timing diagram shows the exact crank positions (degrees crank angle relative to TDC/BDC) at which the inlet and exhaust valves open and close, and (for SI (spark ignition (petrol engine))) when the spark occurs or (for CI (compression ignition (diesel engine))) when fuel injection starts and ends.

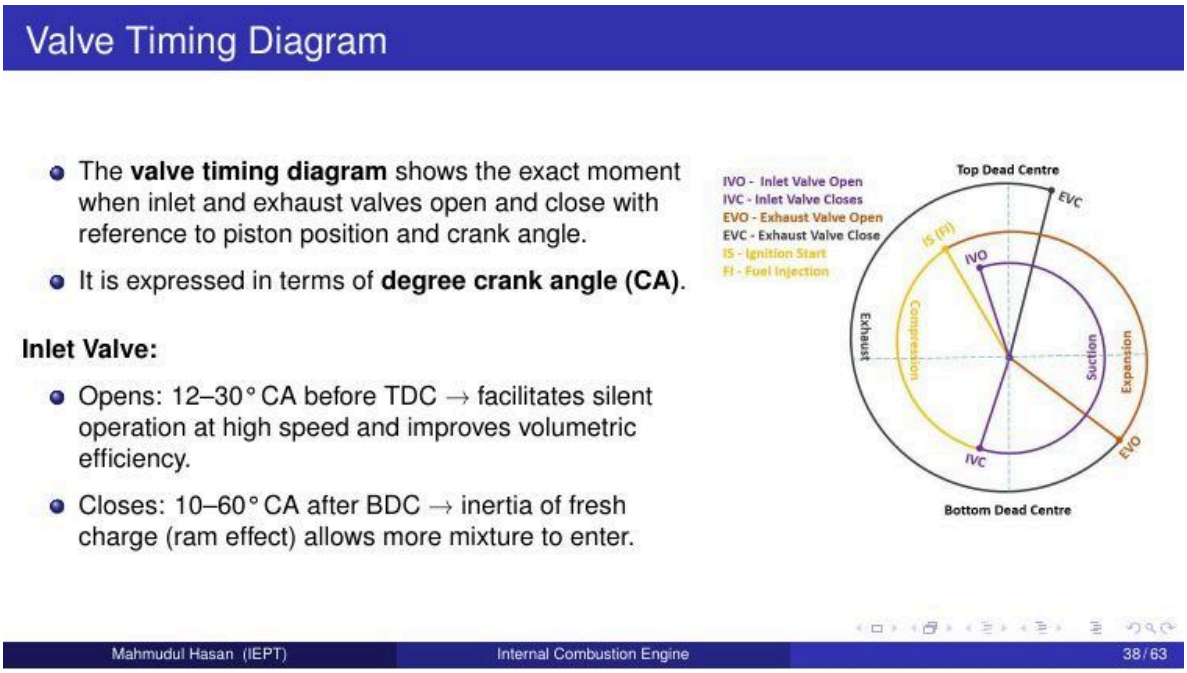


Valve timing: 4-stroke petrol engine (IVO 10-20° bTDC, IVC (inlet valve closes) 30-40° aBDC, IGN 20-30° bTDC, EVO (exhaust valve opens) 30-50° bBDC, EVC (exhaust valve closes) 10-15° aTDC)



Valve timing: 4-stroke diesel engine (FVO 10-15° bTDC, FVC 15-20° aTDC, IVC 25-40° aBDC, EVO 30-50° bBDC)

Event	Typical timing	Reason (function)
Inlet valve opens (IVO)	10-30° before TDC	Valve fully open when the piston starts the suction stroke; uses exhaust momentum to start the inflow; quiet operation
Inlet valve closes (IVC)	30-60° after BDC	**Ram effect**: inertia of the incoming charge keeps it flowing in after BDC → higher η_v
Ignition / injection	20-35° before TDC	Combustion takes time; peak pressure should occur ~10-15° after TDC
Exhaust valve opens (EVO)	30-55° before BDC	**Blow-down**: exhaust leaves under its own pressure, so less work is needed to push it out; little expansion work is lost because pressure near BDC is low
Exhaust valve closes (EVC)	10-20° after TDC	Momentum of outgoing gas helps scavenge residuals and draws fresh charge
Valve overlap	IVO (inlet valve opens) to EVC around TDC (~20-60°)	Both valves open: scavenging of the clearance volume



Valve timing diagram (slide)

Why valves open earlier and close later than the dead centres, i.e. actual vs theoretical timing (2022 7b, 2021 6b)

In the theoretical cycle valves open and close exactly at TDC and BDC. In a real engine: (1) a valve cannot open or close instantly; cam lift is gradual, so to have it fully open during the stroke it must start opening earlier and close later; (2) gases have **inertia**: the incoming charge keeps flowing after BDC (ram effect) and the outgoing exhaust keeps flowing after TDC; (3) **blow-down** of exhaust before BDC reduces pumping work; (4) combustion takes finite time, so ignition is advanced. All these improve η_v, power and scavenging at the design speed.

Valve overlap and its consequences (2021 6b)

- Good at high speed: exhaust momentum pulls fresh charge in and sweeps out residual gas → better scavenging, higher η_v, cooler exhaust valve.

- At low speed and idle (and in throttled SI engines with low manifold pressure): exhaust flows back into the intake, raising residual dilution → rough idle, poor combustion.
- Fresh charge can short-circuit into the exhaust → fuel loss and HC (hydrocarbons (unburnt fuel)) emission in SI engines (not in CI, which scavenge with air).
- Too large overlap risks valve-piston contact; supercharged engines use larger overlap (up to 60° +) to cool the chamber with excess air.

Why inlet valves are made larger than exhaust valves (2022 7b, 2020 1c)

The intake flow is driven only by the small pressure difference between the atmosphere and the cylinder (about 0.1-0.2 bar in an NA engine), whereas exhaust is driven out by the high cylinder pressure at EVO (3-5 bar blow-down) and then by the piston. To fill the cylinder fully (high η_V) the intake needs a large flow area with low velocity and loss. Also, the exhaust valve runs very hot, so a smaller exhaust valve head is easier to cool (shorter heat path to the seat) and less likely to warp. Hence the inlet valve is ~15-25 % larger in diameter.

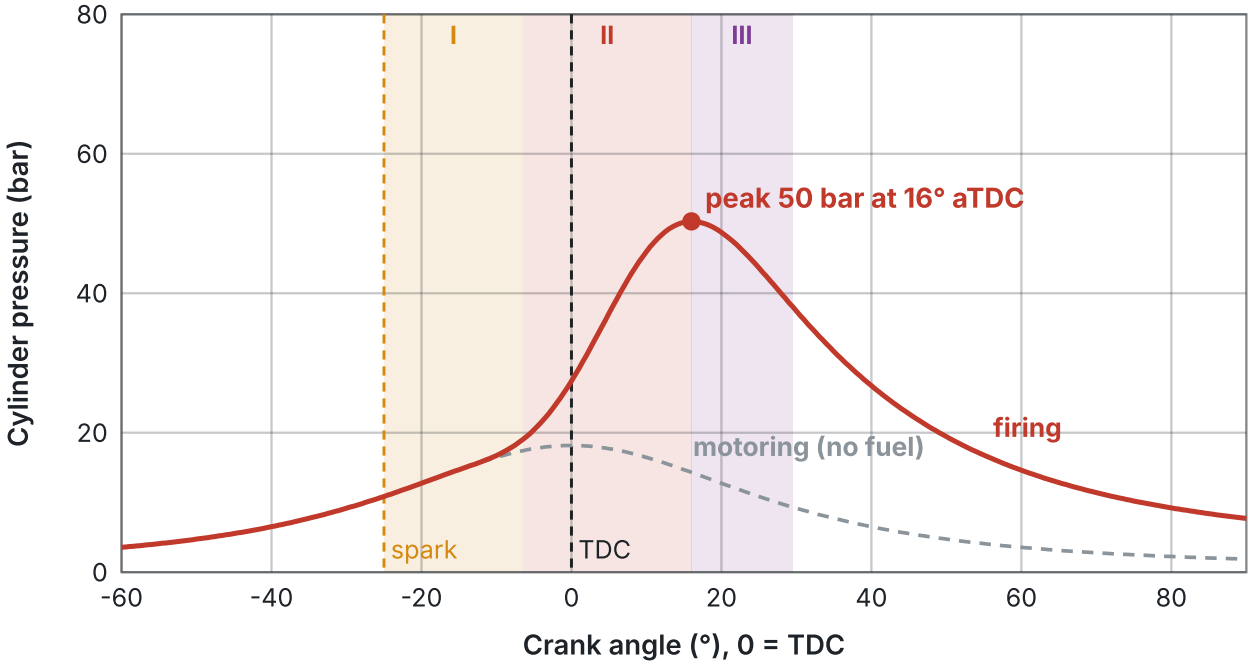
B10 Stages of combustion in SI engine Must study Likely · 73% Easy

PAST-PAPER QUESTIONS

- **2024 Q3(c)** [10] How does combustion occur in SI (spark ignition (petrol engine)) engines? Illustrate and explain the different stages from ignition to complete burning.
- **2023 Q6(a)** [10] Discuss the different stages of combustion in a typical SI engine with suitable diagrams.
- **2021 Q7(b)** [12] Discuss the different stages of combustion in SI engine using appropriate diagrams.

INTERACTIVE **SI engine: pressure vs crank angle**

Dashed = motoring (no fuel burnt). Shaded = the three stages. Change spark timing, speed and fuel octane; the best timing puts the peak about 10–15° after TDC.



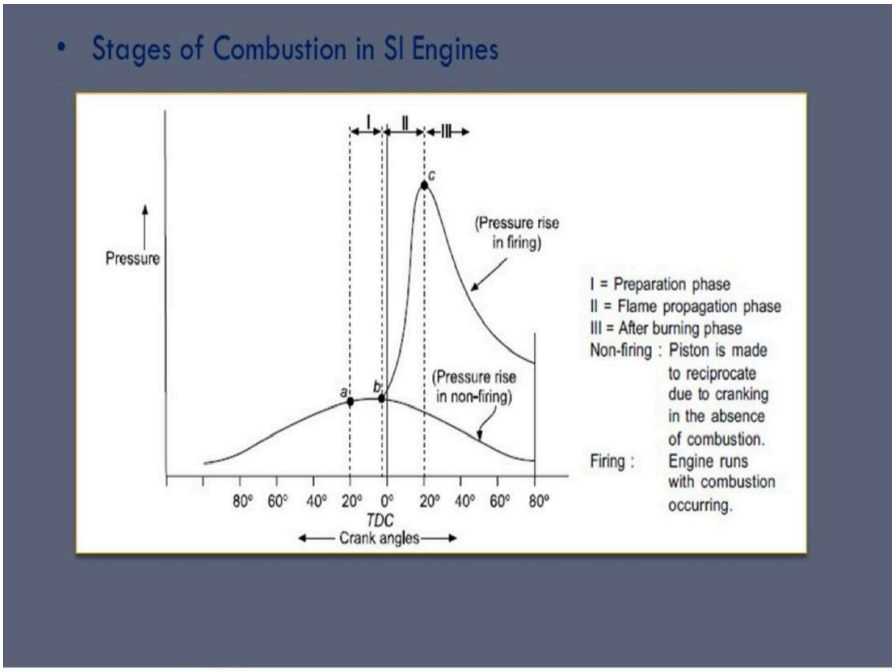
Stage I ignition lag ≈ 19° CA (grows with speed, so spark advance must increase with speed) · **Stage II** flame propagation to the peak · **Stage III** after-burning.
Good timing (near MBT, maximum brake torque): peak 10–15° after TDC.

Before you start: how an SI engine burns

The spark lights a small flame kernel, which grows into a **flame front** that sweeps across the chamber, burning the mixture ahead of it. Pressure rises fastest when most of the charge burns, a few degrees after TDC (top dead centre).

Stages of combustion in an SI engine (2024 3c, 2023 6a, 2021 7b)

In an SI engine a homogeneous mixture of vaporized fuel, air and residual gas is ignited by a spark (> 10 000 °C) near the end of compression; a flame nucleus forms and a turbulent flame front propagates across the chamber. Ricardo divided it into three stages:

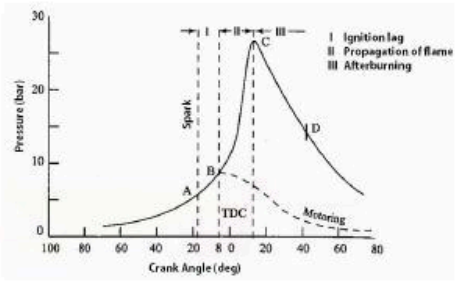


SI engine pressure vs crank angle: I preparation (ignition lag) A-B, II flame propagation B-C, III after-burning C-D; dashed = motoring (non-firing)

Stage	Period	What happens
I. Ignition lag / preparation phase (flame development)	From spark (A) to the point where the pressure curve leaves the motoring curve (B); ~10-20° CA, 0.0015-0.002 s	A chemical process: a self-propagating flame nucleus grows around the plug; ~5-10 % of the charge burns but no visible pressure rise. Depends on fuel, mixture strength (shortest ~10 % rich), temperature and pressure, residual gas, spark energy and gap. In ms it is nearly constant, so in crank degrees it grows with speed → spark advance must increase with speed
II. Flame propagation (rapid combustion)	From B to the peak pressure C (~10-15° aTDC)	Turbulent flame sweeps across the chamber at 10-30 m/s; 80-90 % of the mixture burns; steep pressure and temperature rise. Rate depends on turbulence (increases with speed), A/F (air-fuel ratio), compression ratio, load, chamber shape
III. After-burning	After C, during expansion	Flame reaches walls and crevices and slows; the remaining mixture near the walls burns slowly; pressure falls because of expansion although heat is still released

SI Engine Combustion Overview

- Homogeneous mixture of:
  - Vaporized fuel
  - Air
  - Residual gases
- Ignited by single intense spark ($>$ 10,000°C)
- Generates pre-flame that spreads through mixture
- Three distinct stages of combustion



SI combustion overview (slide): stages I, II, III on the pressure-crank angle diagram

Remember: Spark at A → lag A-B (chemical, nucleus) → rapid propagation B-C (most of the heat, peak 10-15° after TDC) → after-burning C-D near walls.

B5 Stoichiometry, equivalence ratio, rich vs lean; why CI runs lean

Must study | Likely · 67% | Easy

PAST-PAPER QUESTIONS

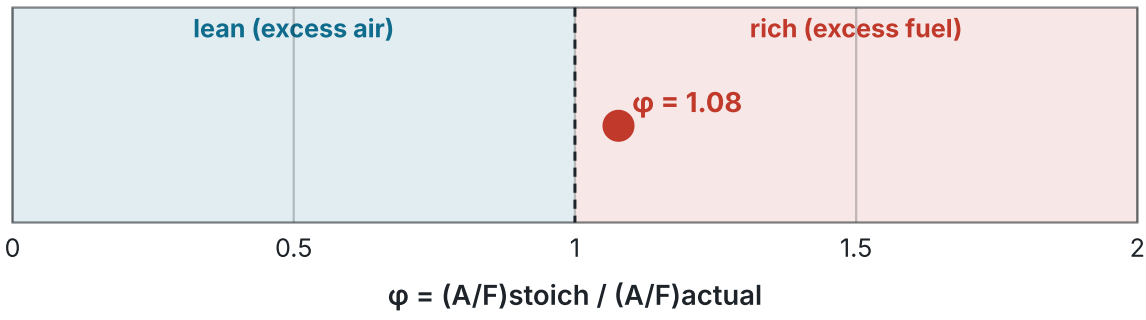
- **2023 Q5(c)** [9] What is meant by "Equivalence ratio"? "Generally a CI (compression ignition (diesel engine)) engine operates with lean mixture" – Explain.
- **2022 Q6(a)** [11] What is meant by combustion stoichiometry? Distinguish fuel rich combustion from fuel lean combustion.
- **2021 Q5(a)** [9] What is meant by "Equivalence ratio"? "Generally a CI engine operates with lean mixture" – Explain.
- **2020 Q4(b)** [9] What is meant by fuel rich combustion and fuel lean combustion? (first part)

Symbols used here

Symbol	Name and meaning	Unit
C_xH_y	A hydrocarbon fuel with x carbon and y hydrogen atoms	
$x + y/4$	Moles of O_2 for complete burning of 1 mol of fuel	mol
3.76	Moles of N_2 that come with each mole of O_2 in air (79/21)	mol
4.76	Moles of air per mole of O_2 (1 + 3.76)	mol
$(A/F)_s$	Stoichiometric air-fuel ratio by mass	kg air / kg fuel
φ	Equivalence ratio = $(F/A)_{actual} / (F/A)_s$	no unit
λ	Relative air-fuel ratio = $1/\varphi$	no unit

INTERACTIVE **Stoichiometric A/F and equivalence ratio**

Pick a fuel C_xH_y and an actual A/F. The bar shows rich or lean.



1. $C_8H_{18} + (x + y/4)(O_2 + 3.76N_2)$: $x + y/4 = 8 + 18/4 =$ **12.50 mol O_2**
2. Air = 12.50 × 4.76 = 59.50 mol × 28.96 = 1723.1 kg; fuel = 114.2 kg
3. $(A/F)_{stoich} =$ **15.08**
4. φ = 15.08 / 14 = **1.077** (λ = 0.93), rich

A CI engine at light load has A/F 50–80 (φ 0.2–0.3) because there is no throttle; at full load it stops at $\varphi \approx$ 0.7–0.8 (smoke limit).

Before you start: stoichiometric and equivalence ratio

Stoichiometric = exactly enough air to burn all fuel (petrol about 14.7 kg air/kg fuel). **Equivalence ratio φ** = (actual F/A (fuel-air ratio)) / (stoichiometric F/A): φ > 1 rich, φ < 1 lean, φ = 1 exact. Diesel engines control power by changing only the fuel (no throttle), so they always run with excess air (lean).

Combustion stoichiometry (2022 6a)

Stoichiometry is the exact chemically correct proportion of fuel and oxidizer for complete combustion, so that all C becomes CO_2 and all H becomes H_2O with no oxygen or fuel left. For C_xH_y in air:

$$C_xH_y + (x + y/4)(O_2 + 3.76\ N_2) \rightarrow x\ CO_2 + (y/2)\ H_2O + 3.76(x + y/4)\ N_2$$

E.g. iso-octane C_8H_{18}: 12.5 mol O_2 → $(A/F)_s$ = 12.5 × 4.76 × 28.96 / 114 = **15.1**; petrol ~14.7, diesel ~14.5, methane 17.2.

Equivalence ratio (2023 5c, 2021 5a)

$$\varphi = (F/A)_{actual} / (F/A)_{stoich} = (A/F)_{stoich} / (A/F)_{actual};\quad \lambda = 1/\varphi \text{ (relative air-fuel ratio)}$$

φ = 1 stoichiometric; φ < 1 **lean** (excess air, % excess air = (1/φ − 1) × 100); φ > 1 **rich** (excess fuel).

Fuel-rich vs fuel-lean combustion (2022 6a, 2020 4b)

	Fuel-rich (φ > 1)	**Fuel-lean (φ < 1)**
Oxygen	Insufficient; some fuel cannot burn	Excess O_2; all fuel can burn
Products	CO_2, H_2O, N_2 + **CO, H_2, unburnt HC (hydrocarbons (unburnt fuel)), soot**	CO_2, H_2O, N_2 + **O_2**
Flame temperature	Peak slightly rich (~φ 1.05-1.1)	Lower as φ decreases
Engine effect	Maximum power (best-power mixture), but low efficiency, high CO and HC	Best economy, low CO/HC; too lean → misfire; NOx (nitrogen oxides (NO + NO_2)) peaks just lean of stoichiometric
Use	SI (spark ignition (petrol engine)) idle, full load, cold start	SI cruise, all CI operation, gas turbines, furnaces

"Generally a CI engine operates with lean mixture" (2023 5c, 2021 5a)

- A CI engine is **quality governed**: there is no throttle, the full charge of air is always drawn in, and power is controlled by the quantity of fuel injected. At part load only a little fuel is injected into the same air, so overall φ is very low (0.2-0.4 at idle).
- Fuel is injected as a spray only 20-40° CA before combustion; it must find oxygen quickly. Mixing is never perfect, so **excess air** is needed for the fuel to burn completely. If overall φ goes above ~0.7-0.8, the rich spray core cannot find oxygen and **black smoke** (soot) rises sharply; so full load is limited lean (smoke limit).
- Lean overall operation also gives higher γ, lower heat loss and higher efficiency, low CO.

- Locally, however, the mixture ranges from very rich (spray core) to very lean (edges); combustion takes place near stoichiometric in the diffusion flame.

Remember: φ = actual F/A ÷ stoichiometric F/A. CI: no throttle + spray needs to find O_2 + smoke limit → always overall lean (φ < 0.8).

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

Stoichiometric A/F (air-fuel ratio) **of iso-octane (if asked):** $C_8H_{18} + 12.5(O_2 + 3.76N_2) \rightarrow 8CO_2 + 9H_2O + 47N_2$

$(A/F)_s$: $(x + y/4) \times 4.76 \times 28.96 / M_{fuel}$

$= 12.5 \times 4.76 \times 28.96 / 114$

$=$ **15.1**

CI runs lean: (1) no throttle, full air every cycle, load set by fuel quantity, so at part load φ is very low; (2) injected spray must find O_2 in a few degrees of crank angle, so excess air is needed for complete burning; (3) above φ ≈ 0.7–0.8 black smoke rises (smoke limit). Hence CI engines always run overall lean.

B12 Surface ignition, pre- vs post-ignition, end gas, flame front

Must study

Likely · 60% Easy

PAST-PAPER QUESTIONS

- **2023 Q6(b)** [7] What are meant by surface ignition? "Pre-ignition is more harmful than post-ignition" – Discuss.
- **2022 Q7(a)** [11] What is meant by surface ignition? "Pre-ignition is more harmful than post-ignition" – Justify the statement.
- **2021 Q7(a)** [11] What are meant by: (i) surface ignition, (ii) end gas, and (iii) flame front? "Pre-ignition is more harmful than post ignition" – Explain.

Before you start: abnormal ignition

Surface ignition = the mixture is lit by a hot spot (glowing carbon deposit, hot valve or spark plug) instead of the spark. If it happens before the spark it is **pre-ignition** (dangerous, advances combustion, overheats); after the spark it is **post-ignition**. The **end gas** is the unburnt mixture ahead of the flame front.

Definitions (2021 7a, 2022 7a, 2023 6b)

- **Surface ignition**: ignition of the fuel-air mixture by a **hot surface** (overheated spark plug electrode or insulator, exhaust valve head, glowing carbon deposits, sharp metal edges, protruding gasket) instead of the timed spark. Its flame front then propagates normally.
- **End gas**: the last part of the unburnt charge, ahead of the advancing flame front and farthest from the spark plug; it is compressed and heated by the burning gas behind the flame and is the part that may auto-ignite (knock).
- **Flame front**: the thin reaction zone (a fraction of a mm thick) that separates the burnt gas from the unburnt mixture and travels across the chamber, where the main chemical reaction and heat release occur.

Surface ignition can be **pre-ignition** (occurs **before** the spark) or **post-ignition** (occurs **after** the spark, from a hot spot elsewhere in the unburnt charge, giving a second flame front). It may be **knocking or non-knocking**; persistent knock overheats parts and causes surface ignition (run-away surface ignition).

"Pre-ignition is more harmful than post-ignition" (2023 6b, 2022 7a, 2021 7a)

- **Pre-ignition** advances the start of combustion uncontrollably, well before the spark, during compression. The piston must then compress burning, expanding gas: negative work, very high peak pressure and temperature, loss of power, rough running. Heat transfer increases, the hot spot becomes hotter, so pre-ignition occurs earlier and earlier (**run-away**), leading to severe knock, melted or holed pistons, broken rings, and seizure in a multi-cylinder engine if only some cylinders are affected.
- **Post-ignition** occurs after the spark has already started the normal flame; it only adds a second flame front that speeds up combustion, slightly raising the rate of pressure rise. Timing is still controlled by the spark, so the damage is small (it may promote knock but no uncontrolled advance).
- Hence pre-ignition is far more harmful. Cures: avoid deposits, use correct spark plug heat range, good cooling of exhaust valve, avoid sustained knock.

B13 Knock in SI vs CI; ignition delay SI vs CI; SI knock limited

Must study

Likely · 67% Medium

PAST-PAPER QUESTIONS

- **2023 Q6(c)** [12] How do ignition delay and knock in SI (spark ignition (petrol engine)) engine differ from those in CI (compression ignition (diesel engine)) engine? Explain with appropriate diagrams.
- **2022 Q7(c)** [10] How knocking in SI engines differs from that in CI engines? Explain with appropriate diagrams.
- **2022 Q8(c)** [13] What is meant by end gas auto-ignition? "SI engine performance is knock limited but CI engine not" – Justify the statement.
- **2021 Q8(a)** [10] How knock in SI engine differs from knock in CI engine?
- **2021 Q7(c)** [12] What is meant by "Knock" in relation to SI engine? "SI engine performance is knock limited but CI engine not" – justify the statement.
- **2020 Q5(b)** [12] How ignition delay in SI engine differs from ignition delay in CI engine? "Factors tend to increase knock in SI engine reduce knock in CI engine" – Justify.
- **2020 Q5(c)** [8] What is meant by abnormal combustion? Explain the origin of knock in SI engine with the help of auto-ignition theory.

LEARN IT FROM ZERO

Knock: the same word, two opposite problems

In a petrol (SI) engine the whole chamber is filled with ready-mixed fuel and air. The spark starts a flame that travels across like a grass fire. As it burns, the hot gas behind it expands and **squeezes and heats the unburnt corner** (the **end gas**). If that corner gets hot enough for long enough before the flame arrives, it **explodes by itself**: two flames collide, pressure waves bounce around and you hear a metallic "ping". This happens at the **end** of combustion. Cure: a fuel that is slow to self-ignite (high octane), lower compression, retarded spark, cooler charge, faster flame.

In a diesel (CI) engine self-ignition is the normal way to burn. The problem is the opposite: if the fuel takes **too long** to self-ignite (long delay), too much fuel piles up and then burns at once at the **start** of combustion (see the spike in the B11 graph). Cure: a fuel that self-ignites quickly (high cetane), higher compression, hotter or boosted air.

So every factor works in **opposite directions**: what helps the SI engine (long delay, high self-ignition temperature, low compression) hurts the CI engine, and vice versa. That one idea answers most of the "justify" statements. On the B10 graph lower the octane number or advance the spark to see SI knock appear.

Before you start: knock in one minute

SI knock: the end gas, squeezed and heated by the advancing flame, self-ignites before the flame reaches it, giving pressure waves (ping). It happens at the **end** of combustion. **CI knock**: too long an ignition delay lets a lot of fuel collect, which then burns all at once at the **start** of combustion. So what prevents SI knock (fuel that resists self-ignition) causes CI knock, and vice versa.

Knock in SI engines; origin by auto-ignition theory (2021 7c, 2022 8c, 2020 5c)

Abnormal combustion is any combustion that departs from the normal spark-initiated flame travelling at normal speed: knock and surface ignition. **Knock (detonation)** in an SI engine: as the flame advances, the burnt gas expands and compresses the unburnt **end gas**, raising its pressure and temperature. If the end gas reaches its **self-ignition temperature** and stays there longer than its ignition delay before the flame arrives, it **auto-ignites** spontaneously in one or more places. This sudden release of energy creates strong pressure waves that oscillate across the chamber (up to ~5000 Hz), heard as a metallic pinking noise; they scour the protective gas boundary layer, raising heat transfer, and cause piston crown erosion, ring and head damage, power and efficiency loss, and pre-ignition.

No knock if t_{flame} (time for flame to consume end gas) < t_{delay} (auto-ignition delay of end gas)

Knock in SI vs CI; ignition delay in SI vs CI (2023 6c, 2022 7c, 2021 8a, 2020 5b)

> **Diagrams to draw:** SI p-θ with sharp high-frequency oscillations **after** the pressure peak (end of combustion). CI p-θ with a very steep pressure rise immediately after the delay at the **start** of combustion (and the ROHR (rate of heat release) spike).

Point	SI engine	CI engine
When knock occurs	Near the **end** of combustion (end gas auto-ignites)	At the **beginning** of combustion (fuel accumulated during long delay burns at once)
Cause	Auto-ignition of the homogeneous end gas before the flame arrives	Too long ignition delay
Nature	Pressure waves, very high rate of pressure rise locally; can wreck pistons	Also rough, but CI already relies on auto-ignition; it is mainly noise and shock loading
Pre-ignition possible?	Yes (charge is premixed)	No (only air is compressed; fuel injected at the desired time)
Ignition delay meaning	Delay of the end gas should be **long** (so the flame consumes it first)	Delay of the injected fuel should be **short**
Wanted fuel	High SIT (self-ignition temperature), high octane	Low SIT, high cetane

Ignition delay in SI vs CI: in SI the delay (ignition lag) is the time from spark to the first measurable pressure rise of the flame kernel; it is a chemical process of flame nucleus growth, ~0.0015 s. In CI the delay is from start of injection to start of combustion and has a large **physical** part (atomization, vaporization, mixing) plus a **chemical** part; it governs the whole combustion and noise.

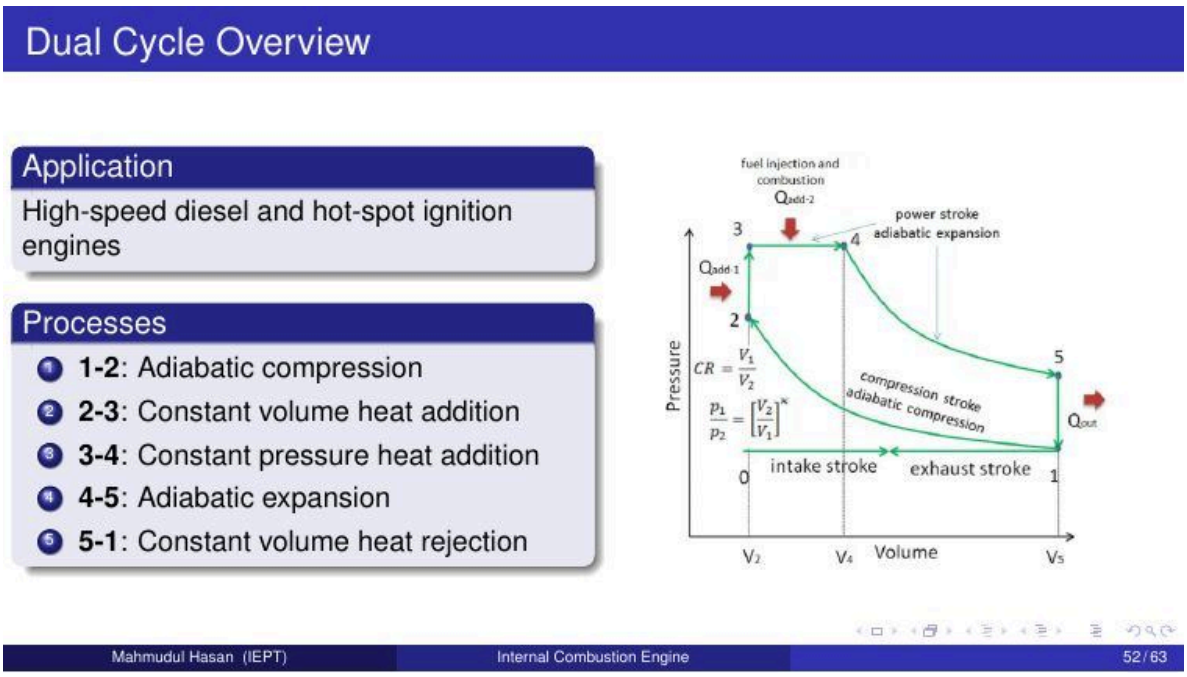
"Factors tending to increase knock in SI reduce knock in CI" (2020 5b)

Factor	To reduce SI knock	To reduce CI knock
Self-ignition temperature of fuel	High	Low
Ignition delay of fuel	Long	Short
Compression ratio	Low	High
Inlet temperature	Low	High
Inlet pressure / supercharging	Low	High
Cylinder wall temperature	Low	High
Speed	High (more turbulence, faster flame)	Low (delay in °CA smaller)
Combustion chamber size	Small	Large

Since knock in SI is avoided by a long delay of the end gas while knock in CI is avoided by a short delay of the injected fuel, every factor acts in opposite directions.

"SI engine performance is knock-limited but CI engine is not" (2022 8c, 2021 7c)

- SI efficiency and power rise with compression ratio, inlet pressure (supercharging) and spark advance, but every one of these raises end-gas temperature and pressure and so causes knock. The **maximum compression ratio (~10-12), boost and timing** are therefore fixed by the onset of knock with the available fuel octane: SI performance is **knock limited**.
- In a CI engine higher compression ratio, inlet temperature and boost **shorten** the ignition delay and **reduce** knock; there is no end gas because only air is compressed. Compression ratio (14-22) is limited by mechanical stress and friction, and full-load fuel by **smoke**, not knock. So CI performance is not knock limited.



Dual cycle (slide): realistic model of modern high-speed diesels

B4 Indicator diagram: indicated and pumping work; pumping loss, WOT, turbocharged engines

Must study · Likely · 67% · Medium

PAST-PAPER QUESTIONS

- **2023 Q5(b)** [6] With the help of indicator diagram, explain: (i) Net indicated work per cycle, (ii) Pumping work per cycle.
- **2023 Q5(d)** [6] How does pumping work in a turbocharged engine differ from that in a naturally aspirated engine? Discuss.
- **2022 Q5(b)** [13] What is meant by pumping loss? Why pumping loss is higher in conventional SI (spark ignition (petrol engine)) engine compared to CI (compression ignition (diesel engine)) engine? How to reduce it?
- **2021 Q5(c)** [10] With the help of indicator diagram, explain: (i) Indicated work/cycle; (ii) Pumping work/cycle.
- **2021 Q5(d)** [6] Why pumping loss is minimum for engine operating at WOT (wide open throttle) condition?
- **2020 Q4(c)** [12] What is meant by "wide open throttle (WOT)" operation? Why pumping loss is higher in conventional SI engine compared to CI engine? How to reduce it?

LEARN IT FROM ZERO

Two loops on one diagram

A four-stroke engine does two jobs: it makes power (compression + expansion strokes) and it **breathes** (exhaust + intake strokes). If you plot cylinder pressure against volume for the whole cycle you get a tall loop on top and a thin loop at the bottom.

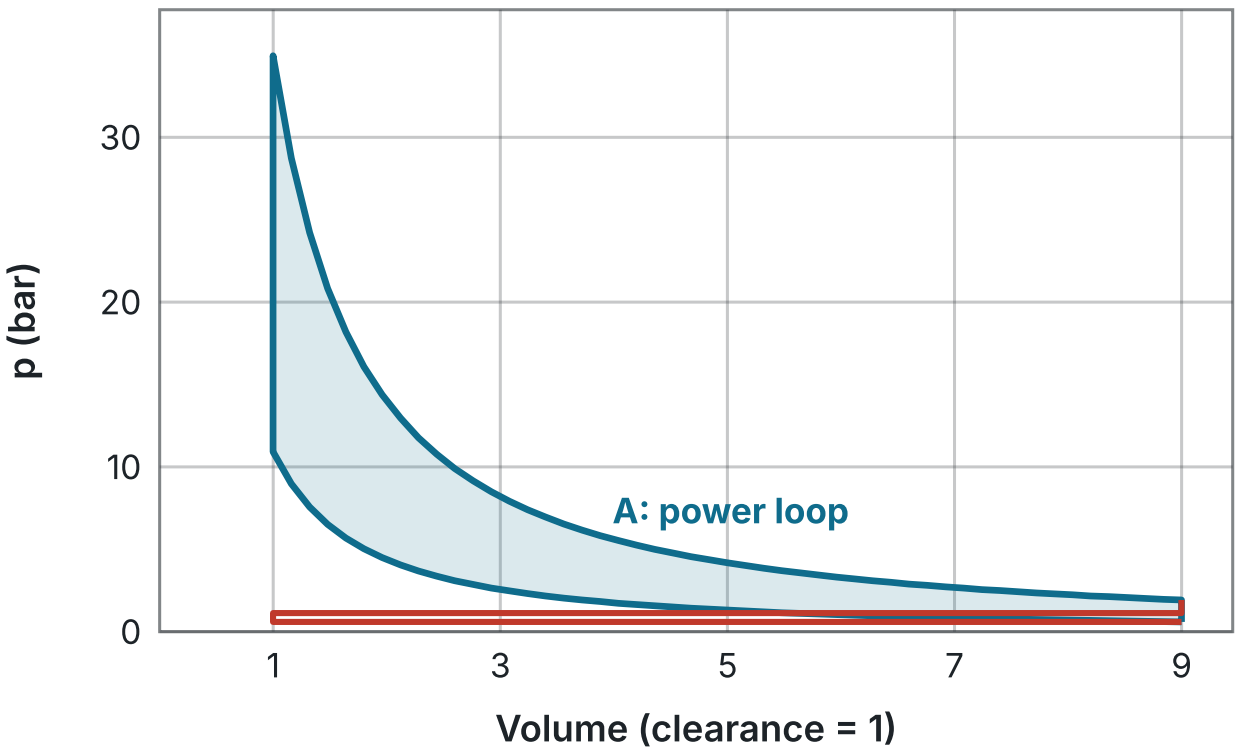
Top loop (area A) = work the gas gives. Bottom loop (area B) = work the piston spends breathing: on the exhaust stroke it pushes gas out against exhaust pressure (a bit above 1 bar), on the intake stroke it pulls fresh charge in against intake pressure (below 1 bar). If intake pressure is lower than exhaust pressure, the piston loses work every cycle: **pumping loss**. Net work = A − B.

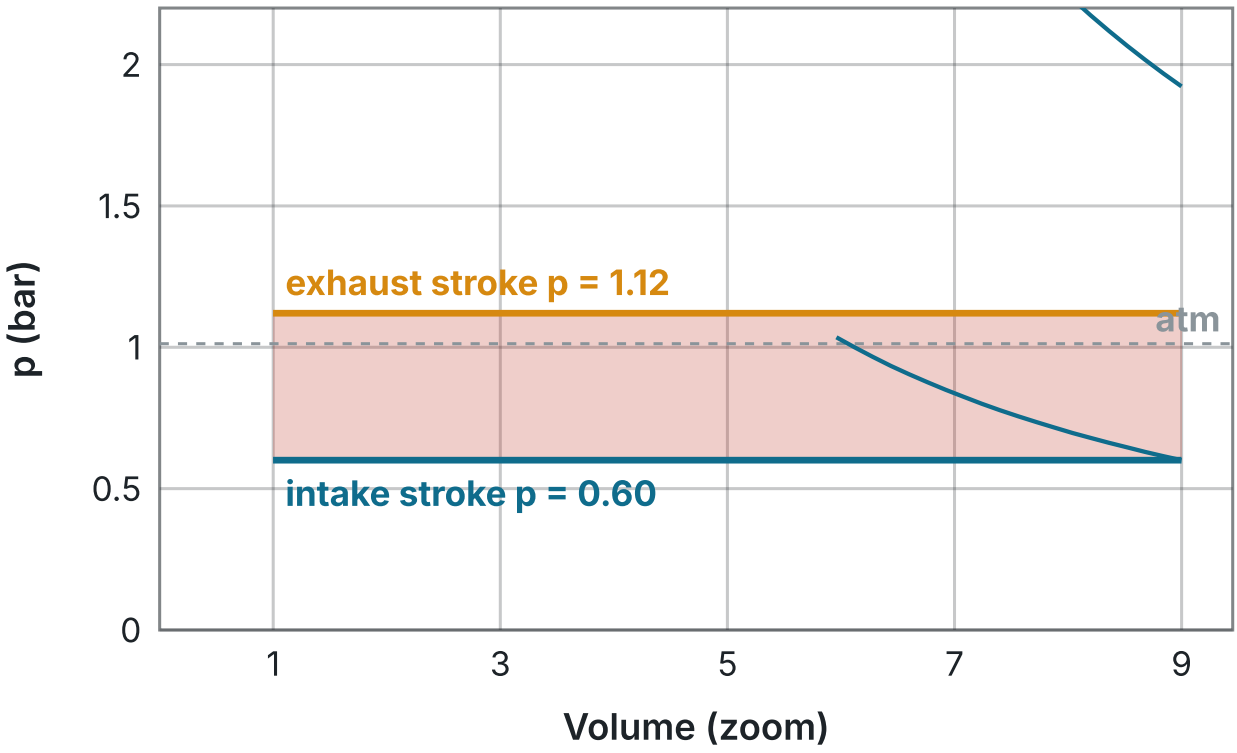
Why SI engines suffer most: a petrol engine controls power with a **throttle** (a flap that chokes the air). At light load the flap is nearly closed, the intake falls to ~0.3–0.5 bar, and the piston must pull against that vacuum: loop B becomes big. A diesel has **no throttle** (it controls power by fuel quantity only), so its intake stays near 1 bar and loop B stays thin. At **WOT** (wide open throttle) the SI intake is also near 1 bar, so pumping loss is smallest. With a **turbocharger** intake pressure can be higher than exhaust pressure, so the bottom loop turns round and gives work instead of taking it.

Try it below: close the throttle and watch the shaded area grow.

INTERACTIVE **Indicator diagram: power loop and pumping loop**

Close the throttle and watch the intake line drop: the lower (pumping) loop gets bigger. Right chart zooms on the bottom 0–2 bar.





Pumping mean effective pressure ≈ $p_{exhaust} - p_{intake}$ = **0.52 bar** ⇒ pumping work is **negative (a loss, area B)**.

Part throttle: the throttle chokes the intake far below 1 bar, so the piston must pull against a vacuum on every intake stroke. That is why SI engines waste fuel at light load. Net indicated work = area A − area B.

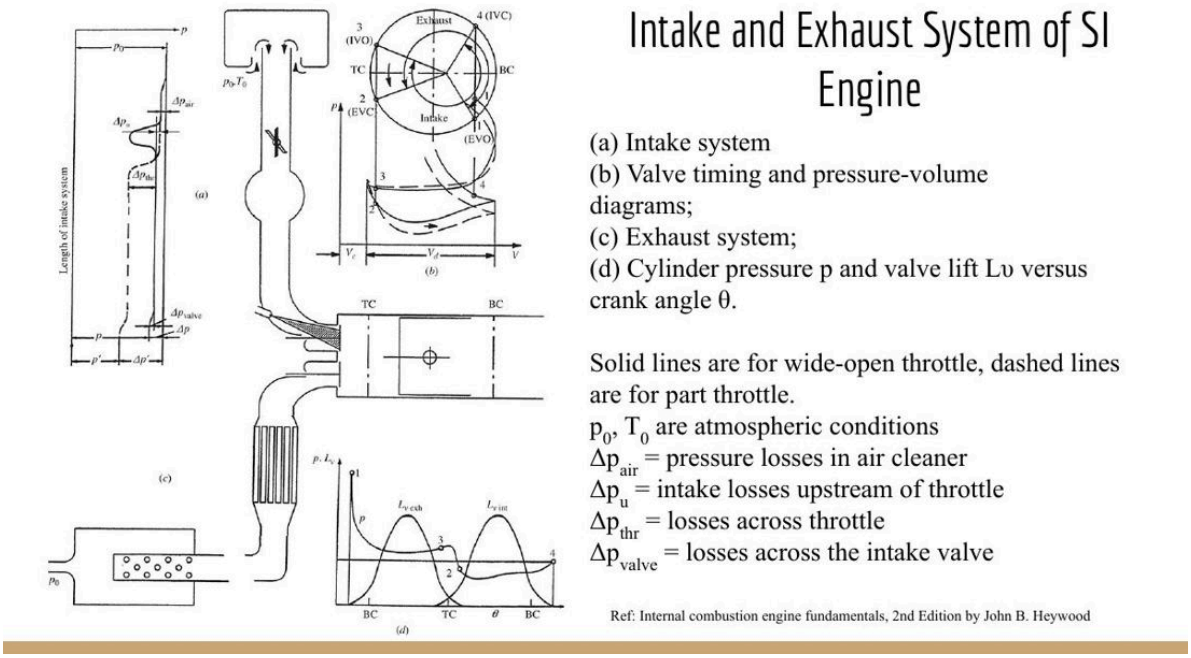
Symbols used here

Symbol	Name and meaning	Unit
$W_{c,ig}$	Gross indicated work per cycle (area A, top loop)	J
W_p	Pumping work per cycle (area B, bottom loop)	J
$W_{c,in}$	Net indicated work = A − B	J
pmep	Pumping mean effective pressure = $W_p/V_d \approx p_e - p_i$	bar
p_i, p_e	Intake and exhaust pressure	bar
WOT	Wide open throttle (full load)	

Before you start: indicator diagram

Plot cylinder pressure against volume for a whole cycle and you get two loops: a big loop (compression + expansion) whose area is the **gross indicated work**, and a small loop at the bottom (exhaust + intake strokes) whose area is the **pumping work** lost to push gas out and suck air in. Closing the throttle in an SI engine makes the intake pressure low, so pumping loss rises at part load.

Indicator diagram: indicated work and pumping work (2023 5b, 2021 5c)



Intake/exhaust system and p-V diagram: WOT (solid) and part throttle (dashed); the lower loop is the pumping loop

The indicator diagram (p-V) of a four-stroke engine has two loops:

- **Upper (power) loop**, traced clockwise over compression and expansion: its area A is the **gross indicated work per cycle** $W_{c,ig}$ (work delivered to the piston over compression + expansion strokes only).
- **Lower (pumping) loop**, traced anticlockwise during exhaust and intake strokes: its area B is the **pumping work per cycle** W_p, the work the piston does to push exhaust out (at $p > p_{atm}$) and draw charge in (at $p < p_{atm}$). In a naturally aspirated engine it is **work done on the gas** (negative).

- **Net indicated work per cycle** $W_{c,in} = W_{c,ig} - |W_p|$ = area A − area B. IMEP (net) = $W_{c,in}/V_d$; pumping mean effective pressure pmep = W_p/V_d.

Pumping loss; why it is higher in SI than CI; how to reduce (2022 5b, 2020 4c)

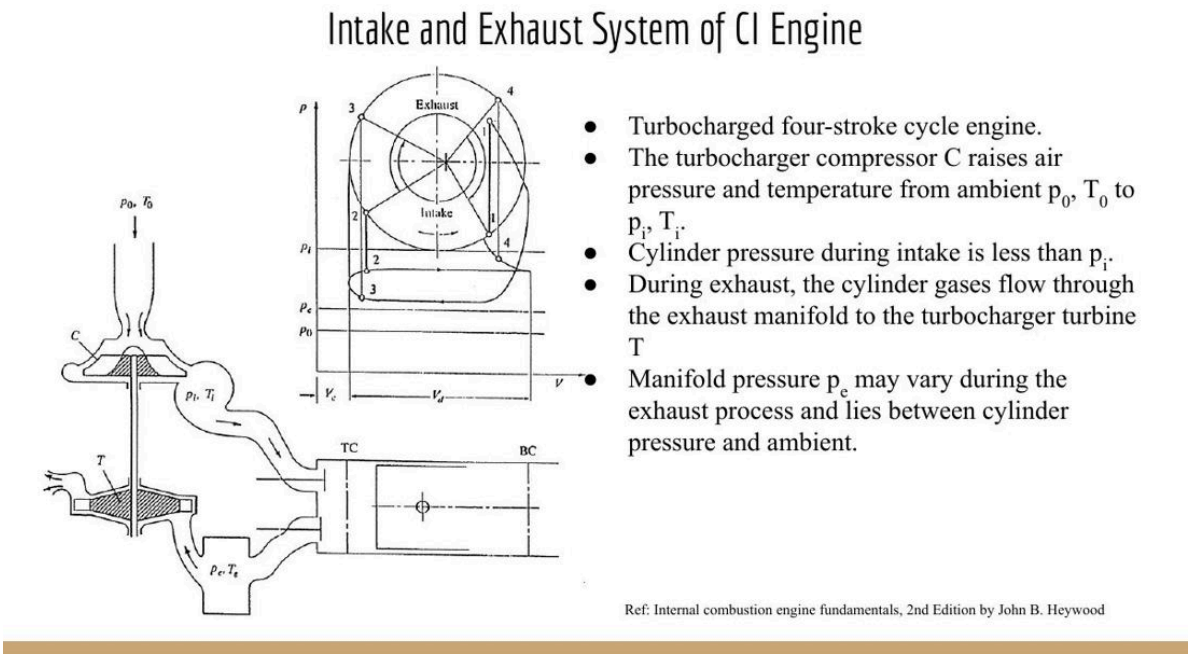
Pumping loss is the work spent per cycle in drawing fresh charge in and expelling burnt gas (area of the pumping loop). In an SI engine the load is controlled by a **throttle** (quantity governing): at part load the throttle is nearly closed, so manifold pressure is far below atmospheric (e.g. 0.3-0.5 bar) while exhaust is above atmospheric; the pumping loop becomes large. A CI engine has **no throttle** (quality governing): load is controlled by the fuel quantity and the cylinder always draws nearly atmospheric air, so its pumping loss stays small at all loads.

Ways to reduce pumping loss in SI engines: run at wider throttle with dilution (lean burn, high EGR (exhaust gas recirculation)), **variable valve timing / lift** (early or late IVC (inlet valve closes), Miller/Atkinson) to control load without the throttle, cylinder deactivation, downsizing with turbocharging, direct injection with stratified charge (unthrottled), larger valves and smooth low-loss intake and exhaust, multi-valve heads.

Wide open throttle and why pumping loss is minimum at WOT (2020 4c, 2021 5d)

WOT operation means the throttle plate is fully open: the engine inducts the maximum charge and gives maximum torque at that speed (full load). At WOT the throttle causes almost no pressure drop, so intake pressure is close to atmospheric (only filter and valve losses) and the intake line of the p-V diagram lies close to the exhaust line; the pumping loop is thin and pumping work is a minimum. At part throttle the intake line drops well below atmospheric and the loop area grows.

Pumping work in turbocharged vs naturally aspirated engines (2023 5d)



Turbocharged four-stroke engine: intake pressure p_i raised by the compressor, exhaust passes through turbine

In an NA engine $p_{intake} < p_{exhaust}$, so the pumping loop is anticlockwise and pumping work is **negative** (a loss). In a turbocharged engine the compressor raises intake pressure, and when boost pressure exceeds the exhaust back pressure (turbine inlet pressure), the intake line lies **above** the exhaust line: the gas exchange loop is traced clockwise and the pumping work becomes **positive**

(the incoming charge pushes on the piston), adding to net work. If the turbine is small and creates high back pressure ($p_{exh} > p_{boost}$), pumping work is negative but the energy is partly recovered in the turbine. A mechanically driven supercharger also gives positive pumping work but takes shaft power to drive.

Remember: Pumping loop area = loss. SI throttles air → big loop at part load; CI has no throttle → small loop. WOT: no throttle drop → minimum. Turbo: $p_{in} > p_{ex}$ → pumping work positive.

B14 CI knock factors and ignition delay; good SI fuel is bad CI fuel; poor fuel in large slow CI; smoke limited

Must study · Likely · 60% · Medium

PAST-PAPER QUESTIONS

- **2023 Q7(c)** [13] Discuss the major factors that influence the tendency of knock in CI (compression ignition (diesel engine)) engine.
- **2023 Q7(b)** [10] "A good SI (spark ignition (petrol engine)) engine fuel is a bad CI engine fuel" – Explain with appropriate diagrams.
- **2023 Q8(d)** [5] Why poor ignition quality fuel can be used in low speed large sized CI engine? Explain.
- **2023 Q6(d)** [6] "CI engine performance is smoke limited" – Justify the statement.
- **2022 Q6(b)** [10] "CI engine performance is smoke limited" – Justify the statement.
- **2021 Q8(c)** [10] Discuss the factors that affect ignition delay in CI engines.
- **2021 Q8(b)** [10] "A good SI engine fuel is a bad CI engine fuel" – justify the statement.
- **2021 Q8(d)** [5] Why low ignition quality fuel can be used in very large, slow-speed CI engine?

UNDERSTAND **Read this to understand; you do not need to write it**

Every statement in this topic comes from one idea: a CI engine wants a **short ignition delay**. Ask of each factor: does it make the air hotter or denser, or the fuel quicker to ignite? Then it shortens the delay and reduces CI knock. See the B11 graph: shorten the delay and the premixed spike shrinks.

Before you start: ignition delay

Ignition delay in a CI engine = time (or crank angle) between start of injection and start of combustion. A short delay = smooth engine. It is shortened by high cetane fuel, high compression ratio (hotter air), high intake temperature and pressure, and good atomisation.

Factors that affect ignition delay and knock in CI engines (2023 7c, 2021 8c)

CI knock is caused by a long ignition delay, so every factor that shortens the delay reduces knock.

Factor	Effect
Fuel cetane number / SIT (self-ignition temperature)	Higher cetane (lower SIT, paraffinic) → shorter delay, less knock. Ignition improvers (amyl nitrate) help
Compression ratio	Higher r → higher air temperature and pressure at injection → shorter delay
Intake air temperature and pressure (supercharging)	Higher → shorter delay
Load (fuel-air ratio)	Higher load → hotter walls and residuals → shorter delay
Engine speed	Delay in ms falls slightly with speed (more turbulence, hotter) but in crank degrees it rises; low speed gives more time
Injection timing	Injecting too early (cold air) or too late lengthens delay; optimum ~15-20° bTDC
Injection pressure, atomization, quantity during delay	Finer spray shortens physical delay; injecting less fuel during the delay (pilot injection, rate shaping) reduces the premixed spike
Turbulence / swirl, chamber design	Better mixing shortens delay; IDI chambers, hot spots help
Coolant / wall temperature, cylinder size	Hotter walls and larger cylinders (less heat loss) reduce knock

"A good SI engine fuel is a bad CI engine fuel" (2023 7b, 2021 8b)

- A good SI fuel must **resist auto-ignition** so that the end gas does not knock: high SIT, long ignition delay, high octane number (iso-paraffins, aromatics).
- A good CI fuel must **auto-ignite readily** so that the delay is short and CI knock is avoided: low SIT, short delay, high cetane number (straight-chain paraffins).
- Octane and cetane are roughly inversely related (high-octane fuels have low cetane). So petrol in a diesel engine gives a very long delay and violent knock (or no ignition), and diesel in an SI engine knocks immediately. Hence a good SI fuel is a bad CI fuel and vice versa.

> **Diagram:** sketch octane number vs cetane number as a falling line (ON ≈ 120 − 2 CN (cetane number)), and SIT high for SI fuel, low for CI fuel.

Why poor ignition-quality fuel can be used in large, slow-speed CI engines (2023 8d, 2021 8d)

Ignition delay is a time (ms) that depends mainly on the fuel and charge conditions. In a slow engine (100-300 rpm) one crank degree takes a long time, so a given delay in ms is only a few **crank degrees**; little fuel accumulates before ignition and the rate of pressure rise stays acceptable. Large cylinders also lose less heat (smaller surface/volume), so compressed air is hotter, and they are usually turbocharged (high pressure and temperature), which shortens delay further. Therefore low-cetane, cheap heavy fuel oil can be burnt without knock in large marine and stationary diesels, while small high-speed diesels need high-cetane fuel.

"CI engine performance is smoke limited" (2023 6d, 2022 6b)

- Power of a CI engine is controlled by the amount of fuel injected into a fixed quantity of air.
- Because fuel is injected as a spray and mixes in limited time, local rich zones exist even when the overall mixture is lean. As more fuel is injected (overall φ above ~0.7-0.8), the spray cannot

find enough oxygen, soot forms faster than it is oxidised and **black smoke** rises steeply; fuel economy also worsens.

- Therefore the maximum fuel delivery, and so the maximum torque and power, is set at the point where exhaust smoke reaches the legal/visible limit, well before stoichiometric. This is the smoke limit, just as SI engines are knock limited. Turbocharging (more air) and better injection raise the smoke limit.

B9 SI mixture formation strategies; HCCI and DISI Good to have Likely · 60% Medium

PAST-PAPER QUESTIONS

- **2024 Q6(c)** [9] Write short notes on: (i) Homogeneous charge compression ignition (HCCI), (ii) Direct injection spark ignition (DISI).
- **2023 Q7(a)** [12] Discuss the three strategies of mixture formation process in SI (spark ignition (petrol engine)) engine with suitable diagrams.

Before you start: mixture formation

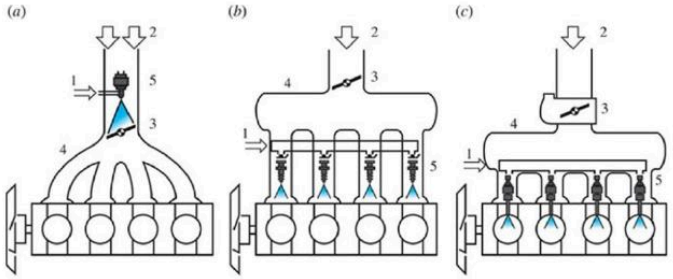
Fuel and air can be mixed outside the cylinder (carburetor, port injection: homogeneous mixture) or inside it (direct injection). **DISI** (direct injection spark ignition) = direct injection spark ignition. **HCCI** (homogeneous charge compression ignition) = homogeneous charge compression ignition: a premixed lean mixture is compressed until it ignites everywhere at once, with no spark and no flame front.

Three strategies of mixture formation in SI engines (2023 7a)

Fuel Injection System

Fuel injection system can be divided into two basic types:

1. Manifold: (a) Throttle body & (b) Port
2. (c) Gasoline direct injection (GDI)



Schematics of different fuel-injection system arrangements: (*a*) single point, (*b*) multipoint (port), and (*c*) direct in-cylinder injection (DI). (1) Fuel; (2) Air; (3) Throttle; (4) Intake manifold; (5) Fuel injector. [2] (*Courtesy Robert Bosch GmbH and SAE.*)

Fuel injection arrangements: (a) single-point (throttle body), (b) multipoint port injection, (c) direct in-cylinder injection

1. **Carburetion**: fuel drawn into the air stream by the venturi depression of a carburetor upstream of the throttle; mixture formed in the intake manifold. Simple and cheap but poor A/F (air-fuel ratio) control, uneven distribution between cylinders, wall wetting; obsolete in cars.
2. **Manifold / port fuel injection**: (a) **single-point (throttle body) injection**: one or two injectors above the throttle; (b) **multipoint port fuel injection (MPFI)**: one injector per cylinder spraying onto the back of the intake valve; ECU meters fuel from air flow, speed and λ-sensor. Precise φ = 1 control, equal distribution, good cold start; homogeneous charge.
3. **Direct in-cylinder injection (GDI / DISI)**: fuel injected at high pressure (50-200 bar) directly into the cylinder, either during intake (homogeneous) or late in compression (**stratified charge**, rich near the plug, lean overall). Charge cooling, higher compression ratio, unthrottled part load.

HCCI: homogeneous charge compression ignition (2024 6c)

A **homogeneous** (premixed, usually very lean or diluted) fuel-air mixture is inducted as in an SI engine but is **ignited by compression** as in a CI (compression ignition (diesel engine)) engine; auto-ignition occurs simultaneously at many points throughout the charge, with no flame propagation.

- **Advantages**: very low NOx (low peak temperature in lean mixture) and almost no soot (no rich zones); high efficiency (high compression ratio, unthrottled, fast heat release close to constant volume); can use various fuels.
- **Challenges**: no direct trigger, so **combustion timing is hard to control** (depends on temperature, composition, pressure); narrow load range (misfire at low load, knock-like high pressure-rise rate at high load); high HC (hydrocarbons (unburnt fuel)) and CO; cold-start difficulty. Control by variable valve timing, EGR (exhaust gas recirculation)/residual trapping, variable compression ratio, fuel blending.

DISI: direct injection spark ignition (2024 6c)

Petrol is injected directly into the combustion chamber and ignited by a spark plug.

- **Advantages**: fuel evaporating inside the cylinder cools the charge → higher η_v and lower knock tendency → higher compression ratio; precise fuel metering, fast response, better cold start (no wall wetting in ports); stratified lean operation at part load with the throttle wide open → less pumping loss, 10-20 % better economy; suits downsized turbo engines.
- **Challenges**: higher PM (soot from fuel films on piston, short mixing time); stratified lean mode makes NOx (nitrogen oxides (NO + NO_2)) after-treatment hard (lean NOx trap needed); injector deposits; high-pressure fuel system cost.

	HCCI	DISI	Conventional SI	Conventional CI
Mixture	Homogeneous	Homogeneous or stratified	Homogeneous	Heterogeneous
Ignition	Auto-ignition	Spark	Spark	Auto-ignition
Load control	Mixture quality	Throttle / stratification	Throttle	Fuel quantity
NOx / PM (particulate matter)	Very low / very low	Medium / medium	High / low	High / high

B8 Scavenging; two-stroke vs four-stroke; why 2-stroke CI is common but 2-stroke SI is not

Must study · Possible · 40% · Easy

PAST-PAPER QUESTIONS

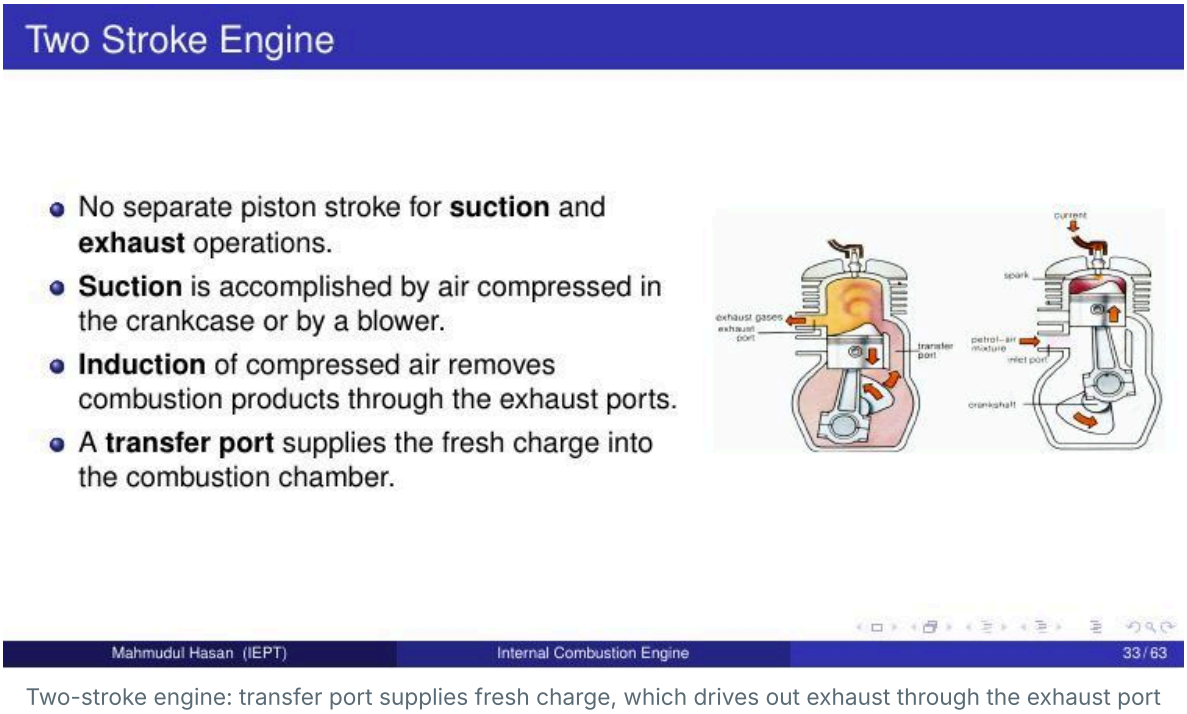
- **2022 Q3(b)** [10] What are the impacts of scavenging? Why scavenging requires more attention in case of 2-stroke engine than that for 4-stroke engine?
- **2022 Q8(b)** [8] Why 2-stroke CI (compression ignition (diesel engine)) engines find wider use whereas 2-stroke SI (spark ignition (petrol engine)) engines are used only to a limited extent?
- **2021 Q6(a)** [7] What is meant by scavenging? Why scavenging is more important for 2-stroke engine than for 4-stroke engine.
- **2021 Q6(d)** [7] Why 2-stroke CI engine is more common than 2-stroke SI engine? (second part)
- **2020 Q4(b)** [9] Why two stroke CI engines find wider use whereas two stroke SI engines are used only to a limited extent? (second part)
- **2020 Q2(b)** [10] "Four stroke IC (internal combustion) engine is economical and less pollutant than two stroke engine" – Justify the statement.

Before you start: two-stroke vs four-stroke

A four-stroke engine needs 4 piston strokes (2 revolutions) per power stroke: intake, compression, power, exhaust. A two-stroke does it in 2 strokes (1 revolution) by using ports in the cylinder wall: fresh charge blows the burnt gas out. That blowing-out is **scavenging**. In an SI two-stroke the fresh charge contains petrol, so some fuel escapes with the exhaust.

Scavenging and its impacts (2022 3b, 2021 6a)

Scavenging is the process of clearing the burnt gases out of the cylinder and replacing them with fresh charge (air or mixture). In a two-stroke engine there is no separate exhaust stroke: near BDC (bottom dead centre) both exhaust and transfer ports are open, and fresh charge compressed in the crankcase (or by a blower) enters and pushes the exhaust out. Types: **cross-flow**, **loop (reverse-flow)** and **uniflow** (most efficient: ports at one end, exhaust valve at the other).



Two-stroke engine: transfer port supplies fresh charge, which drives out exhaust through the exhaust port

Impacts: good scavenging means less residual gas → more fresh charge (higher power), cooler cycle, less knock; poor scavenging leaves residuals (low power, rough running). Over-scavenging sends fresh charge straight out of the exhaust (**short-circuiting**) → fuel loss, HC (hydrocarbons (unburnt fuel)) emission, low efficiency (in SI).

Why scavenging is more important in 2-stroke than 4-stroke (2022 3b, 2021 6a)

- In a 4-stroke engine the piston itself pushes the exhaust out during a full exhaust stroke and draws charge in a full suction stroke; only the clearance volume is left, and the overlap period helps a little. Scavenging is a minor matter.
- In a 2-stroke the whole gas exchange happens in ~120-150° of crank angle around BDC with both ports open at the same time; the only means of removing exhaust is the incoming charge. Its quality decides how much residual remains and how much fresh charge escapes. So power, efficiency, fuel consumption and emissions of a 2-stroke depend directly on scavenging efficiency.

Why 2-stroke CI engines are widely used but 2-stroke SI engines only to a limited extent (2022 8b, 2021 6d, 2020 4b)

- A 2-stroke **SI** engine scavenges with **fuel-air mixture**: 20-30 % of the fresh mixture short-circuits through the exhaust port → high fuel consumption and very high HC emission; mist lubrication also burns oil (smoke). It cannot meet emission laws, so it is limited to small cheap engines (mopeds, chain saws, outboard motors).
- A 2-stroke **CI** engine scavenges with **pure air** (blower or turbocharger); any air that short-circuits carries no fuel, so there is no fuel loss, and the extra air cools the cylinder. Fuel is injected after the ports close. It gives twice the power strokes, high power/weight, and the uniflow design gives good scavenging. Hence it is used widely in large marine and locomotive engines (largest engines in the world are 2-stroke diesels).

"Four-stroke IC engine is economical and less pollutant than two stroke engine" (2020 2b)

Four-stroke	Two-stroke
Separate exhaust and suction strokes; no charge loss	Short-circuiting of fresh charge through exhaust port (SI)
Higher thermal and part-load efficiency, lower BSFC (brake specific fuel consumption)	Lower thermal efficiency, poorer part-load efficiency
Higher volumetric efficiency (more time for induction)	Lower; residual gas dilution
Separate pressure lubrication; little oil burnt	Oil mixed with fuel and burnt → blue smoke, deposits
Low HC and CO; can use 3-way catalyst	High HC, CO, smoke; catalyst overloaded
Lower wear and noise	Higher wear, noisier

So the 4-stroke is economical (less fuel, less oil) and cleaner, at the cost of weight, size and cost.

B11 Stages of combustion in CI engine (ROHR vs crank angle); flames; SIT

Must study · Possible · 53% · Medium

PAST-PAPER QUESTIONS

- **2023 Q8(c)** [11] Describe the different stages of combustion in a CI (compression ignition (diesel engine)) engine with the help of rate of heat release (ROHR) versus crank angle (θ) diagram.
- **2022 Q6(c)** [14] Describe the different stages of combustion in CI engine with the help of rate of heat release (RDHR) verses crank angle (θ) diagram.
- **2022 Q5(c)** [10] What is the physical meaning of SIT (self-ignition temperature)? Explain the key characteristics of premixed flame and diffusion flame.
- **2020 Q5(a)** [10] How flames are classified? Describe the different stages of combustion in CI engine with the help of "rate of heat release-crank angle" diagram.

LEARN IT FROM ZERO

The CI combustion story in four scenes

A diesel squeezes **only air** until it is very hot (about 500–700 °C). Then liquid fuel is sprayed in. Fuel cannot burn as a liquid: each droplet must first heat up, **evaporate** and **mix** with air, and then the chemistry needs a moment to "wake up". That waiting time is the **ignition delay** (scene 1). Meanwhile the injector keeps spraying, so fuel **piles up**.

Scene 2: the piled-up, already-mixed fuel catches fire **all at once**. That is the tall, narrow spike of heat release: **premixed (uncontrolled) combustion**. If the delay was long, the pile was big and the spike is violent: pressure jumps, the engine makes a hammering noise. That is **diesel knock**.

Scene 3: now the chamber is burning, so new fuel burns as soon as it finds air. The speed is set by **how fast fuel and air mix** (swirl, spray), so it is called **mixing-controlled (diffusion) combustion**: a lower, wider hump that lasts until injection ends.

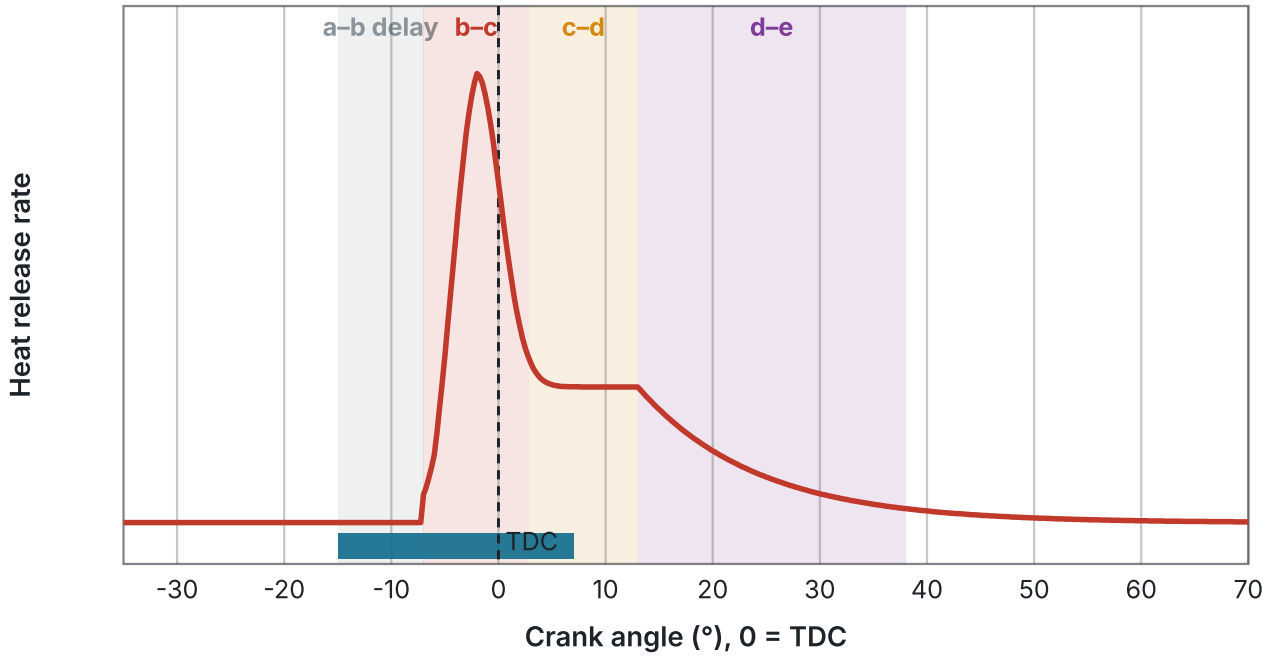
Scene 4: **late combustion**: the last bits of fuel and soot burn slowly as the piston goes down: a long low tail.

ROHR (rate of heat release) = rate of heat release = how many joules are released per degree of crank rotation. Try the graph: make the delay long and watch the spike grow.

INTERACTIVE

CI engine: rate of heat release (ROHR) vs crank angle

Fuel is injected over the blue bar. A long ignition delay lets more fuel pile up, and it all burns at once: a tall premixed spike = diesel knock.



Fuel burnt in the premixed spike ≈ **31 %** · knock risk: **low: smooth engine**

Stages: **a–b** ignition delay (fuel evaporates and mixes, nothing burns) → **b–c** premixed (rapid, uncontrolled) → **c–d** mixing-controlled (burns as fast as fuel and air mix) → **d–e** late combustion tail. Shorten the delay with high cetane fuel, higher compression ratio, hotter or boosted intake air.

Before you start: how a CI engine burns

Fuel is sprayed into hot air. It does not burn at once: there is an **ignition delay** while droplets evaporate and mix. Then the fuel collected during the delay burns suddenly (premixed burning, sharp pressure rise), then the rest burns as it is injected (mixing-controlled), and finally the last bits burn slowly. ROHR = rate of heat release.

Classification of flames; premixed vs diffusion flames (2020 5a, 2022 5c)

	Premixed flame	**Diffusion (non-premixed) flame**
Mixing	Fuel and air mixed before combustion	Fuel and oxidizer separate; mix by diffusion/turbulence at the flame
Rate controlled by	Chemical kinetics and flame speed; flame front propagates through the mixture	Rate of mixing; flame sits where $\varphi \approx 1$
Appearance	Blue, little soot	Yellow, luminous soot
Examples	SI (spark ignition (petrol engine)) engine, Bunsen burner, gas stove	CI engine (after premixed phase), candle, oil burner

Flames are also classified as **laminar or turbulent**, and **stationary or propagating**.

Self-ignition temperature, SIT (2022 5c)

The **self-ignition temperature** is the lowest temperature at which a fuel-air mixture ignites by itself (auto-ignites) without a spark or flame, the heat from chemical reaction exceeding heat loss. Physically it measures how readily a fuel auto-ignites: low SIT → good CI fuel (high cetane, short delay); high SIT → good SI fuel (high octane, knock resistant).

Stages of combustion in a CI engine with ROHR vs crank angle (2023 8c, 2022 6c, 2020 5a)

> **Sketch:** heat release rate (J/°CA) vs crank angle. Mark SOI (start of injection) ~15-20° bTDC; a flat or slightly negative portion (a-b, evaporation cooling) = ignition delay; a tall narrow **spike** just before/at TDC (b-c) = premixed combustion; a lower broad **hump** (c-d) = mixing-controlled combustion, ending around end of injection; a long low **tail** (d-e) into expansion = late combustion.

Stage	Period	Description
1. Ignition delay (a-b)	From start of injection to start of combustion	Physical delay (atomization, evaporation, mixing, heating) + chemical delay (pre-flame reactions). Fuel accumulates in the chamber
2. Premixed / rapid / uncontrolled combustion (b-c)	A few CA degrees	Fuel that mixed to within flammable limits during the delay burns almost at once → very high rate of heat release and of pressure rise ($dp/d\theta$); a long delay makes this phase violent (diesel knock, noise)
3. Mixing-controlled / diffusion combustion (c-d)	Until about the end of injection	Rate set by how fast fuel vapour and air mix; controlled by injection rate, swirl and turbulence. Main heat release; soot forms in rich core and NO in hot diffusion flame
4. Late combustion (d-e)	Well into expansion	Remaining fuel, soot and fuel-rich products burn slowly as mixing continues; rate falls with temperature during expansion

Remember: CI = delay → premixed spike → mixing-controlled hump → late tail. Long delay = big spike = diesel knock.

B3 Volumetric efficiency: significance, mean piston speed, valve timing and speed effects, quasi-static effect

Must study · Possible · 33% · Medium

PAST-PAPER QUESTIONS

- **2022 Q2(c)** [10] How quasi-static effect influences the volumetric efficiency of an internal combustion engine. (second part)
- **2021 Q5(b)** [10] What is the physical significance of volumetric efficiency? "Mean piston speed ranges within 8 to 15 m/s" – Discuss.
- **2021 Q6(c)** [9] How inlet valve opening and closing, and engine speed affect volumetric efficiency?

Symbols used here

Symbol	Name and meaning	Unit
η_v	Volumetric efficiency	%
V_s (V_d)	Swept volume = $(\pi/4)d^2L$ per cylinder	m^3
S_p	Mean piston speed = 2LN (N in rev/s)	m/s (8–15)
L	Stroke	m
IVO (inlet valve opens), IVC (inlet valve closes)	Inlet valve opens / closes (crank angle)	°
p_e/p_i	Exhaust to intake pressure ratio	no unit
F/A (fuel-air ratio)	Fuel-air ratio by mass	no unit
M_a, M_f	Molar mass of air (29) and fuel	kg/kmol

Before you start: volumetric efficiency

Volumetric efficiency η_v = actual mass of air drawn in per stroke / mass that would fill the swept volume at outside conditions. It shows how well the engine "breathes". More air = more fuel can be burnt = more power.

Physical significance of volumetric efficiency (2021 5b)

η_v = actual mass of air inducted per cycle / ($\rho_a \times V_s$). It shows the engine's **breathing capacity**. Because the fuel that can be burned is limited by the air in the cylinder, **maximum power and torque are proportional to η_v**; a high η_v means more power from the same engine size.

Factors: quasi-static effects (fuel vapour, residual gas, p_e/p_i), intake and exhaust flow friction, port and valve flow, heat transfer to the charge, valve timing, choking at the inlet valve, intake/exhaust tuning, supercharging.

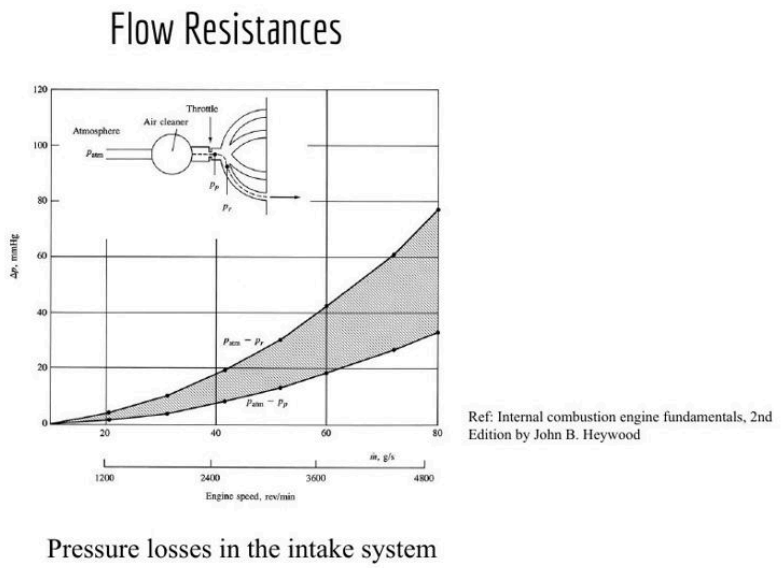
"Mean piston speed ranges within 8 to 15 m/s" (2021 5b)

Mean piston speed S_p = 2LN (L stroke, N rev/s). It is a better measure of engine speed than rpm because flow velocities, inertia stresses and friction all depend on it. Engines of all sizes are designed for maximum S_p of about **8-15 m/s** (large slow diesels ~8, car engines ~15):

- **Upper limit**: gas velocity through the inlet valve rises with S_p; above ~15 m/s flow resistance and **choking** at the inlet valve make η_v fall sharply; inertia forces of reciprocating parts (stress), friction and wear rise with S_p^2; piston ring and lubrication limits.
- **Lower limit**: below ~8 m/s power per unit size is too small and heat loss is large.
- So a long-stroke engine must run at lower rpm and a short-stroke engine can run at higher rpm; large engines run slowly, small engines fast, but at similar S_p.

Effect of inlet valve opening and closing, and engine speed, on η_v (2021 6c)

- **Inlet valve opening (IVO)** before TDC (top dead centre): the valve is well open when the piston starts down, so the inflow is not restricted; with overlap the outgoing exhaust helps pull fresh charge in (scavenging), raising η_v at high speed. Too early at low speed → exhaust flows back into the intake (more residual).
- **Inlet valve closing (IVC)** after BDC (30-60°): the moving charge has momentum (**ram effect**), so it keeps flowing in after BDC (bottom dead centre); late IVC raises η_v at **high speed**. At **low speed** the same late closing lets the rising piston push charge back into the manifold (**backflow**), lowering η_v.
- **Engine speed**: η_v vs speed is a hump. At low speed: backflow and heat transfer (charge heated longer). At high speed: flow friction ∝ speed² and choking at the valve; less time to fill. Valve timing, port design and intake tuning set the speed of maximum η_v (which is also the maximum-torque speed). Variable valve timing tries to keep it high over the range.



Pressure losses in the intake system

Pressure losses in the intake system grow with engine speed

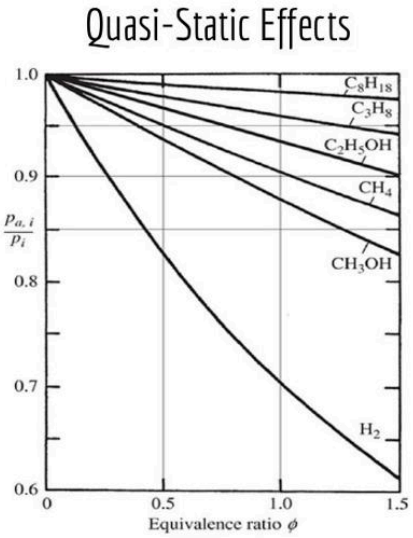
Quasi-static effects on η_v (2022 2c)

Quasi-static effects are those that do not depend on speed or flow dynamics:

- **Fuel vapour (and water vapour) in the intake**: they occupy volume and reduce the partial pressure of air: $p_{a,i}/p_i = 1/[1 + (F/A)(M_a/M_f) + \text{water}]$. For petrol vapour the loss is ~2 %; for gaseous fuels (CH_4, H_2) and alcohols much larger (H_2 ~30 %).
- **Ratio of exhaust to intake pressure** p_e/p_i and **compression ratio**: residual gas left in the clearance volume expands when the intake valve opens and takes space; η_v falls as p_e/p_i rises

(throttled engine) and the effect is smaller at high compression ratio.

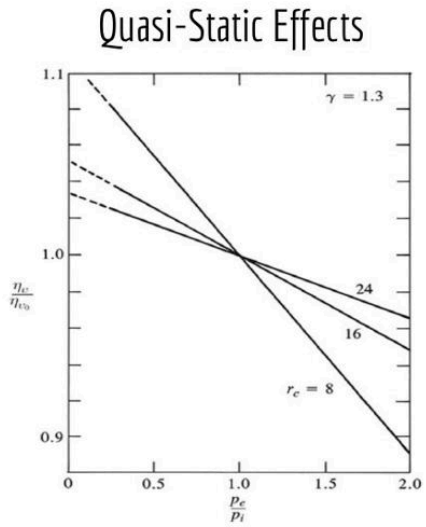
- Charge heating by hot walls/residual and latent heat cooling by fuel evaporation (which raises η_V, e.g. alcohols).



Ref: Internal combustion engine fundamentals, 2nd Edition by John B. Heywood

Effect of fuel (vapor) on inlet air partial pressure

Effect of fuel vapour on inlet air partial pressure (quasi-static)



Ref: Internal combustion engine fundamentals, 2nd Edition by John B. Heywood

Effect of Intake and Exhaust Pressure Ratio and Compression Ratio

Effect of exhaust/intake pressure ratio and compression ratio

B2 Rope brake engine test numerical (BP, IP, ηm, ηbth, BSFC) Skip if short

Possible · 33% Medium

PAST-PAPER QUESTIONS

- **2024 Q6(b)** [20] Single-cylinder four-stroke oil engine with rope brake: brake wheel diameter 600 mm, dead load 400 N, spring balance 50 N, speed 600 rpm, indicator diagram area 420 mm², length 60 mm, spring scale 1.1 bar/mm, bore 100 mm, stroke 150 mm, oil 0.815 kg/h, CV (calorific value) 44000 kJ/kg. Calculate BP (brake power), IP (indicated power), mechanical efficiency, brake thermal efficiency and BSFC (brake specific fuel consumption).

Symbols used here

Symbol	Name and meaning	Unit
D	Effective diameter of the brake drum (rope centre)	m
W	Dead load hung on the rope	N
S	Spring balance reading on the other end	N
N	Engine speed	rpm
a	Area of the indicator diagram (from the planimeter)	mm²
l	Length (base) of the indicator diagram	mm
s	Spring scale of the indicator (pressure per mm of height)	bar/mm
p_{mi}	Indicated mean effective pressure	bar ($\times 10^5$ for N/m²)
L	Stroke length	m
A	Piston area = $\pi d^2/4$ (d = bore)	m²
n	Power strokes per minute: N/2 four-stroke, N two-stroke	per min
$\dot{m}_f$	Fuel consumption	kg/h
CV	Calorific value of fuel	kJ/kg
η_m, η_{bth}	Mechanical efficiency BP/IP, brake thermal efficiency BP/fuel power	%

UNDERSTAND **Read this to understand; you do not need to write it**

Two powers, two measurements. The rope brake measures what comes **out of the shaft** (BP). The indicator diagram measures what the gas does **on the piston** (IP). The difference is lost to friction, so BP must be smaller than IP and η_m must be below 100 %.

BP: rope force difference (W − S) acts at radius D/2: torque T = (W − S)D/2. Power = 2πNT/60 = (W − S)πDN/60 watts.

IP: the average height of the indicator loop (a/l in mm) times the spring scale gives the average gas pressure p_{mi}. Force = p_{mi} × A; work per power stroke = force × L; times power strokes per

second. Memory: **PLAN/60**.

With the 2024 numbers BP comes out bigger than IP. Do every step anyway (marks are for the method) and write one line that the data are inconsistent.

INTERACTIVE **Rope brake test calculator**

Defaults are the 2024 6(b) data. Change any value to practise a new version of the question.

1. BP = (W − S)πDN/60 = (400 − 50) × π × 0.6 × 600/60 = **6.597 kW**
2. p_{mi} = as/l = 420 × 1.1/60 = **7.70 bar**
3. A = π/4 × 0.1² = 0.00785 m²; n = 300 power strokes/min
4. IP = p_{mi}LAn/60 = 770000 × 0.15 × 0.00785 × 300/60 = **4.536 kW**
5. η_m = BP/IP = **145.5 %** **(> 100 %: the data are inconsistent, say so in one line)**
6. Fuel energy = 0.815 × 44000/3600 = 9.961 kW; η_{bth} = BP/fuel energy = **66.2 %**; η_{ith} = 45.5 %
7. BSFC = 0.815/6.597 = **0.124 kg/kWh**; FP = IP − BP = -2.062 kW

Before you start: the rope brake test in plain words

A rope is wrapped round the engine flywheel; one end carries a dead weight W, the other is held by a spring balance reading S. Friction of the rope absorbs the engine power. Net force = W − S, acting at radius D/2, so torque T = (W − S) D/2 and BP = 2πNT/60 = (W − S)πDN/60.

The **indicator diagram** (pressure vs volume drawn by an instrument) gives the mean effective pressure: p_{mi} = area × spring constant / length. A four-stroke engine fires once every 2 revolutions, so power strokes per minute n = N/2.

Rope brake test (2024 6b, 20 marks)

Data: D = 0.6 m, W = 400 N, S = 50 N, N = 600 rpm, a = 420 mm², l = 60 mm, s = 1.1 bar/mm, bore d = 0.1 m, stroke L = 0.15 m, $\dot{m}_f$ = 0.815 kg/h, CV = 44 000 kJ/kg, four-stroke, single cylinder.

Brake power: BP = (W − S) π D N / 60 = (400 − 50) × π × 0.6 × 600 / 60 = 350 × π × 6 = 6 597 W = **6.60 kW**

IMEP (indicated mean effective pressure)**:** p_{mi} = a s / l = 420 × 1.1 / 60 = **7.7 bar** = 7.7×10^5 N/m²

Piston area A = π/4 × 0.1² = 7.854×10^{-3} m²; power strokes per min n = N/2 = 300 (four-stroke)

Indicated power: IP = p_{mi} L A n / 60 = $7.7 \times 10^5 \times 0.15 \times 7.854 \times 10^{-3} \times 300 / 60$ = 4 536 W = **4.54 kW**

Mechanical efficiency: η_m = BP / IP = 6.60 / 4.54 = 1.45 (145 %)

Brake thermal efficiency: η_{bth} = BP × 3600 / ($\dot{m}_f$ × CV) = 6.597 × 3600 / (0.815 × 44 000) = 23 750 / 35 860 = **66.2 %**

BSFC = $\dot{m}_f$ / BP = 0.815 / 6.597 = **0.124 kg/kWh**

Important: with the numbers printed, BP comes out larger than IP (η_m > 100 %) and η_{bth} = 66 % is unrealistic, so the loads in the paper are inconsistent (a real engine of this size would give η_m 70-80 %, η_{bth} 25-30 %, BSFC 0.25-0.35 kg/kWh). In the exam: show every formula and substitution as above (the marks are for the method), then write one line: "η_m > 1 shows the given data are inconsistent." If the examiner took n = N by mistake, IP = 9.07 kW and η_m = 72.7 %; mention this as an alternative only if asked.

Also useful: indicated thermal efficiency η_{ith} = IP × 3600 / ($\dot{m}_f$ CV) = 4.536 × 3600 / 35 860 = 45.5 %; friction power FP (friction power) = IP − BP.

Remember: BP = (W−S)πDN/60. p_{mi} = as/l. IP = p_{mi}LAn/60, **n = N/2** for four-stroke. η_m = BP/IP. η_{bth} = 3600 BP/($\dot{m}_f$ CV). BSFC = $\dot{m}_f$/BP.

Same answer, step by step with reasons

Step (write this)	Why / what it means
BP = (W - S) π D N / 60 = 350 × π × 0.6 × 600/60 = 6.60 kW	Net rope force × rim speed (πDN/60 m/s) = power.
p_{mi} = a s / l = 420 × 1.1 / 60 = 7.7 bar	Average height of the indicator loop (area / length) × spring scale (bar per mm).
A = π/4 × 0.1^2 = 0.007854 m^2; n = 600/2 = 300	Piston area; four-stroke fires every second revolution.
IP = p_{mi} L A n / 60 = 7.7 × 10^5 × 0.15 × 0.007854 × 300/60 = 4.54 kW	Remember **PLAN**: Pressure × Length × Area × Number of power strokes per second.
n_m = BP/IP = 6.60/4.54 = 1.45	Should be below 1; here it is not, so the paper's data are inconsistent (say so in one line).
n_{bth} = BP / (fuel energy per s) = 6.597 / (0.815 × 44,000/3600) = 66.2 %	Fuel energy rate = kg/h × kJ/kg / 3600 = kW.
BSFC = 0.815/6.597 = 0.124 kg/kWh	Fuel per hour divided by brake kW.

Memory trick: IP = **PLAN**/60 (n = N/2 for four-stroke).

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

Given: D = 0.6 m, W = 400 N, S = 50 N, N = 600 rpm, a = 420 mm², l = 60 mm, s = 1.1 bar/mm, d = 0.1 m, L = 0.15 m, $\dot{m}_f$ = 0.815 kg/h, CV = 44 000 kJ/kg, four-stroke

Brake power: BP = (W − S)πDN / 60

= (400 − 50) × π × 0.6 × 600 / 60 = 6597 W

= **BP = 6.60 kW**

Indicated mean effective pressure: p_{mi} = a s / l

= 420 × 1.1 / 60

= **p_{mi} = 7.7 bar**

A = π/4 × 0.1² = 7.854 × 10^{-3} m²; n = N/2 = 300 per min (four-stroke)

Indicated power: $IP = p_{mi}\, L\, A\, n / 60$

$= 7.7 \times 10^{5} \times 0.15 \times 7.854 \times 10^{-3} \times 300 / 60 = 4536$ W

= **$IP = 4.54$ kW**

Mechanical efficiency: $\eta_m = BP / IP$

$= 6.60 / 4.54$

= **$\eta_m = 1.45$ (145 %)**

Brake thermal efficiency: $\eta_{bth} = BP / (\dot{m}_f \times CV / 3600)$

$= 6.597 / (0.815 \times 44\,000 / 3600) = 6.597 / 9.961$

= **$\eta_{bth} = 66.2$ %**

BSFC: $BSFC = \dot{m}_f / BP$

$= 0.815 / 6.597$

= **0.124 kg/kWh**

Comment: $\eta_m > 1$ is impossible (BP cannot exceed IP), so the given data are inconsistent; the method above is the required one.

B7 Air-standard cycles: Otto efficiency derivation; Diesel cut-off ratio and γ

Good to have | Possible · 33% | Hard

PAST-PAPER QUESTIONS

- **2024 Q5(c)** [13] Sketch the P-V diagram of Otto cycle and derive mathematical expression for thermal efficiency of Otto cycle.
- **2024 Q6(a)** [6] How does cut-off ratio affect diesel cycle efficiency? What role does the specific heat ratio (γ) play in cycle efficiency?

LEARN IT FROM ZERO

Why the efficiency depends only on compression ratio

An engine turns heat into work by letting hot, high-pressure gas **push the piston and expand**. The more the gas expands after the heat goes in, the more of that heat becomes work, and the cooler the gas is when it is thrown out. The compression ratio r (biggest volume ÷ smallest volume) is exactly "how many times the gas expands". So a bigger r means more work from the same heat.

The **Otto cycle** (petrol engine idealised) has four steps: **1→2** squeeze the air without heat loss (isentropic); **2→3** add heat instantly at the top, volume fixed (the spark burns everything at once); **3→4** let it expand without heat loss; **4→1** dump the leftover heat with volume fixed (exhaust).

The trick in the derivation: in an isentropic process temperature changes by the factor $r^{\gamma-1}$ when volume changes by r. Compression raises T by $r^{\gamma-1}$ and expansion lowers it by the same factor. So the heat thrown out ($T_4 - T_1$) is exactly the heat put in ($T_3 - T_2$) divided by $r^{\gamma-1}$. Efficiency = 1 − out/in = $\mathbf{1 - 1/r^{\gamma-1}}$. The temperatures cancel: only r and γ are left.

The **Diesel cycle** adds its heat at constant pressure while the piston already moves down (from volume 1 to ρ, the **cut-off ratio**). Heat added late is expanded less, so for the same r the Diesel is less efficient, and the more fuel (bigger ρ) the worse. Diesels still win in real life because they use r = 14–22 while petrol engines are stuck at 8–11 by knock.

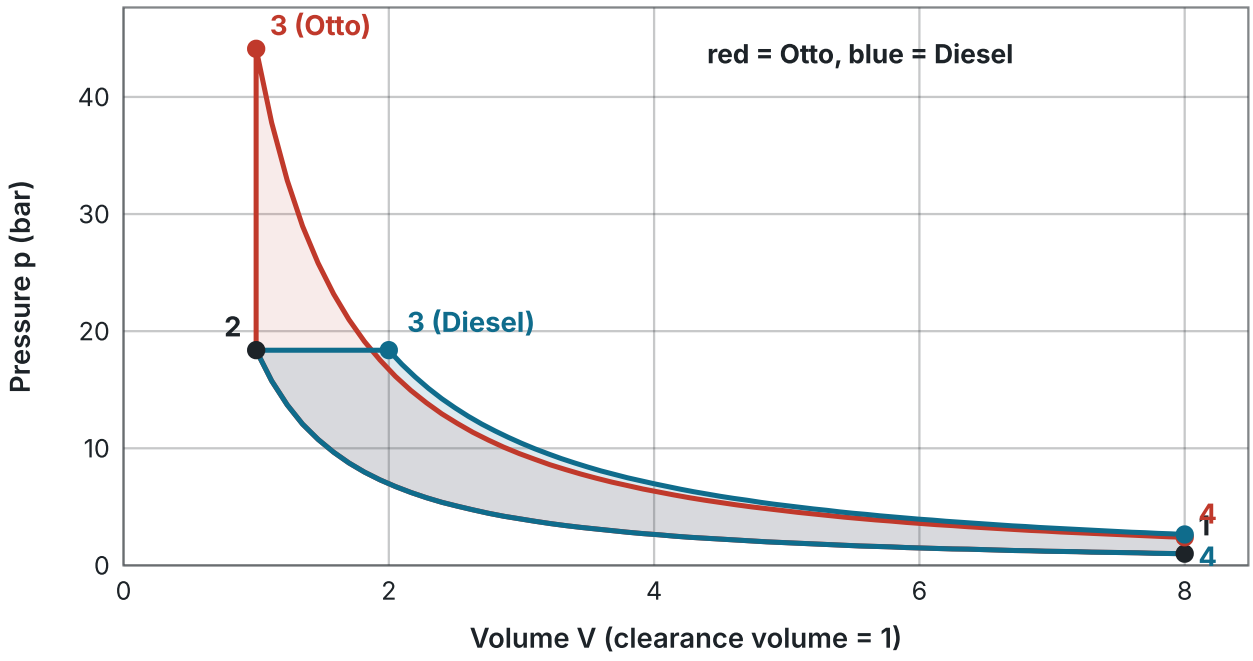
Symbols used here

Symbol	Name and meaning	Unit
r	Compression ratio = V_1/V_2 = (swept + clearance) / clearance	no unit
ρ (r_c)	Cut-off ratio = V_3/V_2, volume when fuel injection (heat addition) stops	no unit
r_p	Pressure ratio during constant-volume heat addition = p_3/p_2	no unit
γ	Ratio of specific heats c_p/c_v (air 1.4)	no unit
c_v, c_p	Specific heat at constant volume / pressure (air 0.718 / 1.005)	kJ/kg K
Q_s, Q_R	Heat supplied / heat rejected per cycle	kJ
$T_1 \ldots T_4$, p, V	Temperature, pressure, volume at the four corners of the cycle	K, bar, m^3
η	Air-standard thermal efficiency = work / heat supplied	%
MEP (mean effective pressure)	Mean effective pressure = net work / swept volume	bar

INTERACTIVE

Otto vs Diesel cycle: p–V and T–s

Both cycles start at 1 bar, 300 K, have the same compression ratio r and get the same heat. Move the sliders and watch the efficiencies.



$\eta_{Otto} = 1 - 1/r^{\gamma-1} = 56.5\ \%$ · $\eta_{Diesel} = 49.0\ \%$ · T_2 = 689 K, p_2 = 18.4 bar

Same r and same heat in: Otto wins, because all its heat goes in at the smallest volume (point 2) and is then expanded the full r times. The Diesel adds heat while the piston already moves away,

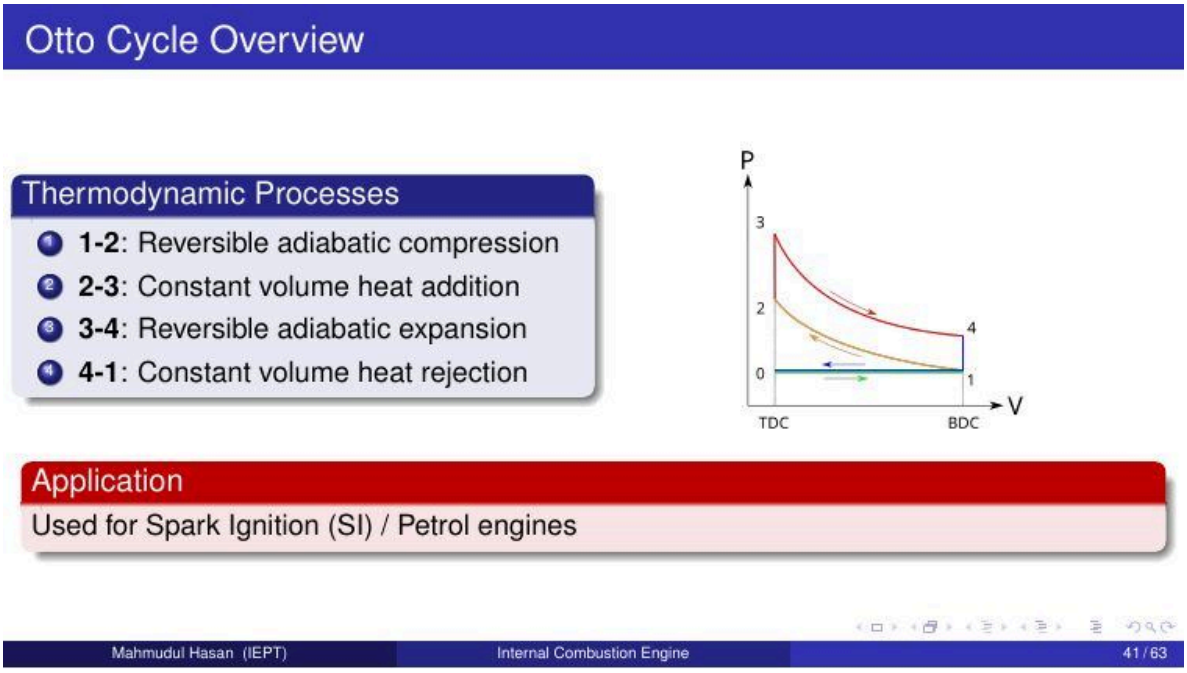
so the later part of the heat is expanded less. Push ρ up: the Diesel efficiency falls. Push ρ to 1: the two become equal.

Before you start: what an air-standard cycle is

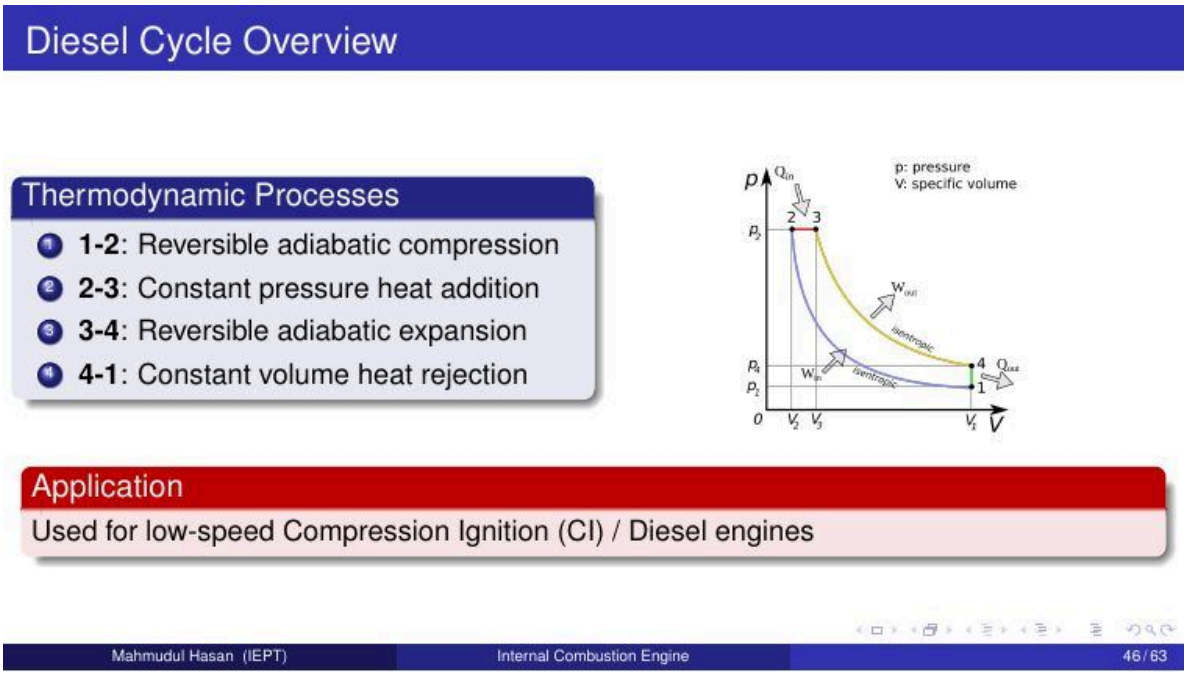
A real engine is complicated, so we study an idealised version: the working fluid is pure air (ideal gas, constant c_p, c_v), compression and expansion are **isentropic** (no friction, no heat loss: pV^{γ} = constant, $TV^{\gamma-1}$ = constant), and burning is replaced by adding heat. Otto cycle = heat added at constant volume (petrol); Diesel cycle = heat added at constant pressure (diesel).

$\gamma = c_p/c_v = 1.4$ for air. Compression ratio r = largest volume / smallest volume.

Otto cycle: p-V diagram and thermal efficiency derivation (2024 5c)



Otto cycle p-V: 1-2 isentropic compression, 2-3 constant volume heat addition, 3-4 isentropic expansion, 4-1 constant volume heat rejection



Diesel cycle p-V: heat added at constant pressure 2-3

Air-standard assumptions: air as working fluid, ideal gas with constant specific heats, reversible processes, heat added and rejected externally. Compression ratio $r = V_1/V_2 = V_4/V_3$.

> Heat supplied (2-3, constant volume): $Q_s = m\, c_v\, (T_3 - T_2)$
> Heat rejected (4-1, constant volume): $Q_R = m\, c_v\, (T_4 - T_1)$
> $\eta = (Q_s - Q_R)/Q_s = 1 - (T_4 - T_1)/(T_3 - T_2)$
> Isentropic 1-2: $T_2/T_1 = (V_1/V_2)^{\gamma-1} = r^{\gamma-1} \Rightarrow T_2 = T_1\, r^{\gamma-1}$
> Isentropic 3-4: $T_3/T_4 = (V_4/V_3)^{\gamma-1} = r^{\gamma-1} \Rightarrow T_3 = T_4\, r^{\gamma-1}$
> $T_3 - T_2 = r^{\gamma-1}(T_4 - T_1)$
> $\therefore \mathbf{\eta_{Otto} = 1 - 1 / r^{\gamma-1}}$

Efficiency depends only on compression ratio (and γ); e.g. r = 8, γ = 1.4: $\eta = 1 - 1/8^{0.4} = 56.5$ %. In SI (spark ignition (petrol engine)) engines r is limited (6-10.5) by knock. Also MEP = $p_1\, r\, [(r^{\gamma-1} - 1)(r_p - 1)] / [(\gamma - 1)(r - 1)]$.

Effect of cut-off ratio on Diesel cycle efficiency; role of γ (2024 6a)

$\eta_{Diesel} = 1 - [1/(\gamma\, r^{\gamma-1})] \times (\rho^{\gamma} - 1)/(\rho - 1)$, cut-off ratio $\rho = V_3/V_2$

- The bracket $(\rho^{\gamma} - 1)/[\gamma(\rho - 1)]$ is always > 1 and **increases with ρ**. So for a fixed compression ratio, a larger cut-off ratio (more fuel injected for more load, heat added over a longer part of the stroke) **reduces efficiency**: the later heat is added, the less it is expanded. As ρ → 1 the Diesel cycle approaches Otto efficiency.
- Hence for the same r, $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$; but diesels use much higher r (14-22), so they are more efficient in practice. For the same peak pressure: $\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$.
- **Role of γ**: efficiency increases with γ because the temperature rise per unit compression ($T_2/T_1 = r^{\gamma-1}$) is greater, so more work is extracted per unit heat. Air γ = 1.4 gives the ideal value; in real engines fuel vapour, H_2O and CO_2 and high temperature reduce γ to ~1.25-1.3, which is one

reason real (fuel-air cycle) efficiencies are lower than air-standard; lean mixtures keep γ higher and so are more efficient.

Same answer, step by step with reasons

Step (write this)	Why / what it means
$\eta = 1 - Q_{out}/Q_{in} = 1 - (T_4 - T_1)/(T_3 - T_2)$	Efficiency = (heat in − heat out)/heat in; at constant volume Q = m $c_v\Delta T$, and m c_v cancels.
$T_2 = T_1\, r^{\gamma-1}$	Isentropic compression: $TV^{\gamma-1}$ = constant, volume shrinks r times.
$T_3 = T_4\, r^{\gamma-1}$	Expansion has the same volume ratio r.
$T_3 - T_2 = r^{\gamma-1}(T_4 - T_1)$	Subtract the two lines.
$\eta = 1 - 1/r^{\gamma-1}$	The bracket cancels. Example: r = 8 gives 1 − 1/2.297 = 56.5 %.

Memory trick: Otto: **"one minus one over r to the 0.4"**. Diesel = Otto × a penalty bracket that grows with cut-off ratio.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

Otto cycle efficiency. [Draw p–V: 1 → 2 isentropic compression, 2 → 3 heat added at constant V, 3 → 4 isentropic expansion, 4 → 1 heat rejected at constant V.] Assume air-standard: ideal gas, constant c_v, reversible processes. $r = V_1/V_2 = V_4/V_3$.

Heat supplied $Q_s = m\, c_v(T_3 - T_2)$; heat rejected $Q_R = m\, c_v(T_4 - T_1)$

$\eta = 1 - Q_R/Q_s = 1 - (T_4 - T_1)/(T_3 - T_2)$ … (1)

$1 \to 2$: $T_2/T_1 = (V_1/V_2)^{\gamma-1} = r^{\gamma-1}$; $3 \to 4$: $T_3/T_4 = (V_4/V_3)^{\gamma-1} = r^{\gamma-1}$

$\therefore T_2 = T_1 r^{\gamma-1}$, $T_3 = T_4 r^{\gamma-1} \Rightarrow T_3 - T_2 = r^{\gamma-1}(T_4 - T_1)$

Put in (1): $\boxed{\eta_{Otto} = 1 - 1/r^{\gamma-1}}$ e.g. r = 8, γ = 1.4: $\eta = 1 - 1/8^{0.4} = 56.5\ \%$

Cut-off ratio (Diesel): $\eta_D = 1 - (1/r^{\gamma-1}) \times (\rho^{\gamma} - 1)/[\gamma(\rho - 1)]$. The bracket is > 1 and grows with ρ, so a larger cut-off ratio (more load) lowers efficiency; ρ → 1 gives Otto. Higher γ raises efficiency (bigger temperature change per compression), so real gases with lower γ give lower efficiency.